Tîm Trawma Cymru

1 - Introduction

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Date: 03/19

Review Date: 03/21

Thank you for registering for the Tîm Trawma Cymru Course. The aim of this course is to develop well-trained, well-practiced, highly efficient trauma teams across the three trauma units in North Wales. It is designed and delivered by clinicians in our own Trauma Network, and is constantly evolving to give you the most up-to-date advice from our local experts. Our goal is for North Wales to have a reputation for excellent trauma training and world-class trauma care.

This course is in-house training that is purposefully designed to train how best to manage trauma patients in our hospitals in North Wales. It is not meant to be an alternative to your own professional training, ATLS, ATACC or European trauma course. It is recommended that all doctors involved with trauma also complete and keep in date for these courses. For the nurses, the complete course will count as level 1 training for your ANTC. The complete course will count as 16 CPD points (See below).

There are 3 parts to the North Wales Trauma Course.

A. Pre-course reading and MCQ – to be completed by all (You can self-accredit CPD points for this if you complete a reflection on it - 1 CPD per hour of work to completely read the manual and complete the MCQ. As a guide this should equate to 6 CPD points)
B. Trauma skills course – to be completed by nurses and ODPs (Accredited for 5 CPD points)
C. Trauma Team Simulation course – to be completed by all (Accredited for 5 CPD points)



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Part A: Pre-course reading & MCQ

The purpose of this pre-course reading is to ensure that everyone gets the most out of the very limited time on the simulation day. We won't have much time to teach the basics of trauma management and so it is important that everyone has a basic working knowledge before they attend. It is also to ensure that you familiarize yourself with our local policies and guidelines – which you should know already anyway!

The chapters are deliberately kept brief, so they are only brief summaries of the topic. We would expect many of you (especially the senior nurses and doctors) to have a much more in depth knowledge and understanding and we have provided some links and references for further reading around the subject if you feel you need it. This manual does not supersede your own departmental guidelines. You must operate within your own departmental guidelines and protocols.

Part B: Trauma skills course

The trauma skills course is aimed at everyone on the trauma team and is designed to cover basic skills and procedures that you would be expected to know how to do as part of a trauma team member. These include an introduction / revision of the following:

1. Application of pelvic binders
2. Preparing for Rapid Sequence Induction and being an emergency airway assistant
3. Intra-osseous access
4. Use of major haemorrhage protocol and blood product administration
5. Use of the rapid infuser (Belmont)
6. Application of splints for limb fractures
7. Application of cervical collars
8. Use of scoops and vacuum mattresses
9. Logrolling
10. Scribing
11. Communication & Handovers

Not every skill is relevant for every role, but you will probably find it useful anyway to have an insight in to them.

We do not include training on every skill that you would be expected to have. There is limited time and we have to be realistic and pragmatic about what you are most likely to need and what we can cover in a day. For example – insertion of an intercostal chest drain is too complicated a skill to teach on this course. If this is a skill that you need, then you need to learn that in your normal workplace or on a specific course.

Part C: Trauma Team Simulation Course

The final part of the course is putting it all together and focusing on how to get our trauma teams

working together like a world class, highly trained, well practiced team – even when they have not had the chance to practice together before.

You will watch a video demonstration of how a trauma team should work together. You will then go through 6 simulations. Three before lunch where you will have a bit more coaching form the faculty, then three after lunch which will be more challenging.

This will be a long and demanding day with challenging scenarios and little time for breaks – just like a normal day at work! It should also be the most fun part of the course.

My advice is to bring tons of enthusiasm, and a sense of humour.

Summary

- To get the most out of this course, be keen: do the background reading, read around the topics and familiarise yourself with your department equipment, guidelines and protocols.
- Supplement this course with regular skills teaching and simulation practice in your departments.
- Look out for and attend trauma governance meetings in your hospital – or go to the EMRTS clinical governance days (see www.emrts.cymru).
- Attend trauma conferences such as Traumacare or the London Trauma Conference to keep up to date.
- We are always looking for fresh, enthusiastic faculty to join our training team. If you are interested please email me at mark.knights@wales.nhs.uk

2 - Epidemiology of Trauma.

Author: Sue O'Keeffe, Critical Care & Trauma Network Manager, Wales

Date: 06/17

Review Date: 06/19

Introduction

Major trauma has traditionally been viewed as a disease of young men caused by high-energy transfer mechanisms of injury and it is the leading cause of death and disability in those under 40 years old.[1]

Major trauma in elderly patients is becoming increasingly recognised as a significant challenge to health systems.[2] In many countries, the elderly comprise the most rapidly expanding section of the population. Advances in medical care mean that many enjoy better health, mobility and independence longer than ever before, increasing their exposure to the possibility of injury. Frailty is also a factor; the likelihood of falls increases with age as does the probability of significant injury as a consequence, while physiological reserve and the ability to recover is diminished. [3]

Background

As noted in the TARN (Trauma Audit and Research Network) chapter an ISS (Injury Severity Score) of >15 is the threshold defining major trauma. In 2012 BCUHB (Betsi Cadwaladr University Health Board) commissioned Royal Stoke University Hospital (RSUH) Major Trauma Centre to be their provider for major trauma i.e. to care for and treat those patients with an ISS >15 who need specialist management.

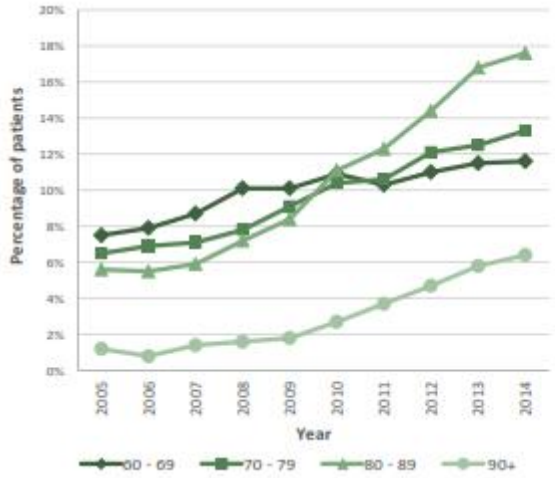
The volume of trauma patients transferred for specialist trauma care has increased year on year. In 2013, the first full year of operation, 108 patients were transferred. In 2018, 188 patients went to RSUH for definitive trauma care. *NB: these figures exclude 'out of region' patients and TARN negative patients.*

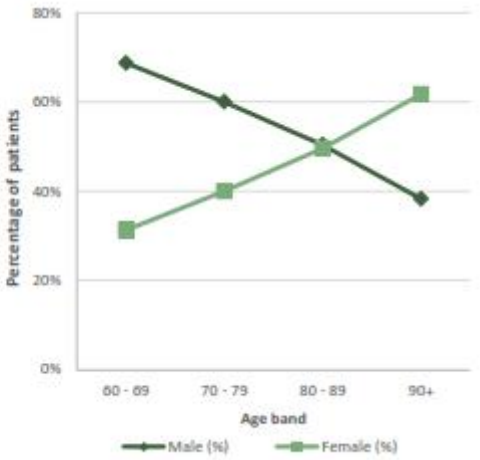
National picture

A recent study[4] demonstrated a profound change in the demographics of major trauma recorded in the UK over the last 20 years.

The information, derived from TARN, suggests that the true incidence of major trauma in the elderly greatly exceeds previous predictions. Rather than being a disease of young men, the majority of patients suffering major trauma identified in the UK are now older than 50 years of age. This age group now accounts for more than 50% of the severely injured patients registered in the Trauma Audit and Research Network database.

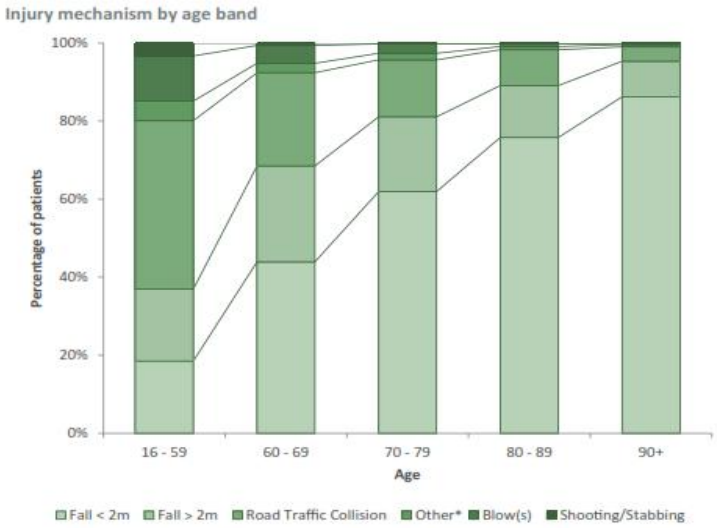
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Mechanism of injury changes with age. Low falls become a much more common cause of major trauma. A fall of <2m is now the most common mechanism of injury within the TARN database (it should be noted that patients with single fractures distal to the elbow or knee or an isolated fractured neck of femur are not included in the TARN dataset even if the patient spends more than 3 days in hospital; the inclusion of these injuries would further increase the number of patients suffering injuries from falls <2m).

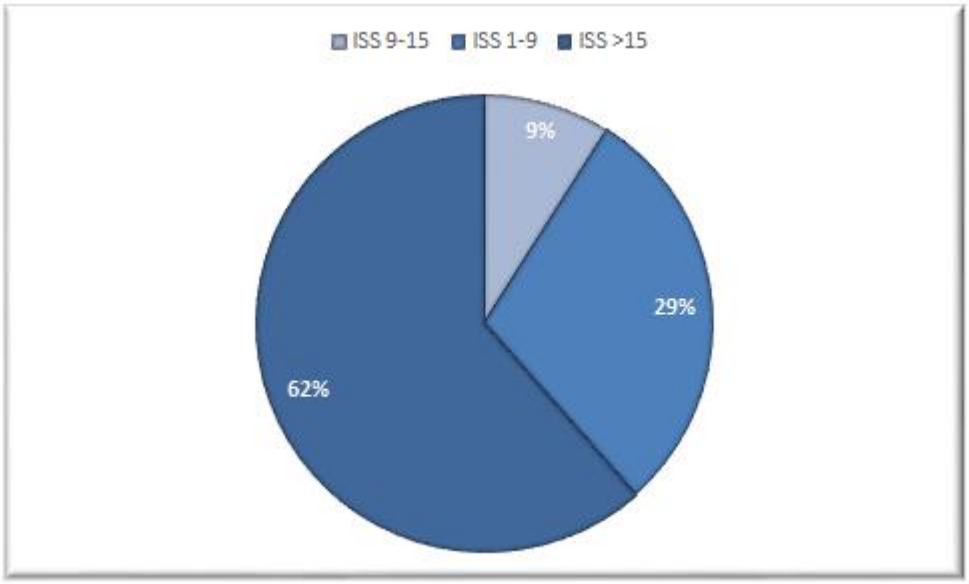
Younger people tend to have their trauma incidences outside, at the weekends and in the early hours whereas older people are very much more likely to be injured indoors. In contrast, older patients present with major trauma after 09:00. However, this statistic might not reflect the true timing of injury in older people, as there may be a time difference between injury and hospital presentation, as older patients may fall at night, but not be discovered until carers arrive in the morning.



Contrary to opinion there is no evidence of seasonal variation in the elderly incidences of trauma, possibly because most of their trauma occurs indoors. There is evidence however that pre-hospital triage appears to be less reliably applied to older people and, because of this, it is more likely that their initial assessment will be by a junior doctor (which then leads to delays in investigations and treatment).[5]

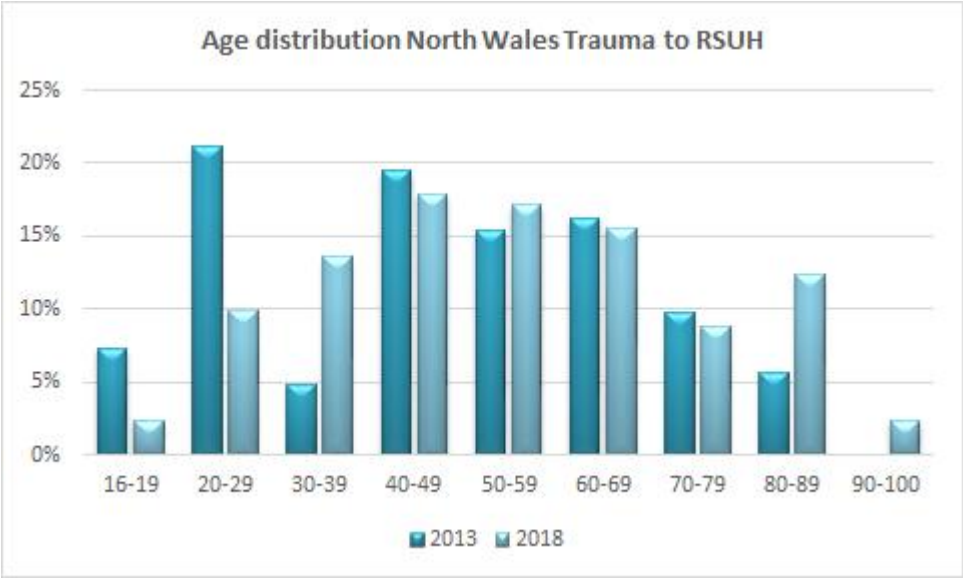
Local picture

Initially it was anticipated that the majority of the patients transferred to RSUH from North Wales would be the younger trauma population with an ISS >15.



North Wales to RSUH, ISS breakdown, Jan 2013-Dec 2018 (1004 patients - excluding TARN negative)
Source: TARN database

There is a noticeable, and expected, spike in the 20-30's transfers then a bell curve around the 50-60 age group. As per national trend North Wales has also seen an increase in elderly trauma as the graph comparing transfers 2013 and 2018 shows.



North Wales to RSUH, age distribution, Jan 2013-Dec 2018
Source: TARN database

Summary

The population in whom major trauma is identified and recorded is becoming more elderly, and the predominant mechanism that precipitates major trauma is a fall from <2 m. Training for those involved in the management of patients suffering trauma has often concentrated on the (young males) aspects of injury.[2] It is therefore important to note:

- There are now two types of major trauma:

  - High energy: young, male, RTC, triage positive, early detection, early intervention, MTC.
  - Low energy: older, female, fall <2m, triage negative, late detection, late intervention, TU.
- Within the next few years, the over 75s will be the single largest group suffering major trauma.
- The predominant mechanism of injury causing major trauma is no longer road traffic collision but low falls.
- Early identification of major trauma in older people is difficult.
- The specific needs of the elderly must be considered in the design of major trauma services.

References

[1]Krug EG, Sharma GK, Lozano R. The global burden of injuries. Am J Public Health 2000;90:523–6.

[2]Grossman MD, Ofurum U, Stehly CD, et al. Long-term survival after major trauma in geriatric trauma patients: the glass is half full. J Trauma Acute Care Surg 2012;72:1181–5.

[3]Bruijns SR, Guly HR, Bouamra O, et al. The value of traditional vital signs, shock index, and age-based markers in predicting trauma mortality. J Trauma Acute Care Surg 2013;74:1432–7.

[4]A Kehoe, J E Smith, A Edwards, D Yates, F Lecky. The changing face of major trauma in the UK. Emerg Med J 2015;32:911–915. Downloaded from http://emj.bmj.com/ on June 1, 2017

[5]Trauma Audit Research Network (TARN). Major Trauma in Older People, 2017.

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3 - North West Midlands & North Wales Trauma Network.

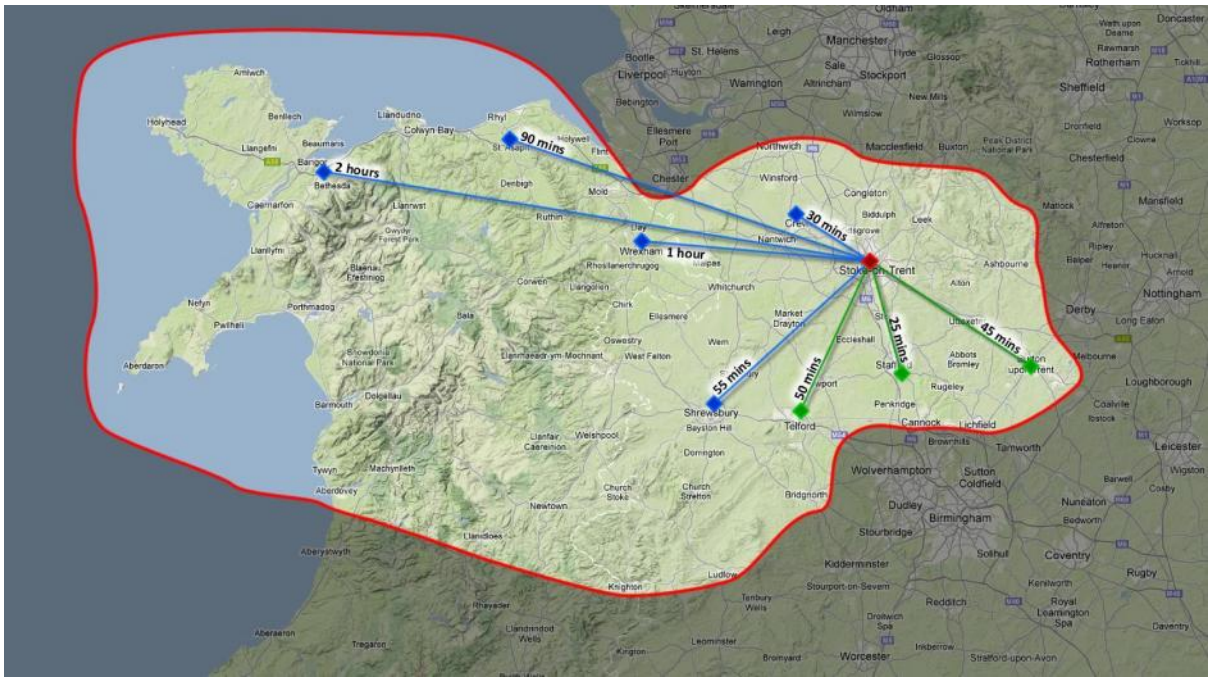
Author: Sue O'Keeffe, Critical Care & Trauma Network Manager, Wales

Date: 05/17

Review Date: 05/19

Introduction

The North West Midlands Trauma Network was established in April 2012. North Wales then joined and became part of the North West Midlands and North Wales Trauma Network in October 2012.



Major Trauma Centres are set up to provide specialised trauma care and rehabilitation. They are hubs that work closely with local Trauma Units. Major Trauma Centres operate 24 hours a day, 7 days a week. They are staffed by consultant-led trauma teams that meet the patient on arrival at the hospital and have immediate access to the best diagnostic and treatment facilities, including blood transfusion, CT scans, interventional radiology and emergency operating theatres.

Background

Every year across England and Wales, 12,500 people die after injury. It is the leading cause of death among children and young adults of 44 years and under. In addition, there are many thousands who are left severely disabled for life. Prior to the establishment of Major Trauma Networks the level of

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care for these patients was shown to be poor, with lack of regional organisation and a poor consultant level involvement in decision-making[1, 2] (National Confidential Enquiry into Peri-Operative Deaths (NCEPOD) 'Trauma who Cares' 2007, National Audit Office 'Major Trauma Care in England 2010').

The National Audit Office report concluded that *"major trauma care in England does not represent value for money because the service is not being delivered efficiently or effectively"*. Subsequently, the NHS Clinical Advisory Group's report of September 2010, 'Regional Networks for Major Trauma', led to the establishment of Major Trauma Centres, Trauma Units and Trauma Networks in England.

Definitions

Hospitals and services within the North West Midlands & North Wales Trauma Network

Major trauma centre	• Royal Stoke University Hospital (part of the University of North Midlands Trust)
Trauma units	• Leighton Hospital • Royal Shrewsbury Hospital • Wrexham Maelor Hospital • Glan Clwyd Hospital • Ysbyty Gwynedd (Bangor Hospital)
Pre-hospital care	• West Midlands Ambulance Service • North West Ambulance Service • Welsh Ambulance Service • West Midlands Air Ambulance • North West Air Ambulance • EMRTS (Emergency Retrieval & Transfer Service/Welsh air Ambulance)

Peer review

Peer Review is mandatory for all Trauma Networks, MTCs and TUs in NHS England. Because north Wales is part of the North West Midlands and North Wales Trauma Network the three TUs have chosen to opt for being Peer Reviewed. This is:

- To ensure services are as safe as possible
- To improve the quality and effectiveness of care
- To improve the patient and carer experience
- To undertake independent, fair reviews of services
- To provide development and learning for all involved
- To encourage the dissemination of good practice

Standards

A vast number of standards exist in relation to the delivery of individual aspects of trauma care such as National Institute for Health and Clinical Excellence (NICE) guidance on management of head injury and the British Orthopaedic Association Standards for Trauma (BOAST) guidelines for open fractures and pelvic fractures. A number of these, together with additional expert opinion, have been assimilated into a performance management framework. Adherence to standards and KPIs form the basis of Peer review.

Evidence Base and Key Publications

Regional networks for major trauma clinical advisory groups report (2010)
http://www.uhs.nhs.uk/Media/SUHTInternet/Services/Emergencymedicine/Regionalnetworksformajortrauma.pdf

NICE Guideline ng39 Major trauma: assessment and initial management (2016)
https://www.nice.org.uk/guidance/ng39/resources/major-trauma-assessment-and-initial-management-1837400761285

NICE Guideline ng40 Major trauma: service delivery (2016)
https://www.nice.org.uk/guidance/ng40/resources/major-trauma-service-delivery-1837446110917

NICE Guideline ng41 Spinal injury: assessment and initial management (2016)
https://www.nice.org.uk/guidance/ng41/resources/spinal-injury-assessment-and-initial-management-1837447790533

NICE Guideline ng37 Fractures (complex): assessment and management (2016)
https://www.nice.org.uk/guidance/ng37/resources/fractures-complex-assessment-and-management-1837397402053

Summary

- Major Trauma Networks, Major Trauma Centres and Trauma Units were established following recommendations from NHS England where it was shown trauma care could be improved.
- North Wales became part of the North West Midlands and North Wales Trauma Network in October 2012.
- A Trauma Network (TN) is the name given to the collaboration between the providers commissioned to deliver trauma care services in a geographical area.
- A Major Trauma Centre (MTC) is a multi-specialty hospital, on a single site, optimised for the provision of trauma care.
- A Trauma Unit (TU) is a hospital in a Trauma Network that provides care for most injured patients but also has the ability to stabilise patients for onward transfer to MTCs for definitive care.
- Local Emergency Hospital (LEH) is a hospital in a Trauma Network that does not routinely receive acute trauma patients (excepting minor injuries that may be seen in an MIU).
- Peer Review is primarily undertaken to improve the quality and effectiveness of care.

References / Further Reading:

[1]National Confidential Enquiry into Peri-Operative Deaths (NCEPOD) 'Trauma who Cares' (2007)
http://www.ncepod.org.uk/2007report2/Downloads/SIP_summary.pdf

[2]National Audit Office 'Major Trauma Care in England' (2010)
https://www.nao.org.uk/wp-content/uploads/2010/02/0910213.pdf

4 - The trauma unit patient pathway and key decisions to make

Author: Dr Mark Knights, Consultant Anaesthetist, YG

Date: 04/19

Review Date: 05/21

Introduction

Trauma teams in North Wales consist of a mixture of different specialties, with different experience, training and background. It is very rare that you will have the group team of people in a trauma team, and on many occasions you will be working with people you have not even met before . Without a standardised approach to managing trauma the situation can quickly descend into chaos. Luckily most patients will follow the same pathway and there are relatively few key decisions that need to be made along this pathway. Therefore it is possible to have a standardised approach or patient pathway [see Appendix A], and the more we standardise, the more efficient and well practiced our teams will become. This chapter looks at the whole process in general. Subsequent chapters will look at each element in more detail.



Prealert

The first key decision on receipt of the Prealert is whether to trigger a full trauma call. Your department should have clear guidelines on this based on the mechanism of the trauma, the injuries sustained and the physiological condition of the patient [see chapter 8]. Other decisions to make at this point include:

- Whether to activate the major haemorrhage protocol and prepare blood products
- Whether to call CT and give them advance warning

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- Whether to call your consultant

As the trauma team arrives remember to brief them on suspected injuries and anticipated course. Make sure everyone has signed in and wearing appropriate PPE. Utilise the whole team to prepare equipment and drugs, blood, etc.

Remember to prepare the bair hugger! Patients arrive cold and keeping them warm is a real challenge.

Patient arrival

See **chapter 11** for details on the team's actions on arrival of the patient. Remember a quick check of the patient to exclude catastrophic haemorrhage, cardiac arrest or obstructed airway.

The patient will then normally be transferred from ambulance trolley onto the ED trolley followed by ATMIST (Age, Time, Mechanism, Injuries sustained, Signs, Treatments) handover from the paramedics to the whole team.

Key decisions to make here are:

- whether to follow this normal routine or
- whether the patient needs an immediate life saving intervention first such as control of major haemorrhage, or commence resuscitation for traumatic cardiac arrest.



Primary Survey

The primary survey must be completed quickly but thoroughly, as it will inform many of the next key decisions. In a trauma team the primary survey can be divided amongst the team members. Typically the anaesthetist will assess airway (and disability). Someone else can assess Catastrophic haemorrhage, Breathing and Circulation.

Findings from the primary survey need to be communicated clearly and loudly back to the TTL

(Trauma Team Leader) and the scribe. This will be covered in **Chapter 12**. The next key decision based on these findings include:

- Whether to commence resuscitation with blood products
- Whether to give Tranexamic Acid (if not already given)
- Whether to perform RSI (Rapid Sequence Induction)
- What other interventions are necessary prior to CT eg Pelvic binder, Chest drains, Fracture reduction and splints, antibiotics.
- Whether the patient requires full body CT, more focused imaging or maybe even theatre.

These decisions should be made jointly between the Team Leader, the senior anaesthetist and any relevant specialists (eg general surgeon). Ultimately the team leader should make the final decision but they should listen to the advice from their colleagues.

Moving the patient to CT

The patient should be having their CT scan within 30 minutes of arrival in resus. The CT will provide a detailed definition of injuries and will then determine what definitive treatment the patient requires. The earlier the patient has CT, the earlier they will get definitive treatment. This may sound unrealistic [our average time is currently 55 minutes] but it is achievable with good teamwork and practice.

In this 30 minutes they will have had their primary survey and potentially RSI, chest drains, blood transfusion, etc. CT should not be delayed for anything that is not essential at that time. This includes FAST (Focused Abdominal Sonography in Trauma) scans, X-rays, 12 lead ECG, arterial lines, urinary catheter, etc. FAST scans and X-rays are only appropriate if CT is not ready or if the patient is not yet stable enough to move (and may assist in the decision to go straight to theatre).

The anaesthetist will normally co-ordinate the transfer of the patient to CT. The team should assist the anaesthetist in preparing for transfer to CT. A post RSI / pre-CT checklist is useful to make sure all the important equipment and drugs are taken [See Guidelines]. If damage control resuscitation is in progress this may need to continue during the CT. Do not wait for the patient to be fully resuscitated before moving to CT (Resuscitation is a concept not a place!) This may require the team to take blood products and the rapid infuser with the patient to CT.

The key people who need to go with the patient are: the TTL, anaesthetist and airway assistant, the scribe, the surgeon and maybe a runner. The rest of the team can stay in resus and start preparing kit for things that will need to happen on the patients return. (eg arterial line, urinary catheter, transfer monitoring, infusions, etc.)

After CT

The choices after CT will be:

- Return to ED Resus (the majority of our cases)
- Go straight to theatres in rare cases (eg intra-abdominal bleeding in an unstable patient)

If the patient returns to ED Resus they will still be moving somewhere! Either they have significant injuries that will require an MTC (in which case it is best to start the ball rolling ASAP by sending CT scans to Stoke via the IEP and referring to the TTL) or they will be staying in your hospital.

Options for destination include:

- MTC (Royal Stoke University Hospital or Alder Hey)
- Theatres
- ICU / HDU bed
- Ward bed
- Observe overnight / Discharge Home

Referring to MTC

All three acute hospitals in North Wales are trauma units and are capable of managing many injuries perfectly well. There are times however when the best place for the patient is the major trauma centre, and the potential benefit they will gain outweighs the risks of a secondary transfer. Our MTC is Royal Stoke University Hospital for patients who have had their 16th birthday, or Alder Hey Hospital for children <16 yrs old.

It is impossible to have a guideline on who to refer to the MTC that covers every circumstance. The general indications for transfer to an MTC are:

- Multiple injured patients who require input from different specialists
- Patients with injuries who require management from specialists not available at the local trauma unit

This is covered in more detail in Chapter 22.

Remember not all trauma goes to the MTC. The relevant specialties should be involved in the decision.

EMRTS Cymru
Emergency Medical Retrieval & Transfer Service

Packaging for transfer

Once a patient has been accepted for transfer to the MTC, the next decision is how to transfer them there. This decision should be taken at a senior level and involve the local TTL, senior anaesthetist and the TTL at the MTC. Planning for the transfer should start as early as possible (even prior to CT) as logistics take significant time. The options are:

1. EMRTS transfer (8am - 8pm). Contact the Air Support Desk on 0300 1232301
2. Road transfer with anaesthetic escort (Level 3 critical care patients (ie intubated), Level 2

patients at risk of deterioration on route)
3. Road transfer without anaesthetic escort (Level 1 and stable level 2 patients as deemed appropriate)
4. HM Coastguard with anaesthetic escort (very exceptional circumstances only)

Regardless of the method of transfer the TTL will need to co-ordinate the team to expedite this process as much as possible, whilst still keeping overall situational awareness of the patient's condition, as the team members are likely to become task focused. See **Chapter 23** for details on packaging.

Taking a patient to theatres for damage control surgery

A decision that the patient needs damage control surgery locally should be made jointly between the local TTL, senior anaesthetist and senior surgeon (in a "Command Huddle"). An example for this would be intraabdominal bleeding in a cardiovascularly unstable patient not responding to blood product resuscitation.

Once the decision is made transfer to theatres must not be delayed. The on-call theatre manager or bleep holder should be contacted and requested to make available a theatre ASAP. If the emergency theatres are in use and cannot be vacated, then the next available theatre should be commandeered. The on-call ODP should be sent to prepare the theatre by:

1. Increasing theatre temperature and preparing bair huggers and fluid warmers
2. Preparing anaesthetic machine and intubation equipment
3. Preparing induction drugs
4. Preparing rapid infuser and trauma lines
5. Prepare autologous cell saver
6. Retrieve ultrasound to assist with lines
7. Run through transducers for arterial line and central line

Theatre staff should prepare surgical equipment for an emergency laparotomy.

The patient should be transferred swiftly to theatres with the anaesthetist and TTL, along with any blood products that have not yet been administered.

On arrival of the patient in theatres the TTL should assist in transfer onto the theatre table and then give a brief ATMIST handover to the whole theatre team so that the theatre team know the main relevant injuries and treatment so far.

At this point the surgeon should announce to the whole team briefly what their plan is. The goal of the surgery should always be to do the minimum required to stop bleeding and stop gross contamination to allow continued resuscitation. Now is not the time for definitive bowel repairs, etc.

Finally the anaesthetist should give a brief update (SitRep) on the patient's current condition,

amount of blood product transfused, ROTEM (Rotational Thrombelastogram) result and confirm how much blood product is in the room or needs to be requested / fetched.

The TTL from ED should not leave until this handover has been completed and the surgery has commenced. If the anaesthetist is overloaded with tasks it is reasonable to ask the TTL to stay and continue to assist in being team leader.

Ideally at least two anaesthetists and two ODPs should be present. One pair to manage the anaesthetic and one pair to manage the blood product administration. A runner should be delegated to manage communication with blood bank.

The anaesthetist should aim to repeat the SitReps every 15 minutes until the surgery is complete.

Priorities:

1. Manage Catastrophic Haemorrhage
2. Secure the airway if necessary
3. Commence damage control resuscitation if the patient is shocked
4. Aim to be in CT within 30 minutes



5. Think about the patient's destination early

Always try and stay one step ahead of the game. Plan ahead and prepare for the likely next step in the pathway. Call for help early.

Summary

- Trauma management involves complex tasks and decision making, often under significant time pressure within a multi-disciplinary team of mixed experience.
- The key to improving performance is standardizing our approach by simplifying the pathway, limiting the decisions and practicing in our teams often.

References / Further Reading:

1. NCEPOD “Trauma who cares?” 2007.

5 - Human Factors

Author: Dr Linda Warnock, Consultant Anaesthetist, YG

Date: May 2019.

"Enhancing clinical performance through an understanding of the effects of teamwork, tasks, equipment, workspace, culture and organisation on human behaviour and abilities, and application of that knowledge in clinical settings" Ken Catchpole 2010

Introduction

Human factors is the science explaining the inter-relationship of humans to their environment and to each other. Understanding this is seen as essential to improving safety and efficiency in other safety critical industries. *Human factors* helps us to identify how we can optimise the system to make it easier for people to do the right things and to do them consistently. Failings in human factors, systemic or individual, are the cause of a majority of incidents in aviation and have been a significant feature in high profile disasters in other industries, such as Piper Alpha, Chernobyl, etc. It is now clear that it is also a significant factor in healthcare disasters.

So how do human factors affect trauma teams?

Trauma teams have a particular set of challenges that make the care of the patient especially challenging.

- Communication may be challenging. A busy resus room is noisy: the patient or relatives may be screaming, which makes it difficult to focus on the task. There may be lots of people trying to talk over each other making it difficult to get your message across.
- Team members may be unfamiliar to each other, and never worked as a team before. They may not know names or levels of experience of their fellow team members. This makes it very difficult to delegate appropriately.
- Team members may also be relatively junior and inexperienced. Some may have never been involved in a major trauma before, and unsure of their role.
- Team members may not wish to speak up due to perceived hierarchies within the team.
- Different attending specialties will have different clinical priorities. It is easy to concentrate on just your area and become fixated, losing overall situational awareness.
- There is a significant time pressure and lots of procedures to do. Some of them carry significant risk.
- The stress of all of the above can be overwhelming and make quick decision making very difficult.

By closely considering these challenges we can start to see how we can focus on developing good habits for our non-technical skills within our trauma teams.

These are as important, if not more so, than our clinical knowledge and skills:

1. **Situational awareness**- The Team Leader standing back, seeing the bigger picture, checking "mental pictures", following SOPs and checklists.
2. **Communication**- minimising unnecessary communication. All communication going through the team leader. Use of closed loop communication (repeating what was said back).
3. **Decision making**- Evaluating options, using cognitive aids such as checklists, checking with the relevant team members and referring to their expertise.
4. **Team work**- Team members all doing their bit competently and efficiently, mutual support for their colleagues. Team leader allowing each member of the team to have a voice.

When looking at teamwork we need to consider two aspects in more detail:

Leadership

What makes a good leader?

The role of trauma team leader is a dynamic one, which will change with the need of the patient and the team. A good team leader should be able to enhance communication, foster the team, keep the team focussed on the goal, be supportive, acknowledge and respond to team member limitations. A good team leader should be able to strengthen the skills of the team members.

However!

" He who cannot be a good follower cannot be a good leader"

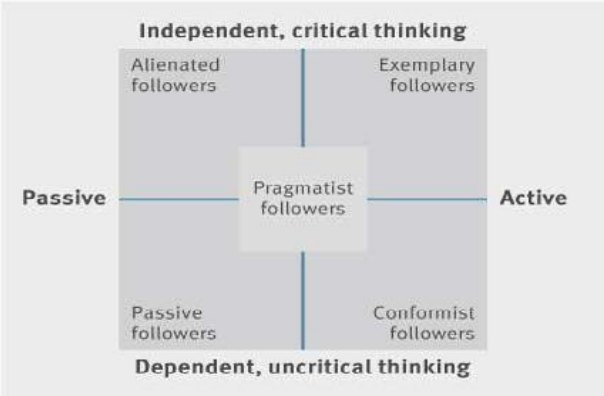
Aristotle

Followership

"Good leadership is the stuff of countless courses, workshops, books and articles. Everyone wants to understand just what makes leaders tick – the charismatic ones, the retiring ones and even the

crooked ones. Good followership, by contrast, is the stuff of nearly nothing." Barbara Kellerman

Followership is the study of how other team members act within a team, all team members have a responsibility to act in an independently thinking manner without blindly following the leader. This should not be destructive but supportive and active. We should all aim to be exemplary followers:



Trauma management is a team approach. A well-trained and well-practiced team is the key to efficient, quality care. We all need to concentrate on our non-technical skills just as much as our clinical skills.

Practical Tips:

1. Always introduce yourself by name and role when arriving at a trauma team.
2. Get everyone in the team to wear an apron and ask them to write their name and role in big letters on the front of it. It's noticeable how much better communication within a team is when you can call someone by their first name.
3. The TTL should brief the whole team on arrival, and also regularly summarise findings and verbalise their thinking. This helps the team maintain situational awareness but also invites the team to offer differentials and alternatives. These should be welcomed.
4. The TTL should remain hands off.
5. If people are having to raise their voice to be heard then the noise level is too high. Minimise unnecessary chat.
6. Always be polite and respectful to the prehospital team. Listen carefully to the handover.
7. Decision Making is often complicated. Involve seniors early and share the burden - Discussion between specialties and referrals to other hospitals should ideally be consultant to consultant discussions.

Summary

1. Training in *Human factors* enhances patient safety.
2. The team leader should keep overall situational awareness as other team members will become too task focussed
3. The team members (followers) should act and think in a way that contributes to the team, supports the team leader and enhances patient safety.

References / Further Reading:

1. Department of Health Human Factors Reference group report 2012
2. Patient safety first: Implementing Human Factors in Health Care
3. " A life in error- from little slips to big disasters" Professor James Reason

6 - Pre-alert

Author: Dr Mark Knights, Consultant Anaesthetist, YG and EMRTS.

Date: 05/19

Review Date: 07/21

Introduction

The ambulance service will normally give a prealert for most major trauma patients. The standard format for this is the ATMIST (+ER). The purpose of this prealert is to allow the team to form, plan and prepare for key interventions that may need to be done in resus.

When to Prealert.

A prealert should be given by the prehospital service on all patients who fit the criteria for a trauma team call. Significant mechanism, significant injuries or abnormal signs or symptoms (eg raised heart rate or reduced conscious level). If in doubt a pre-alert should be given.

Format

The standard format used by pre-hospital teams is the ATMIST. This stands for:

Age.

Time of Injury.

Mechanism of Injury.

Injuries sustained.

Signs and Symptoms.

Treatments given.

It is also very helpful to have an ETA (Estimated time of arrival) and any specific Requirements (eg have blood ready or have an obstetrician) (hence ATMISTER).

Delivery

Ideally, the crew should give the ATMIST directly, but this is not always practical and so often it is given second hand by ambulance control. If this is the case there is a risk that some details will be missed or left out. The prealert should be succinct (about 30 seconds) and get important information across. It is unreasonable to expect a detailed description of the patients injuries to be given at this stage. The pre-hospital crew will be busy with the patient and working in difficult circumstances so give them a break if they don't get all the details correct or forget some things. It is always best to plan for a patient that is worse than anticipated from the prealert information than vice versa.

The person taking the prealert should record all the details and confirm an ETA, then decide on the next course of action according to the trauma call out policy (see Chapter 8). The default should be to call a trauma call. Think carefully before deciding that a trauma call is unnecessary: If the ambulance crew were concerned enough to send a pre alert then there is probably a good reason.

The next big decisions to make are whether to activate the major haemorrhage protocol, request O negative blood in advance, and whether to alert the CT radiographers. These decisions should be made by the trauma team leader who must be of appropriate experience and seniority. If you are considering these questions then you must also call the on-call ED consultant.

Priorities:

1. Someone senior should answer the phone.
2. Have pen and paper ready with "ATMISTER" printed down the left margin.
3. Record the details in the standardised format.
4. Seek out the ED co-ordinator and Trauma Team Leader ASAP.
5. Decide on putting out a trauma call and calling consultants.
6. Consider activating major haemorrhage protocol.

The pre-hospital team may be working very hard in difficult circumstances. The patient's condition may also be changing rapidly, so always plan for worst case scenario.

7 - Kinematics

Author: Dr Mark Knights, Consultant anaesthetist, YG & EMRTS.

Date: 05/19

Review Date: 07/21

Introduction

Kinematics in a nutshell is trying to predict the pattern of injuries from the mechanism.

Understanding the forces generated in collisions and how quickly they act on casualties involved in an incident allows us to understand potential mechanisms of injury and have a high index of suspicion for potential occult (hidden) injuries.

It is important to pay attention to the description of the mechanism of injury given by the prehospital team in order to try and understand the pattern of injuries to look for. Ask the prehospital team if they have pictures of the scene to get a better understanding.

In North Wales most of our trauma is blunt trauma from falls <2m, followed by Road Traffic Collisions (RTC) and falls from height.

Physics

Newtons first law of motion states that an object moving at a constant velocity (or at rest) will continue at that velocity unless acted on by an external force. The force (F) required is proportional to the product of the mass (m) times the acceleration (a).

$$F = ma$$

Consider a person in a car travelling at 60mph, the car strikes a tree providing a force large enough to stop the car, the person however is still travelling at 60mph and requires a force to stop them (typically from the seat belt) and finally even when the person is stopped their internal organs continue to move, giving rise to shearing injuries.

The kinetic energy of a moving object can be found from:

$$KE = ½ mv^2$$

(where KE = kinetic energy, m = mass and v = velocity)

Basically, by doubling the speed we quadruple the kinetic energy liberated – the road safety message “Speed Kills” is a valid one. However, a very heavy object moving slowly (such as a tanker) also has an enormous amount of energy.

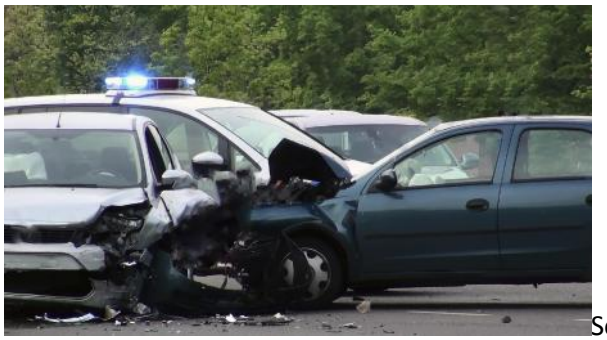
Finally when evaluating the incident it is important to assess the rate that the momentum changed either by acceleration or deceleration:

$$F = \Delta P / \Delta T$$

An example of this can be considered by thinking of two different situations, firstly where a person falls onto a concrete surface and secondly the same person landing on a soft mattress. In the first example the deceleration is very rapid and therefore a lot of force is generated as the change of momentum (ΔP) occurs over a very short time (ΔT). In the second example the mattress cushions the fall - in effect slowing the rate at which momentum is lost. As a result, less force (F) is applied to the victim.

Airbags and vehicle crumple zones are an attempt to extend the impact and change of momentum time thus reducing the force generated on an object. Seat belt pre-tensioners pull tight against the body to make sure that the body is immediately restrained and does not accelerate forward and then stop suddenly. Seatbelts are now even constructed with rip apart sections to extend the length whilst decelerating and therefore prolong the amount of time to stop the body.

An interesting consequence of these safety innovations is that a newer car will look more damaged following an impact when compared to a structurally sound older one, the important difference though is that the newer car has extended the impact deceleration time and absorbed a lot of the energy.


Source: Google

Road Traffic Collisions

Road traffic collisions involving vehicles will broadly fall into the following categories;

1. Car versus: Pedestrian / Cyclist / Motorcyclist / Other road user (e.g. horse & rider)
2. Car versus Car or other large vehicle:
    a. Head on collision
    b. Side impact (T-Bone)
    c. Rear Impact
3. Car/Cycle/Motorcycle versus Stationery Object (e.g. tree):
    a. Head on collision
    b. Side impact

Pedestrian

During an impact with a pedestrian, the vehicle will usually suffer minimal damage but the pedestrian (who has little effect on the speed of the vehicle) suffers from a large transfer of momentum with massive acceleration from near zero. This force can obviously cause massive trauma to any internal structures.

The sequence of events is important to understand as the pedestrian goes through (usually) 3 separate collisions:

1. The initial impact with the vehicle e.g. front bumper causing lower limb trauma (in an adult)
2. Secondary impact as they strike the bonnet of the vehicle causing head, neck and chest injuries
3. Tertiary trauma is caused when they contact the ground resulting in head, spinal and abdominal injuries

N.B. This assumes a standard sized adult being hit by a small to medium sized car. The pattern of injury in the paediatric population or in an adult hit by a large vehicle or truck will change, as the angle of impact and the amount of energy transferred as a result will differ. An analogy is hitting a golf ball with an iron versus a wood club. The iron (in this case, a standard size car) gets under the ball and elevates it. In the same way, a frontal pedestrian impact will result in the victim rolling over the bonnet and the roof of the car. In contrast if a golf ball is hit with a wood, a higher amount of energy is imparted and the trajectory of the ball is flatter. In the case of a pedestrian victim, this equates to a greater forward momentum and different pattern of injury. In the same way that an adult will not roll over the roof of a lorry if they are hit by one, a small child who is hit by a normal sized car will also be thrown forward rather than rolled over the bonnet/roof.

Summary of Typical Pedestrian Injury Patterns:

- Sheer injuries to kidneys, liver and spleen
- Lower limb injuries
- Hyperextension/hyperflexion injuries to the neck
- Head Injuries
- Chest injuries
- Pelvic fractures

Cyclists and Motorcyclists

Cyclists and Motorcyclists may fall, collide with other vehicles or collide with stationary objects. The nature of injuries sustained by riders of cycles/motorcycles varies considerably. It is also very dependent on the level of personal protection worn by the rider. Without the presence of crumple zones or protective airbags to decrease the amount or rate of energy transmission, the rider is subject to the entire kinetic energy of the collision. Without restraints the rider is also exposed to

rotational forces, which can have a devastating effect. If the energy is dissipated by the rider sliding or rolling down the road then they are likely to suffer less severe injuries than if they have a sudden deceleration from hitting a wall or a tree for example.

Almost any type of injury is possible within this group and as much detail about the kinematics should be gained immediately. It goes without saying therefore that this group of road users are at very high risk and require an appropriate high index of suspicion.



Frontal / Head-on

Approximately 70% of all collisions are frontal resulting in rapid deceleration and large forces transmitted to the occupants of the vehicle.

Unrestrained passengers in the vehicle move forward, their knees striking the dashboard, head hitting the roof and then the windscreen (look for bulls-eye on the inside of the windscreen) with a high chance of ejection from the vehicle, through the windscreen. Thoracic and upper abdominal injuries occur from collision with the steering wheel. The prevalence of airbags in vehicles reduces the effects of some blunt force trauma, but not all and the passengers still tend to move forward and can receive serious injuries unless wearing a seatbelt. If the A post deforms during the impact then this can also cause head injury.

The airbag deployment itself can cause injuries, the talcum powder used to cover the air bag results in a fine dust which can aggravate respiratory problems. Burns and grazes can occur on arms and faces plus arm injuries can occur as the airbags deploy. Casualties may also be temporarily deafened from the noise of the explosive deployment in the confined space of the care. With an impending collision the driver tends to brace themselves for the impact holding their breath. This can result in the paper bag effect type injury where pulmonary injury occurs, as well as fractures or dislocations to the wrists and/or forearms.

Summary of injury patterns:

- Shear injuries to kidneys, liver and spleen
- Hyperextension of the lumbar spine
- Chest injury
- Pelvic fracture
- Leg injuries
- Hyperflexion of the neck
- Axial compression of the spine
- Ejection if unrestrained
- Chance fracture and associated injuries

A Chance fracture is an anterior wedge fracture commonly occurring at the thoracolumbar junction (T12 to L3) or midlumbar region. This fracture is traditionally associated with lap belt restraints as a sudden forward flexion of the spine causes a wedging of the spine, occasionally associated with a posterior transverse fracture of the spine. Whilst the anterior wedge fracture itself is managed non-operatively unless there is either associated spinal cord compression, disruption of spinous ligaments or column damage, they are associated with mesenteric, pancreatic or duodenal injuries in 50% of cases in some series, and should raise the index of suspicion of occult intra-abdominal injury.

This also assumes that any seatbelt restraint systems are worn correctly and the patient is the appropriate size to sit in the seat. The effect of "submarining" (sliding down and forward, out of a restraint system) can also lead to prolonged entrapment, lower limb and pelvic fractures and axial loading fractures of the spine.

Rear Impacts

Rear collisions are less common but present a higher risk of cervical spine injury than frontal collisions. The initial collision forces the occupants backwards stretching the body backwards and upwards leading to neck injuries especially in cars without (or with badly adjusted) head rests. When the car stops moving the occupants are then forced forwards and may have similar injuries to the frontal collision.

Summary of injury patterns:

- Hyperextension of the lumbar spine
- Hyperextension and hyperflexion of the neck
- Shear injuries to kidneys, liver and spleen

Side / T bone Collision

Despite the increase in strength in modern vehicles the side of a car has less room and mass with which to absorb energy and we still see side impacts resulting in far more serious injuries and more entrapments. The close proximity of the vehicle construction means the occupant will more likely suffer a head injury, typically from the B post. Other injuries are typically all down the side of the impact but may also occur from hitting the other passengers or fittings inside the car. Of course the exact injury pattern depends on the impact point and the location of the occupant (driver/passenger or front/rear seat). During a side impact the occupant initially moves towards the direction of the incoming force. Immediately after this the reaction forces then move the occupant in the opposite direction. Remember it is possible that two occupants may impact upon each other in the event of a side impact.

Summary of Injury Patterns:

- Head injury through impact with the side of the car (then possibly with adjacent occupant)
- Shear injuries to kidneys, liver and spleen

- Pelvic and long bone injuries
- Pulmonary contusion and diaphragmatic injury
- Lateral hyperextension of the neck
- Arm and leg injuries

Rollover

Although statistically rare, rollovers often produce the most severe injuries. Only 2.4% of crashes involved rollovers but they accounted for one third of occupant deaths. Restrained passengers will normally remain in their seat but will be struck by unrestrained objects/occupants in the vehicle. If casualties are unrestrained they may be ejected from the vehicle leading to a higher rate of mortality.

Rollovers can occur end over end, following a frontal/rear collision or side rollover following a T-bone collision. It is important to remember that a car that has rolled over may come to rest on its wheels, roof or side, and this is where the principles of kinematics are extremely valuable. Look for deformation of the roof, multiple body panels and damage to the environment. Due to the nature of rollover incidents almost any injury is possible and occupants require a high index of suspicion.

Falls

Falls from height may produce major injuries, and are not infrequently involved in successful suicides. Clearly the most important factor influencing the severity of injury is the height fallen. However, as described above, the rate of deceleration is also important, as are the parts of the body striking the ground. Falls from greater then six metres are a major cause for concern. In addition, serious injury has been reported in falls from much lesser heights, and many triage systems use 2 metres or three times the patient's body height as a cut off for a 'major' response.

In elderly patients even a fall from standing height can result in major injuries. A high index of suspicion is needed during triage, assessment and decision making. Have a low threshold for requesting a full trauma team and a whole body CT scan. **See Chapter 26.**

Injuries associated with a fall from height onto feet (eg jump from bridge) typically include calcaneal fractures, lumbar spine blowout fractures and possibly unstable vertical shear pelvic injuries. Thoracic injuries are also common from the knees hitting the chest wall.

Blunt trauma

Blunt trauma is essentially any traumatic injury that does not involve an object entering the body. In contrast to penetrating trauma which may have a single bleeding point or involve only a limited area, patients who are involved in a severe blunt trauma such as being runover or an ejection RTC will present with a multitude of injuries involving remote regions of the body. Blunt traumatic injuries usually have a worse outcome in comparison to penetrating injuries as there is a higher kinetic energy transfer in most cases, and a higher resultant area of tissue damage.

If there is a large amount of bleeding and tissue damage, the microcirculatory system is damaged leading to hypoxic/hypotensive cell death. This may result in poor tissue perfusion despite adequate resuscitation, which may cause a significant systemic inflammatory response syndrome (SIRS) and multiple organ failure. All we can do at present is resuscitate the patient optimally and hope to



restore tissue perfusion.

Penetrating Trauma

Penetrating trauma from stabbing or shooting can represent a vast spectrum of pathology depending on what cavity is penetrated, with what instrument, with what degree of force and what organs or tissues are injured. With low velocity trauma such as stabbing surgical management is more straightforward. Once the appropriate vessel is ligated and systemic flow restored, as long as the period of hypoperfusion and poor oxygen delivery has not been too extreme, then the consequences are not too severe. The obvious caveat to this is that if a vital structure such as the heart, major thoracoabdominal or peripheral blood vessels have been injured then chance of survival declines significantly.

High velocity penetrating trauma (ie Ballistics, gunshot wound, etc.) is thankfully very rare in North Wales. Remember that due to the much higher velocity the energy transfer is very significant and the tissue damage can extend well beyond the track of the ballistic projectile (eg bullet). Such patients should be transferred to an MTC if stable enough for extensive surgical exploration and debridement of all wounds.

Summary

By understanding the magnitude of the forces involved in collision and the direction they occur we can deduce the potential injuries that the casualty may suffer.

In particular:

- Pedestrians are at high risk of serious injury if struck by moving vehicles and should be considered to have such injuries until proved otherwise.
- Patients ejected from their vehicle or involved in a roll over are at increased risk of death.
- Falls from height carry a risk of major injury, especially from above six metres.
- The external wounds sustained by victims of shootings may not reflect the internal damage.

References:

1. ATACC manual. https://www.ataccgroup.com/atacc-uk/

8 - Calling the Trauma Team and Responding to the call

Lead author: Dr Rhiannon Talbot – Consultant in Emergency Medicine, YG.

Date: 05/19

Review Date: 05/21

Introduction

Seriously injured patients should be cared for by the most senior personnel available, and those people should have a good understanding of trauma and how trauma works in their hospital. In order to achieve this, there is a hospital trauma team, which can be activated at any time by calling 2222 and asking switchboard to put out a trauma call to Resus. The initial trauma call will go out to the duty 'middle grade' for each specialty. However, it is expected that relevant consultants will also attend wherever possible.

Trauma Calls

The ED often initiates trauma calls when a pre-alert is phoned through from the ambulance or other pre-hospital service. Your department will have clear guidelines on the criteria for a trauma team call out. These are guidelines only. If you are not sure whether a trauma call is warranted, check with your senior, or put the call out anyway.

A level 1 call mandates a full trauma team activation.

A Level 2 call requires a senior (ST4+/equivalent or Consultant) EM doctor should assess the patient immediately on arrival, who will then decide whether full trauma team activation is required.

If you are unable to attend:

If you are unable to attend you are responsible for finding a replacement. This may mean phoning your consultant or another colleague. Do not simply ignore the bleep!

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Attendance

The trauma team is activated by dialling '2222'. The following table outlines those expected promptly at a Level 1 Trauma Call, and those who will be informed.

Immediate Attendance	To be informed;	Attendance within 30 minutes on request;
ED Consultant or Associate Specialist (notified by telephone if at home and to attend within 30 minutes)	ED radiographer	Senior Orthopaedic Surgeon on-call (St3+ or equivalent)
ED Middle Grade	Pathology Lab technicians – Transfusion/Haematology & Biochemistry	Senior Doctors from other specialities e.g. ENT or O&G etc
Senior Anaesthetic (or ITU) Doctor on-call (ST3+ or equivalent)	Bed Manager	Consultant Anaesthetist
Senior Surgeon on-call (ST3+ or equivalent)	Porter	
Surgical SHO on-call	Nurse in Charge of Theatres	
Theatre ODP on-call		
Resuscitation room nursing staff		
ED SHO		
Orthopaedic SHO on-call		
ED Senior Nurse		
Senior Paediatrician on-call* (ST3+ or equivalent and for paediatric trauma only)		

*Paediatrics

If a child attends with serious injuries, the senior paediatrician on-call will attend, and be responsible for notifying the on-call paediatric consultant once the child has been assessed.

Ground Rules

Many of us have attended the ED Resus room to see a large cluster of people, without clear leadership, surrounding the patient and getting in each other's way.

We have Resus ground rules to prevent this. Please follow them.

- Sign in when you arrive
- Wear a tabard if available - or write your name and role on the front of your apron.
- If you are fulfilling a role in the team, confirm this with the TTL
- Stay behind the foot of the bed unless invited forward by the TTL

Personal Protective Equipment (PPE)

Personal safety is important. Aprons, gloves and, where appropriate, eye protection are a minimum and must be worn by everyone in front of the line. Remember correspondingly coloured lead aprons as well.

Leaving the Trauma Call

Please DO NOT just walk off

As a member of the trauma team you have a responsibility for the patient and a responsibility to adequately hand over care. This may just mean telling the TTL that you are needed elsewhere and checking you are no longer required.

This includes when X-rays are done in Resus. **Wear lead or come straight back in the room.**

The Trauma Team is responsible for the patient's care until they are accepted under the care of another team; this includes transfers to CT or other hospitals.

Summary

- Seriously injured patients require a team with a good understanding of trauma and how to works as part of the trauma team
- Different mechanisms of injury will require different levels of Trauma Team response, and this is a dynamic changing situation. Be familiar with the Level 1 and Level 2 Trauma activation criteria
- Call switchboard using '2222' to activate the trauma team
- Attendance at a trauma call is mandatory, until you are released by the TTL
- Attend promptly, sign in and put on your PPE and Tabards
- The TTL will brief you on your role when the team has arrived
- Stay behind the line unless invited forward by the TTL

Don't forget: To sign in and direct all requests or queries via the TTL

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References / Further Reading:

1. Operational Policy for the Management of Patients with Serious or Multiple Injuries in Ysbyty Gwynedd. Dr Rob Perry. October 2016.

2. Trauma Team Call Out Criteria. 2011. Dr Linda Dykes, Dr Rob Perry & Dr Mark Anderton.

9 - Trauma Team Leadership and Team Roles

Author: Dr Nikki Sommers, ST4 EM, YG

Lead author: Dr Rhiannon Talbot, Consultant in EM,

Dr Mark Knights, Consultant Anaesthetist, YG & EMRTS

Date: 05/19

Review Date: 05/21

Introduction

In order that trauma calls run smoothly and effectively, the team is lead by a Trauma Team Leader (TTL). Their role is to oversee the resuscitation of the patient, from pre-arrival, until after they have left the department. The TTL leads the team brief and allocates roles to other members of the team.

The TTL role is normally undertaken by a consultant in Emergency Medicine, although it may be delegated to another member of their team under supervision. If there is no consultant in the ED at the time a trauma call is put out, then the consultant should be contacted immediately. Until their arrival, the TTL role will be taken on by the duty EM middle grade.

Team Brief

Having activated the trauma team, the TTL will brief the team with the pre-alert details when they arrive. If the department has a whiteboard in resus it is a good idea for the TTL to write the pre-alert details on there.

They will identify roles within the team, including ensuring that there is a scribe and that all members sign in. As part of the team brief, they will then co-ordinate preparation of equipment (as required).

Role Allocation

All Trauma Team members should report to the TTL and sign the attendance sheet. Expect the TTL to confirm with you what level of experience you are and whether you can do the procedures expected of you unsupervised. Don't be offended by this; it is important that the roles are delegated

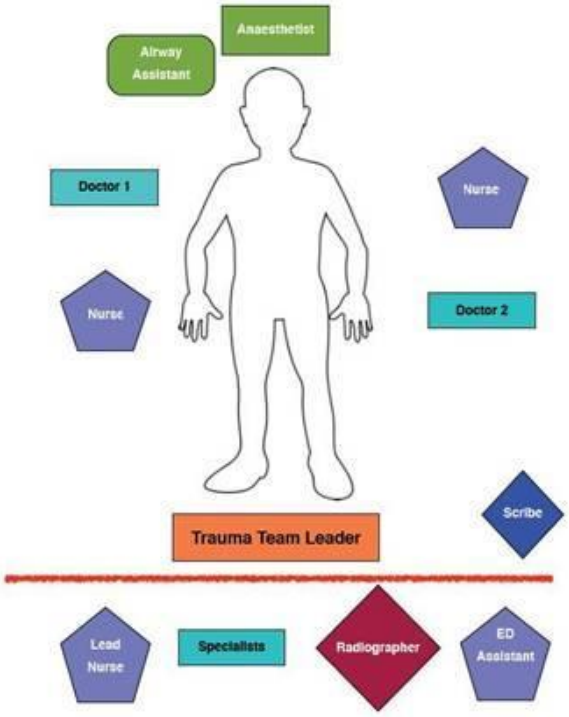
to the most appropriate people. If you haven't done a procedure before (eg chest drain) but would like to be supervised then say so. Once allocated a role, wear the appropriate tabard to aid identification. You will need to wear your plastic apron over your tabard, so make sure the writing on the tabard is on the back. An alternative to tabards is simply to use a marker pen to write your name and role on the front of your apron (Large enough to be read easily from a few metres away).

These are the expected roles for the trauma team members. There will need to be some flexibility according to the skill mix of the team, so be prepared to assist your team members in any of the procedures or roles. **See Chapter 30 for more detailed action cards for each team member.**

Role	**Who?**	**What?**	**Where?**
TTL	ED Cons (or delegated Middle Grade)	Co-ordinate team, receive reports back, summarise injuries, make key decisions in conjunction with team and "behind the line" specialties, prescribe and instruct Nurse 2 re blood and drugs, Keep a watch on time.	At foot of the patient
Doctor One	ED Middle Grade or SHO	Primary Survey (**B**reathing, peripheral **D**isability), interventions for **B** (chest drains) and **C** (pelvic binder, splints)	Patient's right side, next to chest
Doctor Two	Surgical or Orthopaedic SHO	Assess **C**irculation, interventions for **C** (pelvic binder & splints, Vascular/ IO access,) **E**xpose patient	Patient's left side
Airway Doctor(s)	Anaesthetics 1st on-call +/- ICU on-call	Assess and secure **A**irway, GCS and pupils (**D**isability) and other parts of primary survey as appropriate, take AMPLE history. Administer analgesia, look after intubated patient including intra-hospital transfer.	Head End
Airway Assistant	Operating Department Practitioner	Assist with assessment and securing of airway, support Airway Doctor with care of intubated patients. Assist with monitoring.	Head End, on right
Nurse 1	ED Nurse	Initial Exposure, prepare kit for Doctor One & Two.	Floating
Nurse Two	ED Nurse	Attach monitoring +/- initial exposure, manage blood & level 1 infuser, prepare & administer drugs as prescribed &	Floating

		instructed by TTL.	
Scribe	ANP or ED nurse; Ideally someone who has been trained for this role. Not suitable for students.	Complete the ED trauma documentation pack, including the time line, vital signs and administered drugs. Liaise with the TTL to confirm information, and listen out for requests from team members to record specific times, events & doses.	Lectern adjacent to the TTL.
Contact Person	Any nominated person (HCA, ED Doctor, other)	Make phone calls to e.g. blood bank, theatres, required specialties either directly or "get hold of" for TTL	Behind the line and with access to the phone & a bleep.
Specialties & Observers	Trauma Team members not directly treating the patient; general surgery, orthopaedics, Paediatrics, anyone else	1) Make any concerns known to TTL 2) Advise/ decision making as appropriate together with TTL 3) Assist if requested	Behind the line

Trauma Team Positions:



Leadership

As team leader, the TTL will lead the resuscitation from start to finish, as follows:

PRE-ARRIVAL:

- Identify RESUS bay to be used and write pre-alert details on whiteboard.
- Wear tabard and introduce yourself by first name and "Team Leader" to team members.
- Identify trauma team roles and responsibilities and brief the team on the pre-alert info.
- Ensure scribe is coordinating all team members signing in.
- Coordinate preparation of equipment (as required).
- EARLY DECISION: Contact ED Consultant & organise CT.
- EARLY DECISION: Activate massive haemorrhage policy (if required).
- Just before patient arrives (eg when you hear sirens or helicopter) brief the team again, making sure every team member is listening. Check team members are ready and brief them on what will happen at reception of patient, handover and transfer.

PATIENT RECEPTION:



- Do not touch the patient. Confirm with a 10 second check that there is no catastrophic haemorrhage, the patient has a pulse and an airway. Start the timer.
- Coordinate transfer & handover from pre-hospital team. It is vital that the TTL ensures that the whole team are focused and listening to the handover. Be assertive, but don't shout.
- Manage the trauma team response in a stepwise CABCDE approach. Although there may be different people working concurrently to assess different parts of the primary survey make sure you ask for and receive the information back in a logical manner. Ensure the full primary survey is complete and life-threatening problems are treated along the way. Keep noise levels to a minimum, and ensure all communication is coming through you. Resist the temptation to be hands on. If team members are struggling then tactfully ask if they require assistance and swap people accordingly.
- At the end of the primary survey summarise the findings back to the whole team and ensure the scribe has documented them all. Then make a plan in conjunction with the anaesthetist and any relevant specialists who are there (eg surgeon). You need to decide:
    - transfer to CT (or theatre, angio suite or just X-rays).
    - what needs to occur before transfer (eg RSI, blood products, chest drains, TXA, antibiotics) and what can wait.
    - Is the patient likely to need a transfer to MTC (consider a phone call to EMRTS Air Support Desk).
- Brief the whole team what the plan is.
- Keep track of time since arrival; aim for CT within 30 minutes.
- Set goals for your team. Eg *"OK so John can you place a chest drain on the left side, Sarah can you transfuse 1 unit of blood and tell me when that is done, and the rest of the team prepare the patient ready for CT. I would like to be on our way to CT by 16:30...."*
- Ensure appropriate utility of all staff. If team members are overloaded or struggling make

sure they get the assistance they need.

- Ensure blood returned to blood bank if not used and don't forget to stand down CT if not required.



PATIENT TRANSFER:

- Accompany patient to CT or theatre.
- Continue to maintain overall situational awareness and continue resuscitation. Even if the patient goes to theatre you remain the team leader until you have formally handed over this responsibility.
- Make sure you delegate appropriate tasks to those who are not needed to transfer patient (eg set up arterial line, urinary catheter, prepare for transfer to MTC).
- Decide on whether trauma team members can leave trauma team environment.
- After CT, keep the team focused and maintain the momentum.
- If the patient needs referral to the MTC then this should be done by the TTL. In order for you to do this you should handover leadership to another team member (eg anaesthetist) for that time. Remember they may not have a complete picture of everything so summarise what has happened so far, discuss and agree a plan of action before you make the call.
- If the patient needs transfer to MTC get the whole team assisting to facilitate this. Remember the patient is still your responsibility until they leave your department. The anaesthetist and transfer team are at risk of becoming task focused, therefore you need to keep overall situational awareness and watch the patient's vital signs.
- Make sure documentation is accurate and complete.
- Handover to the transferring team yourself. Don't delegate this as you are the only one who has had the full picture.
- Liaise with relatives and police

- Lead the team debrief. This is not the time to dissect and critique. Make it short summarise events and thank everyone for their efforts. If a more formal debrief is likely to be needed take peoples contact details to arrange this.

Summary

The Trauma Team Leader will:

- Lead the team brief.
- Identify team members and allocate roles appropriately.
- Lead the trauma call from start to finish, including debrief afterwards.

References / Further Reading:

1. Refer to your departmental Operational Policy for the management of patients with serious or multiple injuries.

10 - Scribing and Documentation

Author: Dr Rob Perry, EM Consultant, YG,

Dr Mark Knights, Anaesthetic Consultant, YG & EMRTS

Date: 05/19

Review Date: 05/21

Introduction

The scribe is one of the most important members of a trauma team. It is probably also the most difficult role in the team. Even the most efficiently run trauma call involves several different people, and a number of significant events often take place within a very short space of time. The task of the scribe is to ensure that all of these events are accurately recorded and timed. They are also there to be the team leaders second pair of eyes - and to remind the team of anything that has not been checked or done (eg document pupil size, TXA given, etc.). For this reason the scribe needs to be experienced, assertive and also a good leader.

All Trauma Units (and Major Trauma Centres) are judged by the data they submit to the TARN database. The quality of this data is determined entirely by the quality of what is written in the patient's notes, including timings. Ultimately we are judged on only what is documented. This is especially so if the case ends up in a legal claim. Inadequate scribing will make good trauma care look poor!

The role of the Scribe

This role can be undertaken by any staff member who has sufficient experience and clinical knowledge to correctly record the necessary information and to be the TTL's wingman (or wingwoman). Ideally they should have undergone specific training for the role. Most commonly this will be a senior member of the ED nursing staff. **Students (medical or nursing) should not be used for the Scribe.**

The Scribe cannot perform any other role in the resuscitation. If they are required to do so for any reason, then another member of staff must take over from them. The Scribe should position themselves right next to the Trauma Team Leader, so that information can be relayed directly from one to the other.

The main responsibilities include:

- Ensure trauma documents are available.

- Ensure all staff present have been recorded (use sign-in sheet).
- Document the ATMIST handover and all pre-hospital interventions.
- Document the primary survey findings. Make sure a GCS and pupil size is recorded. (If the patient is about to be anaesthetised stop the team and request the GCS first.)
- Record an accurate timeline of events, including all interventions.
- Document observations regularly (every 5 minutes until determined to be stable).
- Log all drugs and fluids given, and times given. Team members should call these out as they are given, but the scribe must also maintain overall awareness of what the team members are doing and challenge team members if they are not calling them out.
- Complete trauma documentation (with the TTL).
- Ensure a copy of the notes is made prior to patient transfer to the MTC.
- Be prepared to challenge the TTL and team if they have not reported any of the above to you.

Documentation

Each of the North Wales hospital sites currently has their own specific documentation. This should be completed for every trauma call. It should also be completed if a senior ED doctor elects to manage a trauma case in-house, and not to put out a full trauma call (a 'level 2' trauma call).

The trauma documents are designed to capture as much of the essential data as possible, but may be difficult for staff to follow if they are unfamiliar with them.

Timings

It is especially important to record the following timings:

- Time of trauma call
- Time of patient arrival
- Name and grade of trauma team leader
- Time of first consultant involvement
- Time Tranexamic acid administered
- Time CT requested, performed, and report available
- Time of first contact with MTC
- Time of decision to transfer patient
- Time of departure

TTL

The TTL should ensure the timer clock is started on the patient's arrival to assist in accurate timings.

They should ensure that general noise level is kept to a minimum and all team members speak clearly so that the scribe is able to hear them. If the scribe misses any information, they should inform the TTL. Conversely, if the TTL has missed something they should check with the scribe.

The TTL should make sure all critical information (injuries found and interventions done) are documented, and that the documentation is accurate. They must document all conversations with the MTC or other specialties and document any advice given, rationale for transfer, choice of transport, etc.

The TTL should also record a short summary of the patient's signs, injuries and all interventions in the trauma notes.

Before the notes are copied and sent on transfer with the patient the TTL should check them for accuracy and completeness and that the crucial timings (above) are recorded.

Remember to start the clock when the patient arrives. Not only does this enable accurate timings to be recorded, it makes the team more aware of the passage of time during the resuscitation.

Anaesthetist

Ensure that you document any airway interventions including the indication for the intervention, grade of intubation, any adjuncts used and any complications. Also document drug doses and timings or check them for accuracy. Don't forget to also prescribe your drugs and sign for controlled drugs!

Document any procedures such as arterial lines, trauma lines, etc.

Summary

- The scribe role is vitally important
- It should be performed by an appropriately trained person
- Trauma paperwork is ultimately what we are judged on. Make sure it is complete and accurate.
- Important data should be recorded specifically

References / Further Reading:

1. Read your own departments operational policy on management of trauma.
2. http://www.trauma.org/archive/resus/traumateam.html

11 - Handover and transfer to the ED trolley

Author: Dr Rob Perry, EM Consultant, YG

Date: 05/19

Review Date: 05/21

Introduction

The handover from pre-hospital to hospital personnel is the only opportunity to pass on vital information. It should be conducted formally, and with the attention of the entire trauma team.

Patients may arrive packaged on a scoop stretcher, in a vacuum mattress, or neither, dependent on the course of events prior to arrival. The TTL is responsible for directing their safe transfer onto the ED trolley using the standard equipment used in that department.

Key priorities

- Handover should be planned and made to a 'silent team' wherever possible
- Transfer should be achieved safely and whilst maintaining control of the cervical spine
- The TTL is responsible for directing both handover and transfer

Handover

A good handover of care is of paramount importance in a trauma case, and should be prioritised ahead of all interventions, other than an urgent need to secure the airway, treat catastrophic haemorrhage, or manage a traumatic cardiac arrest. When the patient arrives in the resuscitation room, the TTL should check that the patient has no catastrophic haemorrhage and has a pulse. They should then confirm with the anaesthetist that they are happy for handover to proceed.

The exact sequence of events will depend on local unit policies, the condition of the patient and how the patient has been packaged. In a stable patient on a scoop for instance the team leader may opt to transfer the patient to the resus trolley and then receive the handover. In an unstable polytrauma patient who presents in wet clothes in a vacuum mattress the team leader may prefer to receive the handover and then deflate vac mat, remove clothes and transfer the patient. Both of these approaches are fine as long as the team leader makes it clear to the whole team what the plan

is, and it is done efficiently and safely. We describe the latter approach here.

The lead paramedic (or PHEM doctor) should address the handover to the TTL, but loud enough for the whole team to hear. ALL team members need to give their full attention to the handover. Ideally a trauma handover should be brief (about 30 seconds). The standard format used is the ATMIST:

- Age
- Time of Injury
- Mechanism
- Injuries Confirmed & Suspected
- Signs & Symptoms
- Treatment given

Any relevant past medical history, medications or allergies should also be included, but if they are not directly relevant the details can wait till later. There should be an opportunity at the end of the ATMIST for the team to ask brief questions.

Once the handover is complete, the team, led by the TTL, should proceed to transfer the patient onto the hospital trolley.

Transfer to hospital trolley

The trauma trolley should be set up prior to the arrival of the patient wherever possible. The exact configuration will vary from hospital to hospital, but may include:

- A padded trauma mattress (Wolverson or similar) which can be moved to and from CT
- Warming mats
- Head restraints

The Trauma Team Leader should direct transfer, unless the patient is intubated, in which case it will be directed by the airway doctor under the TTL's supervision. Whoever leads it must make sure all team members are in position and ready. Use clear, assertive, unambiguous commands "Ready, Brace, Roll"; "Ready, Brace, Slide"; etc. Don't use "1,2,3"

The airway doctor is also responsible for protecting the cervical spine from movement during transfer. Transfers should be performed in a controlled fashion.

Patients who arrive on a scoop stretcher may be moved directly across to the trolley (at least two

team members should lift each side) and the scoop removed with the patient in position.

Patients who arrive on a vacuum mattress should have it deflated (with a team member providing in-line stabilisation to the cervical spine) and be placed onto a scoop stretcher. The placement of the halves of the scoop should be achieved by means of a partial log roll (ideally 10 degrees only) directed by the TTL (unless the patient is intubated). The patient may then be kept on the scoop or managed on a padded transfer mattress according to departmental protocol.

Once transfer is complete, the cervical spine should be immobilised and the team member protecting it released. Monitoring may then be attached and the resuscitation can proceed.

All of the above can be done slickly and quickly, but only with lots of practice. It will be practiced as part of the course, but it is something that you should regularly practice in your departments.



12 - Primary Survey

Author: Dr Ash Basu, ED Consultant, Wrexham Maelor Hospital.

Date: 05/19

Review Date: 05/21

Introduction

The purpose of the Primary Survey is to ensure that all immediately life-threatening injuries are diagnosed and treated quickly and all the major injuries are picked up. Resuscitation and stabilisation should be done at the same time if necessary. Realistically, it is the CT scanner that picks up most injuries and so the emphasis should be on getting to the CT scan quickly (within 30 minutes of arrival), and safely [1].

Numerous time critical decisions are required to be made with a critical decision made every 72 seconds[2.]

This reception and resuscitation by the trauma team should be undertaken in a 'horizontal approach', i.e. multiple tasks performed simultaneously by the various trauma team members rather than sequential 'vertical approach', as is traditionally advocated by ATLS[3].

This 'vertical approach' is the least efficient method of proceeding, and where more than one member of staff are available, a team approach should be employed, with the 'horizontal approach being shown to lead to significant reductions in resuscitation times[3]. The *European Trauma Course* teaches the horizontal approach and it is also the method used by most Major Trauma Centres and the military trauma teams.



C-ABCDE Assessment of the Primary Survey

Assessment of the trauma team should follow the **C-ABCDE** model, with priorities being in order of the most immediate threats to life.

Once these have been addressed, the key priority is to ensure that the patient undergoes CT scan within 30 minutes of arrival[4].

C - Catastrophic haemorrhage

This is the first priority referring to any obvious, major bleeding points such as stab wounds where direct pressure, or indirect pressure in the form of a tourniquet can be applied to try and stem the flow of blood, whilst activation of the Major Haemorrhage Pathway (MHP) if deemed necessary

Any evidence of massive external haemorrhage or traumatic cardiac arrest should have been evaluated prior to formal handover as part of the immediate reception '10-second round'.

See Chapter 11 for actions on arrival of patient.

A - Airway assessment with cervical spine protection

The airway doctor (usually the anaesthetist on the trauma team) is responsible for assessing the airway and communicating to the team leader. Any complete obstruction should be dealt with immediately by the use of basic and advanced airway management. If there are indications for RSI (eg partial obstruction or reduced GCS, then this should be communicated to the Team Leader immediately so that planning for the RSI can proceed. ***(see chapter 14 – Airway management including RSI).***

Cervical spine protection should be considered for all patients, but it is generally accepted that if a patient is fully alert, conscious, co-operative, and is able to understand instructions, they will be able to protect their own cervical spine by keeping their head still.

If, however the patient has a reduced conscious level, either on presentation, or as a result of a Rapid Sequence Induction, then care must be taken to maintain cervical spine immobilisation. Traditionally this is in the form of cervical collar with head blocks, straps or tape.

Consideration, however, should be given those patients with suspected head injuries, as the use of cervical collars may impend venous return by constriction of the neck, therefore causing unwanted rises in intracranial pressure, so in these situations the collar may be omitted.

Once airway assessment has been undertaken, and presuming no airway interventions are required the airway practitioner should assist the trauma team in the following:

- Glasgow Coma Scale (GCS) assessment and examination of pupils
- Analgesia administration after discussion with the TTL

- Obtaining an AMPLE history (**A**llergies, **M**edications, **P**ast Medical History, **L**ast eaten (time of), and **E**vents leading up to injury)
- Supporting other members of team if no airway intervention required

B - Breathing assessment

If using the horizontal approach, one team member should be appointed the role of assessing for and treating breathing problems. This should be someone with good level of clinical examination skills and the ability to insert chest drains or perform thoracostomy if needed (or at least has the support to do these if needed).

The assessment should examine breathing pattern using a combination of inspection, feeling and auscultation of the chest looking specifically for evidence of chest wounds, broken ribs, flail chest, surgical emphysema, pneumothorax or haemothorax. Pulse oximetry (SpO_2) monitoring should be attached as soon as possible and if available nasal end tidal CO_2 should be considered as additional monitoring of ventilatory function, respiratory pattern, and conscious level.

If clinically indicated, a thoracostomy may required to decompress a tension pneumothorax following discussion with the TTL, with subsequent chest tube insertion.

Once breathing and ventilatory assessment and interventions, inspection & palpation of the neck should be undertaken evaluating for: Tracheal deviation, wounds, bleeding, surgical emphysema, laryngeal crepitus / injury, and distended neck veins

See Chapter 15 for more details on chest injuries.

C - Circulatory assessment with control of haemorrhage

The primary role of this clinician is to evaluate for signs and location of haemorrhage with a view to stemming any overt haemorrhage if possible. Their secondary role is to obtain adequate access to the circulation and obtain blood for tests.

The premise of 'on the floor, and four more' should be applied as the five main areas for haemorrhage:

1. External haemorrhage in the form of wounds, ensuring the back of the patient is inspected if there is significant concern regarding the site of external haemorrhage in the particular context of suspected penetrating trauma
2. Thoracic cavity
3. Abdominal cavity including the retro-peritoneum
4. Pelvis

5. Long bones, particularly of the lower limbs

See Chapter 13 for management of major haemorrhage.

Concurrently ECG & Blood pressure monitoring should be commenced (On regular 2.5 minute cycles) and Capillary Refill Time (CRT) should be checked centrally.

Remember that clinical evidence of occult haemorrhage may be subtle or even non-existent, especially in the young and fit, so make every effort to actively exclude it and keep reviewing the patient to look for any changes.

Vascular access should also be obtained using large bore peripheral access, and when this is not possible, intraosseous access should be considered early.

Bloods should be taken including Group & Save and/or Cross Match, ROTEM, FBC and Clotting. Venous blood gas sampling including a lactate is also a useful marker for circulatory insufficiency.

If the Trauma Team have the appropriate skills, they can perform a Focussed Assessment with Sonography in Trauma (FAST), however this should not delay CT. If there is likely to be a significant delay to CT and the patient is deemed cardiovascularly unstable then FAST may offer an additional argument for transfer to theatre. FAST scans are a rule-in rather than rule-out assessment.

D - Disability and neurological assessment

This can be carried out by the airway clinician and consists of:

- a GCS score (see Appendix A),
- assessment of pupils for size, symmetry, and reactivity to light response
- whether there is a difference in motor response in either side (lateralising signs).

It is very important that an accurate assessment of these are documented prior to the patient being anaesthetised for RSI or sedated.

GCS Score marks the patients best Eye, Motor and Verbal Responses.

See Appendix A for guide on most up to date method of recording GCS. For further information and video demonstration visit **www.glasgowcomascale.com**

See Chapter 18 for further details on management of head injuries.

E - Exposure

Finally, complete full exposure is required to complete the primary survey. All wounds, suspected fractures and peripheral circulation should be documented. Clothes and debris should be removed. Wounds should be dressed.

Active measures must be taken to maintain body temperature, and patient dignity.

Remember the priority is to get the patient to CT scan, so this can be postponed if CT is ready, however do not forget.

Log roll

In order to complete the exposure assessment of the primary survey, a log roll is required.

This facilitates the removal of clothes, debris, and to check for any wounds to the back. Only the minimum amount of rolling required to achieve this should be done (about 15 degrees tilt is sufficient)

Extra care, however must be taken in:

- Cardiovascularly unstable patients
- Pelvic trauma
- Spinal trauma

In time critical patients, the team leader may decide to transfer the patient (to theatre or to the MTC) without completing the log roll or other aspects of exposure and secondary survey. Should this occur then this should be clearly communicated to the MTC TTL verbally, and in the clinical record.



Summary

The Primary Survey should address immediately life-threatening injuries whilst ensuring CT imaging is achieved ideally within 30 minutes of arrival.

This progress to CT should only be interrupted for:

- Emergency Airway Management in patients requiring airway, oxygenation or ventilatory support.
- Thoracostomy +/- chest drain insertion in suspected tension pneumothoraces.
- Control of haemorrhage by use of:
    - Tranexamic Acid administration
    - Direct wound pressure and / or tourniquet application
    - Pelvic & long bone splinting
    - MHP activation, and blood product resuscitation
    - Operative management
    - Management of traumatic cardiac arrest

The patient should be moved as little as possible prior to the CT scan and Log rolls should be restricted to 15 degrees tilt.

References

1. Georgiou and Lockey. The performance and assessment of hospital trauma teams. Scandinavian Journal of Trauma, Resuscitation and Emergency Medicine 2010, 18:66
2. Fitzgerald et al. Trauma Resuscitation Errors and Computer-Assisted Decision Support. Arch Surg. 2011 Feb;146(2):218-25.
3. http://www.trauma.org/archive/resus/traumateam.html
4. NHS England Quality Surveillance Team. Major Trauma Services Quality Indicators. 2015 p.60
5. National Confidential Enquiry into Patient Outcome and Death. Trauma: Who cares?. 2007

Appendix A. Guide to Recording GCS.

www.glasgowcomascale.com

CHECK	OBSERVE	STIMULATE	RATE
For factors Interfering with communication, ability to respond and other injuries	Eye opening , content of speech and movements of right and left sides	Sound: spoken or shouted request Physical: Pressure on finger tip, trapezius or supraorbital notch	Assign according to highest response observed

Eye opening

Criterion	Observed	Rating	Score
Open before stimulus	✔	Spontaneous	4
After spoken or shouted request	✔	To sound	3
After finger tip stimulus	✔	To pressure	2
No opening at any time, no interfering factor	✔	None	1
Closed by local factor	✔	Non testable	NT

Verbal response

Criterion	Observed	Rating	Score
Correctly gives name, place and date	✔	Orientated	5
Not orientated but communication coherently	✔	Confused	4
Intelligible single words	✔	Words	3
Only moans / groans	✔	Sounds	2
No audible response, no interfering factor	✔	None	1
Factor interferring with communication	✔	Non testable	NT

Best motor response

Criterion	Observed	Rating	Score
Obey 2-part request	✔	Obeys commands	6
Brings hand above clavicle to stimulus on head neck	✔	Localising	5
Bends arm at elbow rapidly but features not predominantly abnormal	✔	Normal flexion	4
Bends arm at elbow, features clearly predominantly abnormal	✔	Abnormal flexion	3
Extends arm at elbow	✔	Extension	2
No movement in arms / legs, no interfering factor	✔	None	1
Paralysed or other limiting factor	✔	Non testable	NT

Sites For Physical Stimulation

Finger tip pressure | Trapezius Pinch | Supraorbital notch



Features of Flexion Responses

Modified with permission from Van Der Naalt 2004
Ned Tijdschr Geneeskd

Abnormal Flexion
Slow Sterotyped
Arm across chest
Forearm rotates
Thumb clenched
Leg extends



Normal flexion
Rapid
Variable
Arm away from body

13 - Management of Catastrophic Haemorrhage & Damage Control Resuscitation

Author: Dr Stephan Clements, Consultant Anaesthetist, YG & EMRTS

Date: 05/19

Review Date: 05/21

Introduction

Massive bleeding is the commonest cause of death within the first hour after injury and remains an important cause in the next few hours. The severely injured and haemorrhagic patient has a limited amount of physiological reserve before irreversible organ derangement and collapse occur. Damage control resuscitation (DCR) is an approach that aims to minimise the depletion of this reserve. It ideally begins at time of injury and continues in the emergency department, through the operative theatre into the ICU. Here we outline the principles and interventions that encompass DCR.

Pathophysiology

The initial pathophysiologic response to severe injury is characterised by the classically described lethal triad of hypothermia, coagulopathy (ie blood doesn't clot properly) and acidosis in association with complex disordered immune-inflammatory processes.

Coagulopathy is present at the time of admission to the emergency department in up to 25–35% of trauma patients. This is associated with an up to 8 fold higher mortality within the first 24 hours after injury.

Although, acidosis, hypothermia and iatrogenic resuscitative dilution of coagulation factors may contribute to trauma-induced coagulopathy, they are now not considered to be the primary drivers. Instead tissue injury +/- hypoperfusion itself initiates a systemic haemostatic derangement, characterised by hyperfibrinolysis. **In other words your blood clots become unstable, your blood doesn't clot and you bleed more.** This is termed, Acute Traumatic Coagulopathy (ATC) and is evident within only a few minutes of injury.

Preserve first, Replace second

 Email stephan.clements@wales.nhs.uk

Catastrophic Haemorrhage

Firstly, we need to rapidly identify and aggressively control catastrophic haemorrhage.

The first clue that there may be bleeding will be in the pre-alert information. If the pre-alert mentions massive blood loss OR the patient has signs and symptoms of shock OR the pre-hospital team requests blood is ready, you should prepare your team:

1. Activate the major haemorrhage protocol.
2. Collect O negative blood and have it in resus.
3. Prepare the rapid infuser (with saline unless you definitely intend to give blood).
4. Prepare equipment for good I.V. / I.O. access (ideally trauma line or rapid infuser catheters) and nominate one person responsible for gaining access and sending blood samples.
5. Prepare pelvic binder, tourniquets, haemostatic dressings and pressure dressings.
6. Call for a consultant anaesthetist and surgeon to be present and ask them to alert theatres.
7. Dedicate two people in your team to manage blood product administration and someone to liaise with blood bank.
8. Prepare blood bottles for X-match, ROTEM, Blood gas, FBC and clotting (and prioritise in that order).

Management of external bleeding:

Use the **DIPT** approach to severely haemorrhaging limbs:

D- Direct Pressure +/- limb elevation: Expose the source of bleeding and apply firm direct pressure at the site immediately. If soaking through, quickly check you are pressing in the correct place then apply one more dressing. Avoid repeated looks and multiple dressings.

I- Indirect pressure: Applying pressure to the artery proximal to the bleeding point. These measures if applied correctly, will control up to 90% of external bleeds.

P- Packing: Tight packing with haemostatic gauze is appropriate for neck, axillae, abdominal and groin wounds.

T- Tourniquet: Apply ~ 5cm above the site of massive haemorrhage, avoiding joints. Tighten until bleeding stops. N.B. this will cause extreme pain in the conscious casualty. A second proximal tourniquet is often required for lower limbs.

Non-compressible (internal) bleeding:

Minimal handling and movement of the blunt trauma patient is paramount to minimise clot disruption. Anatomically reduce limb fractures (+/- traction) and apply a pelvic binder as a priority, when appropriate. If intra-abdominal or intra-thoracic bleeding is suspected an early decision needs to be made whether to take the patient straight to theatre for damage control surgery (a laparotomy / thoracotomy) or even to angio suite for interventional radiology (if available). This should be a

joint decision between the TTL, anaesthetist and senior surgeon. Damage Control Resuscitation should still be commenced.

Damage Control Resuscitation

Not all injured patients need DCR. So what triggers can use to identify those patients who would benefit? **Often the most reliable trigger is the gut feeling that you get when you see a patient who looks really sick**. (A bit pale, sweaty, cold, confused or agitated, breathless, tachycardic, etc). This combined with a significant mechanism and / or any of the following with should prompt you to start DCR:

Systolic blood pressure (SBP) <90mmHg, Estimated massive blood loss, Hb <110g/L, Base deficit >6, Temperature <35°C.

It is quite possible to have lost a significant amount of blood and have normal or even high blood pressure, especially when you are young, due to patients' adrenaline and stress response to the injury. Blood pressure is not a reliable sign of shock, and should not be the sole trigger for whether to start DCR. DCR may need to be commenced even with a high BP.

Once we have chosen to adopt DCR, there are 3 key elements:

1. Permissive (controlled) Hypotension
2. Haemostatic Resuscitation
3. Damage control Surgery

Controlled Hypotension:



This is a compromise. We allow the blood pressure to fall low enough to slow exsanguination and avoid disrupting unstable clot, but in so doing we compromise tissue perfusion. Suggested targets:

- Penetrating trauma: SBP 60mmHg

- Blunt trauma: SBP 80-90mmHg
- Bleeding Trauma with head injury: SBP 100mmHg
- After 60-90 minutes post injury, target SBP 110mmHg until haemostasis is achieved.

Limit Permissive Hypotension to maximum 60-90 minutes from time of injury

- If BP below target 🡒 give blood products as per Major Haemorrhage protocol
- If BP above target 🡒 check perfusion (strong pulse, warm peripheries, no acidosis on ABG)
    - good perfusion 🡒 continue to monitor.
    - poor perfusion 🡒 give fentanyl 25-50 mcg boluses to decrease catecholamine release and cause peripheral vasodilation. When BP drops give blood products as per major haemorrhage protocol.

Haemostatic Resuscitation: Supporting the coagulation system.

- Activate the Massive haemorrhage protocol (Appendix A). Ask for FFP and platelets early. Aim for 1:1 initially, but then use the ROTEM to guide.
- Only use blood products to resuscitate severely injured & bleeding patients.
- Administer 1g TXA early followed by 1g over 8 hours.
- Think anticoagulants (eg warfarin), especially in the elderly. Give Prothrombin complex concentrate before obtaining lab INR results in severe haemorrhage or if suspected severe head injury in anticoagulated patients.
- Most major trauma patients have a low ionized Ca^{++} on arrival. Give Calcium chloride (not gluconate) after giving the 4th blood product.

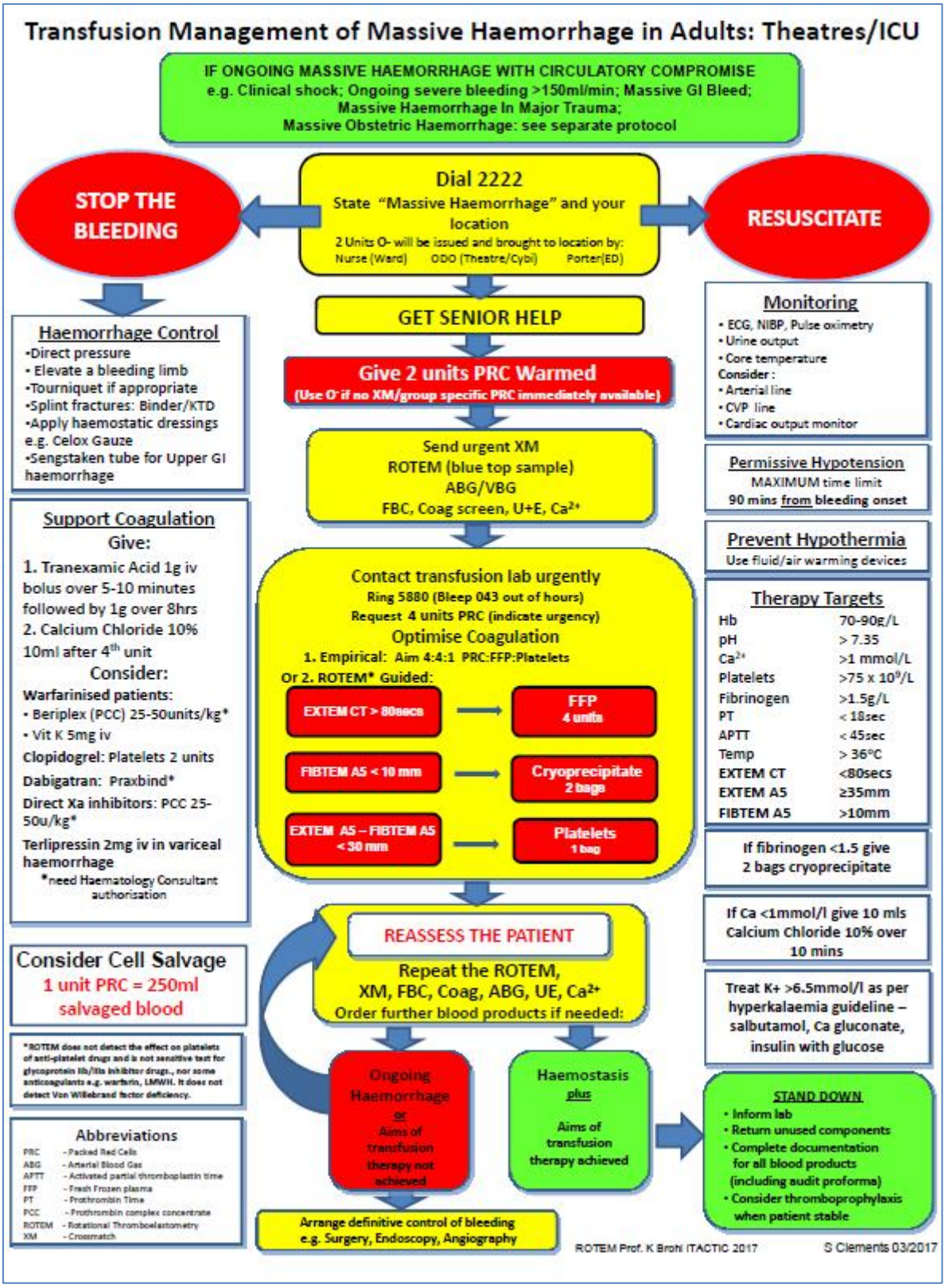
Summary:

1. Coagulopathy is present in up to 35% of severely injured patients on admission and it is bad news.
2. If the pre-alert information hints at bleeding, activate the major haemorrhage protocol and prepare your team.
3. Control external bleeding and apply a pelvic binder quickly to preserve blood.
4. If the patient looks sick commence damage control resuscitation. Don't be reassured by a normal BP.
5. If the patient does not respond and you suspect internal bleeding, make an early decision to go to theatres.
6. Follow the Major Haemorrhage protocol and use ROTEM to guide blood product administration.

References / Further Reading:

1. Read your departmental major haemorrhage protocol.
2. Kushimoto et al. Acute traumatic coagulopathy and trauma induced coagulopathy: an overview. Journal of Intensive Care (2017) 5:6
3. Giannoudi et al. Damage control resuscitation: lessons learned. Eur J Trauma Emerg Surg (2016) 42:273–282
4. Anaesthesia Trauma and Critical Care (ATACC) course manual 2014.

Appendix A – example of a major haemorrhage protocol



14 - Airway management including RSI

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Date: 05/19

Review Date: 05/21

Introduction

Good airway management is a crucial factor in the management of the trauma patient. Poor outcomes in trauma patients often stem from poorly managed airways. The airway will normally be managed by an experienced anaesthetist as part of the trauma team, but it is important that the whole team are aware of how to recognise an airway at risk, when an intervention will be required, how to assist in airway interventions and crucially what to do if the airway intervention starts to go wrong. Airway interventions such as rapid sequence induction (RSI) in the ED are high-risk procedures with a real risk of causing severe harm or death to a patient. The key to minimising this risk this is a well trained, well rehearsed team working together using a standardised approach and communicating their strategy.

Pathophysiology

Possible reasons for a compromised airway in trauma include:

1. Reduced level of consciousness (either from head injury, drugs, alcohol, hypotension or any other reason),
2. Blood, vomit or foreign body in the airway,
3. Direct trauma or burns (thermal or chemical) to the face or neck

Signs and Symptoms

Obvious signs of a compromised airway include noisy breathing (stridor, snoring), inadequate breathing, hypoxia, "see-saw" movement of chest and abdomen, blood or vomit around the airway, trauma or swelling to the face and neck, unconscious and responding only to painful stimulus.

Management

The trauma team should try and predict possible airway compromise and requirement for intervention (including RSI) as much as possible from the pre-alert information. The team leader should brief the team on what the confirmed and potential injuries are and what interventions may be required so that the team can prepare equipment (and RSI drugs, etc.) if necessary.

On arrival of the patient in Resus, the first action should be a quick 10-second check of the patient to check that they have a pulse, there is no catastrophic bleeding and they have an airway. The anaesthetist will usually do the airway assessment. This should be a quick assessment of whether an immediate intervention is required (eg airway is completely obstructed). If it is obvious that the patient will need an airway intervention but it can wait until handover this should be communicated briefly (eg "Airway is fine" or "Suboptimal airway - will need an RSI" or "No airway, Needs intervention now"). Unless immediate intervention is required the team would proceed with receiving ATMIST handover and transferring patient to trolley.

Basic Airway management.

Compromised airways should be managed initially with basic airway opening techniques. These include jaw thrust, insertion of oropharyngeal / guedel airway) or nasopharyngeal airways (with care not to cause further trauma or bleeding). If the plan is to proceed to RSI these temporary measures can help to buy time to prepare properly for the intervention.

Indications for RSI.

The indications for RSI can be broken down into:

- Airway – Compromised airway for any reason or impending compromise.
- Breathing – Poor effort of breathing, inadequate oxygenation or ventilation.
- Neuro – Reduced consciousness for any reason (GCS <9 or not responding to voice).
- Severely agitated or combative patients where sedation is required in order to treat them safely.
- Head injury where tight control of arterial CO_2 is required to prevent secondary brain injury.
- Humanitarian – Patient is in severe pain eg multiple fractures requiring reduction
- Expected clinical course – Eg patient is likely to need onward transfer, and will require intubation for this, or severely shocked patients where sedation and ventilation is required as part of their damage control resuscitation.

The decision to RSI should be taken jointly between the TTL and the anaesthetist.

Rapid Sequence Induction.

The aim of an RSI is to pass an endotracheal tube in to the trachea and inflate the cuff, isolating the lungs in order to protect them from aspiration and allow optimal oxygenation and ventilation. It is very much a team activity.

Preparation

Good preparation of the team, equipment and drugs is essential, as well as a solid plan consisting of a logical sequence of interventions in the event of failure. (6P's and all that). Equipment and drugs should be checked by the anaesthetist and ODP. This must include checking the presence of back up

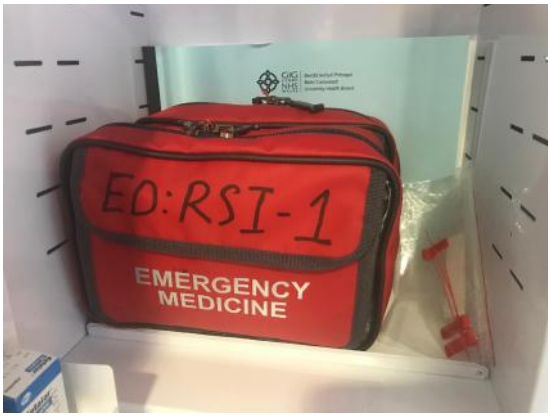
devices such as supraglottic airway device and criciothyroidotomy kit. A checklist should always be used, such as the one in Appendix A.

Always remember to optimise positioning, use apnoeic oxygenation and use a bougie by default to increase your chances of success at first attempt.

Remember to document the GCS and pupil sizes prior to RSI. Not only is this clinically vital information, it is also required for our TARN documentation.

Cervical collars and tapes should always be undone for RSI, with one person nominated to do manual in line stabilisation (MILS) if necessary. Minimal standards of monitoring recommended by the AAGBI should be used (ECG, NIBP on regular intervals 2.5 minutes, SaO2 and ETCO2).

Patients who have bled and are hypovolaemic may be compensating and may have normal or high blood pressure. It is important to assess whether the patient is shocked and consider a fluid bolus or potentially giving a rapid transfusion of blood products prior to RSI, as well as adjusting drug doses appropriately so as not to cause severe hypotension post induction.



RSI Checklist

When the airway team are ready for the RSI this should be communicated to the Trauma Team Leader. The TTL should then ensure the whole team are focused on this intervention with noise minimised to essential communication only. The airway team should then proceed with the pre RSI checklist (read out loud so that the whole team can hear). The checklist should be read as a challenge response. The plan for failure must be verbalised so that the whole team are aware of the plan and where the relevant equipment is if required.

During RSI

Time of starting RSI should be called out followed by the drug doses as they are given so that the scribe can document this.

During the RSI noise must be kept to a minimum. The TTL should keep watch of the patient's vital signs. It is generally unhelpful to call out the oxygen saturations every 10 seconds, however the TTL should be alert to any loss of situational awareness by the anaesthetists and should prompt (then

mandate) the team to pause intubation attempts and move to plan B (eg revert to bag mask ventilation) if the saturations drop significantly.

Post RSI

As soon as the intubation is complete correct positioning should be confirmed by $ETCO_2$ and bilateral breath sounds. Be alert to the possibility of tension pneumothorax in cases of chest injuries. A sudden reduction in ventilation compliance and / or hypotension together with clinical suspicion should prompt urgent thoracostomy to decompress the chest. (See Chapter 15).

Connect the patient to the ventilator as soon as possible and set a target for the $ETCO_2$ (3.5-4.0 kPa in head injury patients which should correspond roughly with arterial CO_2 of 4.5-5.0kPa). Commence sedation and proceed to post intubation care – checking the blood pressure and ensuring adequate MAP (80-90 mmHg for head injuries)

Ensure that the grade of intubation, as well as any difficulties or complications, are documented in the ED trauma notes.



Failed intubation

In short in the event of a failed intubation the anaesthetist should follow the Difficult Airway Society guidelines.

In the event of a poor view the intubator should take 30 seconds to consider what they can do to improve their view before reattempting. This should include a change in head / neck position (unless C-spine injuries suspected), external laryngeal manipulation, release of cricoid pressure, use of a different laryngoscope (eg Videolaryngoscope), use of a stylet, or change in intubator. After two

attempts, a second anaesthetist should take over and have one further attempt. If this fails it should declared as a failed intubation and immediately proceed to plan B, which in most cases will be insertion of a supraglottic airway device (SAD) (preferably 2nd generation such as pro-seal). If not already present, a consultant anaesthetist should be called immediately.

Can't intubate Can't Oxygenate.

If all attempts to intubate and oxygenate have failed and the oxygen saturations are dropping significantly the team should proceed to emergency surgical airway or surgical cricothyroidotomy. It is not appropriate here to dictate who or how this should be done, however each department should have a clear policy on equipment that should be used for this scenario. This intervention should be done by whoever has been trained with the kit available. It is unlikely that anyone will have much experience in this technique. Every department should ensure their staff are regularly trained for this eventuality.

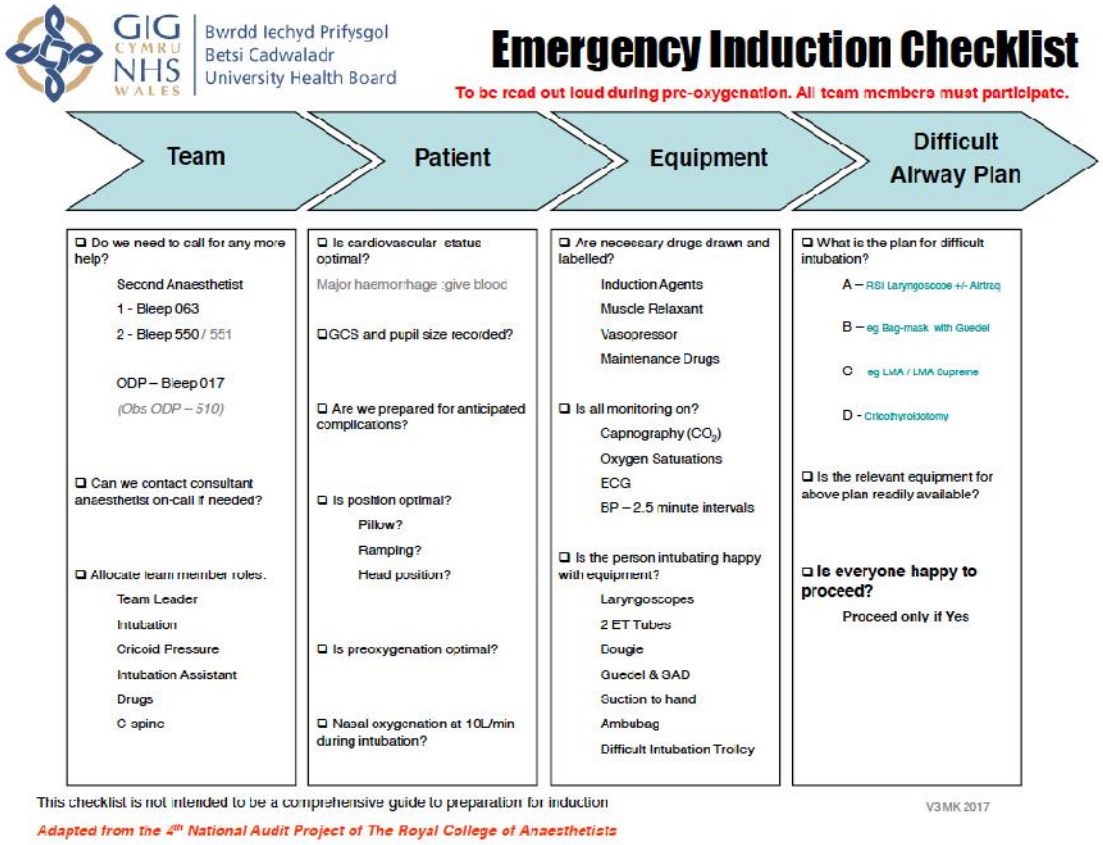
Summary

1. **Predict airway problems and plan ahead**
2. **RSI should never be rushed. Use the checklist and prepare well to reduce the risk of disaster.**
3. **Always have a plan for failure and ensure that the team are aware of it.**
4. **Train and rehearse RSI regularly in your departments including drills for failed intubation.**

References / Further Reading:

1. Familiarise yourself with your departmental RSI checklists
2. NAP4 Royal College of Anaesthetists National Audit Project on complications of airway management
3. www.emergencyinductionchecklist.blogspot.com
4. www.uptodate.com
5. https://www.youtube.com/user/DrAMBHeardAirway

Appendix A – Examples of Pre and Post RSI checklists



GIG CYMRU NHS WALES
Bwrdd Iechyd Prifysgol Betsi Cadwaladr University Health Board

Emergency Induction Checklist

To be read out loud during pre-oxygenation. All team members must participate.

Team

- ❑ Do we need to call for any more help?
  - Second Anaesthetist
  - 1 - Bleep 063
  - 2 - Bleep 550 / 551
  - ODP – Bleep 017
  - *(Obs ODP – 510)*
- ❑ Can we contact consultant anaesthetist on-call if needed?
- ❑ Allocate team member roles:
  - Team Leader
  - Intubation
  - Cricoid Pressure
  - Intubation Assistant
  - Drugs
  - C spine

Patient

- ❑ Is cardiovascular status optimal?
  - Major haemorrhage :give blood
- ❑ GCS and pupil size recorded?
- ❑ Are we prepared for anticipated complications?
- ❑ Is position optimal?
  - Pillow?
  - Ramping?
  - Head position?
- ❑ Is preoxygenation optimal?
- ❑ Nasal oxygenation at 10L/min during intubation?

Equipment

- ❑ Are necessary drugs drawn and labelled?
  - Induction Agents
  - Muscle Relaxant
  - Vasopressor
  - Maintenance Drugs
- ❑ Is all monitoring on?
  - Capnography (CO_2)
  - Oxygen Saturations
  - ECG
  - BP – 2.5 minute intervals
- ❑ Is the person intubating happy with equipment?
  - Laryngoscopes
  - 2 ET Tubes
  - Bougie
  - Guedel & SAD
  - Suction to hand
  - Ambubag
  - Difficult Intubation Trolley

Difficult Airway Plan

- ❑ What is the plan for difficult intubation?
  - A – RSI Laryngoscope +/- Airtraq
  - B – eg Bag-mask with Guedel
  - C eg LMA / LMA Supreme
  - D - Cricothyroidotomy
- ❑ Is the relevant equipment for above plan readily available?
- ❑ **Is everyone happy to proceed?**
  - Proceed only if Yes

This checklist is not intended to be a comprehensive guide to preparation for induction

V3MK 2017

Adapted from the 4th National Audit Project of The Royal College of Anaesthetists

GIG CYMRU NHS WALES
Bwrdd Iechyd Prifysgol Betsi Cadwaladr University Health Board

Post Induction & Pre CT Checklist

To be read out loud post intubation. All team members must participate.

Airway & Breathing

- ❑ ETCO2 trace
  Target ETCO2 =
  [3.5 to 4.0kPa for head injuries / intracranial bleed]
- ❑ Bilateral breath sounds
  Check tube length.
  Thoracostomy / Chest drain required?
- ❑ C-spine resecured
  Blocks and tape
- ❑ Sedation connected & running
  If delayed consider rebolus of ketamine or midazolam

Circulation & Disability

- ❑ Is cardiovascular status managable?
  Target MAP 85mmHg for neuro.
  Penetrating trauma: titrate to radial pulse
  Major haemorrhage : commence damage control resuscitation according to MH protocol.
  Sepsis : start vasopressors as indicated.
- ❑ Do we have enough iv access?
  Easy access to flush line.
- ❑ Pupil size checked & recorded
- ❑ Are we prepared for anticipated complications?

Equipment

- ❑ Are necessary drugs drawn and labelled?
  - Extra Muscle Relaxant
  - Vasopressor
  - Maintenance Drugs
- ❑ Is all monitoring on?
  - Capnography (CO_2)
  - Oxygen Saturations
  - ECG (3 lead)
  - BP 2.5 minute interval
- ❑ Equipment to take with us.
  - Laryngoscope
  - BVM
  - Bougie & ET Tube
  - Guedel & SAD
  - Suction
  - Drugs
  - Oxygen & Ventilator
  - Blood / Fluids

Packaging

- ❑ What is the plan after CT?
  Consider calling EMRTS for transfer
- ❑ Patient packaged on Wolverson mattress
- ❑ Lines accessible
- ❑ Are CT ready ?
- ❑ What equipment needs to be prepared for return?
  Eg Art line, Urinary catheter, syringe drivers, transfer trolley.
- ❑ **Is everyone happy to proceed?**
  Proceed only if Yes

This checklist is not intended to be a comprehensive guide to patient care.

V3. MK2017

15 - Chest injuries and procedures

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Originally written: April 2017

Most recent review: March 2019

Introduction

North Wales trauma units do not have thoracic surgeons on site. Fortunately, the majority of chest injuries do not require a thoracic surgeon. Of those who do, the majority are stable enough to undergo transfer to Stoke, or they wouldn't have made it to hospital in the first place. In terms of chest procedures, we should be able to site a chest drain, and (very rarely) we may need to do a full-on thoracotomy. For patients with rib fractures, we should be able to offer suitable regional anaesthetic blocks.

Classification

Most people divide chest trauma into **Blunt** (fell off a roof and landed on chest) and **Penetrating** (got stabbed in the chest). This is because the injuries you are likely to see with blunt trauma are a bit different from penetrating trauma. BUT this is not always the case, so be alert anyway!

Chest injuries seen with blunt trauma

- Broken ribs
- Pneumothorax/Haemothorax
- Injuries to the lung: Bruising (pulmonary contusion); or lower airway injuries.
- Cardiac contusion (bruising)
- Aortic dissection
- Diaphragm rupture

Chest injuries seen with penetrating trauma

- Pneumothorax/Haemothorax
- Cardiac tamponade
- Injuries to the larynx/upper trachea
- Oesophageal (gullet) injuries

1. Broken Ribs

Broken Ribs - Introduction

Broken ribs aren't usually a big problem in and of themselves: the problem is that they often come as a package with pneumothorax, lung contusions and haemothorax. The pain they cause means patients don't breathe properly and end up with chest infections.

There is a special type of rib fracture though, called a flail chest. This happens when so many ribs are fractured in so many places that the chest wall doesn't work properly. Feel your own chest for a minute: when you breathe in deep, the chest moves out, and air gets sucked in through your nose.

Now imagine if your ribs are all broken in several places. When you try to breathe in deep, your chest will get sucked in instead, and no air will go up your nose. This is bad.

Experts now say that the description given above of a flail chest is actually out of date, and what is important is the fact that the injury impacts on breathing. But it is important that you understand the 'flail' concept whether or not you believe it!

Broken Ribs - Diagnosis

Patient has a lot of pain, particularly when they cough or breathe. Sometimes you can feel that the ribs are broken (this is called crepitus).

Chest x-ray isn't actually that good at picking up rib fractures. They will show up nicely on CT.

Broken Ribs - Management

Identify and treat the associated problems.

For rib fractures that aren't flail, the main treatment is pain relief. Involve an anaesthetist early before the patient gets a chest infection. The patient may need a PCA, an epidural, or other anaesthetic block.

For flail ribs, there may be a surgical option. This depends on a lot of things, so speak to Stoke.

Nobody straps ribs anymore.

2. Pneumothorax

Pneumothorax – introduction

A pneumothorax happens when air escapes into the space between the lung and the chest wall. The lung collapses down. The air might have got in from the outside (e.g. a hole being made in the chest wall by a knife), from the inside (a tear in the lung as a result of blunt trauma) or both (a hole being made in the chest wall and the lung by a knife). Pneumothorax can be partial, complete, open, or tension.

Partial and **complete** pneumothoraces are what they say – the lung is partially, or totally, collapsed.

An **open pneumothorax** is also called a sucking chest wound. Put your hand on your chest again and breathe in deep. The air whistles up your nose and into your lung. Now imagine a big hole in your chest: the air would whistle into the hole! As it does so, the lung collapses down, and your breathing efforts are pulling more air in through the hole than up your nose. This is bad.

Now imagine a hole in your chest where there is some tissue covering it so that air can get in, but not out. This actually happens. Every time you take a breath the pneumothorax gets bigger, until it starts squashing your heart and your other lung. This is a **tension pneumothorax.** It is rapidly fatal unless something is done.

Pneumothorax - Diagnosis

Patients will be short of breath, and may have chest pain. The classic signs are actually hard to find in real life, but are (on the affected side): poor air entry, loud percussion note.

If patients have surgical emphysema (air under the skin – feels like bubblewrap) then they almost always have an underlying pneumothorax.

In a tension pneumothorax the patient is often periarrest. They look desperate, have low sats, and low blood pressure. There will be some sign of chest injury.

The diagnosis is usually obvious on chest x-ray, as long as you look for it. It is also seen on CT.

It is also possible to use ultrasound to find pneumothorax – it is actually quite good. But don't waste time.

Pneumothorax – Management

If the patient is in a bad way, you can buy some time by doing a needle decompression of the affected side. This involves putting a big cannula in the front of the chest, halfway along the clavicle and a little below it. See procedures in appendix.

Other patients should have a chest drain inserted. See procedures in appendix.

If a patient has been brought in intubated by EMRTS, they may have had a thoracostomy. This is the same as a chest drain, but without the actual chest drain. This can also be done in ED if the situation is critical. A chest drain can be inserted in through the existing thoracostomy. Give the site a good clean with betadine / chlorhexidine as it won't be sterile. Our colleagues in Stoke recommend the patient has a dose of antibiotics (Co-amoxiclav) if they have had a prehospital thoracostomy, although it is fair to say this is based more on expert advice rather than robust evidence. If the patient doesn't need a chest drain then the thoracostomy can be cleaned, sutured and covered with a dressing. Monitor for signs of an infection.

Sometimes, tiny pneumothoraces are seen on CT: these don't need a chest drain.

Pneumothorax – a note about ventilated patients

If a patient has a pneumothorax and gets intubated, the chances are it will turn into a tension pneumothorax. This can happen really quickly. So if a patient who has (or is suspected of having) a pneumothorax is to be intubated, you need a plan. This may mean putting the chest drain in first, or (more humanely) giving the general anaesthetic but having someone prepped to do the chest drain as soon as the tracheal tube is in. You may need someone prepped on each side.



3. Haemothorax

Haemothorax - Introduction

A haemothorax is blood in the chest cavity; in the same place you find the air in a pneumothorax. It often comes from intercostal vessels (from broken ribs). It can be massive.

Haemothorax – Diagnosis

Clinically, it can be similar to a pneumothorax: poor air entry, poor chest movement. In addition the patient may be shocked from blood loss into the chest.

The diagnosis is often made when blood comes out of a chest drain. It may also be seen on chest x-ray (not great if patient lying flat), and on CT. Ultrasound is very good at showing fluid in the chest.

Haemothorax – Management

Management centres on what happens when you put in a chest drain. If more than 1500mls comes out (or if the bleeding doesn't stop) then that is generally considered an indication for thoracotomy. Clamping the chest drain probably doesn't help: the bleeding will continue but will just be hidden. However, if the patient can get cell salvage in Stoke, then leaving the blood in the chest may be sensible. Discuss with the Stoke TTL.

4. Injuries to Airways

Injuries to Airways – Introduction

Airway injury generally happens due to either penetrating neck injury, or severe deceleration injury causing the lungs to part company with the trachea. If such a patient makes it to hospital, they will probably need to go to the MTC for surgery.

Less commonly, blunt injuries can damage the airway. This may be from hanging, or from a blow to the neck disrupting something important.

Injuries to Airways – diagnosis

Clinically, these injuries are suspected if the patient is hoarse, has signs of airway obstruction (e.g. stridor), coughs up blood, or if there is the bubblewrap sign of air under the skin (subcutaneous emphysema). Finally, if a chest drain keeps on bubbling but the pneumothorax doesn't improve, that suggests an air leak, possibly from an airway injury.

Air in the centre of the chest (pneumomediastinum) seen on CXR or CT may be from an airway injury. CT will help with the exact diagnosis.

Endoscopy/bronchoscopy helps visualise the injury, and can help with plan for treatment.

Injuries to Airways - management

In upper airway trauma, it may be wise to secure the airway with a tube before the swelling blocks the airway. An ENT surgeon should attend if possible.

If the trauma is further down the airway, it may be necessary to use a double-lumen tube to isolate the bad lung. In a crisis it may be necessary to simply push a standard ET tube into the good lung (using a fibreoptic scope to guide it).

5. Cardiac Contusion

Cardiac Contusion – Introduction

This usually happens due to rapid deceleration. The heart basically bounces off the front of the chest and gets bruised. It can be fatal. There isn't a great deal can be done about it, so don't delay a CT to get an ECG or echo.

Cardiac Contusion – Diagnosis

Suspected if there is a sternal fracture. ECG may be abnormal. Echocardiogram may show part of the heart not moving properly (a regional wall motion abnormality). There is probably no point checking troponin, as nobody knows what to make of the result.

Cardiac Contusion – Management

If ECG is normal, then there probably isn't too much to worry about. If ECG abnormal, then the patient should have cardiac monitoring for 24 hours. If the patient needs surgery for something important, this should go ahead, as most trauma deaths are due to bleeding not arrhythmia.

6. Aortic Dissection

Aortic Dissection – Introduction

This is the term used for damage to the aorta. It usually happens due to rapid deceleration (fall, car crash). Most patients die before hospital. If a patient with this condition makes it as far as hospital, then it must only be a partial injury, and there is a reasonable chance they will survive, as long as we can package them up and get them to the MTC fast.

Aortic Dissection – diagnosis

In North Wales we are likely to make this diagnosis on the trauma CT.

Aortic Dissection – Management

The dissection actually takes second place to other life threatening injuries. But the basic idea is to get the patient safely to the MTC as quickly and as safely as possible. In the unlikely event that the blood pressure is too high, then the patient may need intravenous drugs to keep it under control. (Target MAP <80mmHg, can use beta blockers or vasodilators).

7. Diaphragm Rupture

Diaphragm Rupture – Introduction

This can happen from penetrating trauma, but big ones usually happen due to a big rise in abdominal pressure, for example from a seat belt in a high-speed collision. Basically the abdominal contents get forced into the chest, tearing through the diaphragm. This is usually seen on the left, because the liver is next to the diaphragm on the right and protects it.

Diaphragm Rupture – Diagnosis

This will generally be seen either on the CT, or when someone notices that the NG tube appears to be in the chest on an x-ray.

Diaphragm Rupture – Management

This needs surgical repair in the MTC. We should:

- Intubate/ventilate if respiratory distress
- Be super careful about chest drains – don't want to accidentally stick something in the bowel.

8. Cardiac Tamponade

Cardiac Tamponade – Introduction

The heart is enclosed in a shiny bag called the pericardium. It is a tight bag, and a small amount of blood getting into it can compress the heart and stop it pumping. This usually happens due to a stabbing.

Cardiac Tamponade – Diagnosis

The patient will be in a bad way. The problem is that the signs and symptoms are similar to those of tension pneumothorax and haemorrhage. The diagnosis will be made with ultrasound.

Cardiac Tamponade – Management

A patient with tamponade will not make it to the MTC for definitive management. We will have to sort it out here. This basically means doing a thoracotomy in the ED. See procedures.

9. Oesophageal Injuries

Oesophageal Injuries – Introduction

These generally happen due to penetrating trauma, but it is possible to rupture the oesophagus with blunt trauma if the abdomen is compressed e.g. by a seat belt in high-speed collision. Another way to damage the oesophagus is with a swallowed foreign body, or with swallowed caustic substances like bleach.

Oesophageal injuries are bad news. They have a high mortality due to gastric content entering the chest, and due to the difficulty in repair.

Oesophageal Injuries – Diagnosis

If the patient is conscious, they are likely to complain of chest pain.

There is often fluid and air in the chest cavity (pneumothorax/hydrothorax). When a chest drain goes in, there may be gastric content.

CT scan often gives the diagnosis, but not always.

Oesophageal injuries – Management

This is a specialist field, and will probably require chest drains, surgery to wash out the chest, possible surgical repair, and possibly stenting of the oesophagus. It is a job for the MTC.

Do not insert a gastric tube unless advised by the MTC.

Chest Trauma Summary

Priorities

1. Look after the patient's Airway, Breathing, and Circulation as per any trauma
2. CT will usually give the diagnosis so work on getting patient to CT ASAP.
3. Be able to insert (or assist with) chest drain.
4. Patients with significant chest trauma need to go to Stoke ASAP: liaise with the TTL.

Don't forget

Adequate pain relief for rib fractures

After intubation, if the patient deteriorates, consider tension pneumothorax

Massive haemothorax needs to go to the MTC ASAP.

Final thoughts

- Most chest injuries can be stabilised in ED before transfer to Stoke.
- Have a low threshold for intubation, as deterioration on route to Stoke is possible.
- Pneumothorax/haemothorax need a chest drain.

References / Further Reading:

1. ATACC Manual 2014
2. ATLS manual 9th Edition
3. http://www.uptodate.com

Appendix: Procedures.

The guidance here is aimed primarily at staff who are assisting with procedures. For those undertaking procedures, please read the British Thoracic Society guidelines on pleural disease. (available at https://www.brit-thoracic.org.uk/standards-of-care/guidelines)

Needle thoracocentesis

This is a rapid way of decompressing a suspected tension pneumothorax, prior to treating definitively with a chest drain. The idea is to relieve the pressure to buy time.

All that is required is something to clean the skin, and a large cannula. Dedicated devices (e.g. the Russell Pneumofix) are better, because standard cannulae are often too short, and easily kink.

The needle goes in the affected side, in the mid-clavicular line, at the second rib space. If it works, then you may hear a 'hiss' of air escaping, and the patient should get better right away.

Thoracostomy/Chest drain

A thoracostomy is when we put a hole in the chest. A tube thoracostomy is when we then put a chest drain into the hole. Most clinicians in hospital perform a tube thoracostomy, which they call a chest drain.

The procedure must be done in a proper sterile fashion: infection in the pleural space is a disaster.

There will be a drawer of dedicated equipment for this procedure in resus. It may be in the form of a pack. You will need:

- Procedure trolley
- Sterile field to open equipment on
- Sterility for procedure: gloves and gown; skin prep (e.g. chloraprep); drape with a hole in it
- Procedure equipment: gauze; local anaesthetic with syringe/needle; scalpel; big metal forceps; chest drain; suture; dressing
- Connecting equipment: Drainage bottle; connecting tube; sterile water
- Check with the operator what size of drain they need.

Thoracotomy

This is 'opening the chest', and is done in two circumstances:

- Patient arrested/periarrest, if thoracostomies on both sides have not helped, and there is a suspicion of cardiac tamponade. In cardiac arrest, the patient must have had vital signs within the previous 10 minutes or the procedure is futile.
- Chest drain keeps on draining masses of blood. In this case the patient should be moved to theatre for the procedure.

In North Wales EDs, it is important to bear in mind the following:

- The operator will probably not have done this for quite a while.
- We have not cardiothoracic backup.
- Our geography means that patients are less likely to arrive with us in a timeframe where they will benefit from the procedure.
- If the procedure is undertaken and the patient achieves ROSC, we then have the challenge of moving someone >100 miles with massive instability and an open chest and heart.

For this reason, the threshold for the procedure will be quite high. It is likely that the TTL will want to see evidence of tamponade on ultrasound before proceeding.

Key things to remember are:

- If the patient has arrested, chest compressions are unlikely to help. If they are going to get in the way of important procedures, the TTL will ask for them to be discontinued.
- Anaesthetist intubates and ventilates.
- Large bore IV access is obtained and O neg blood given (Massive Haemorrhage protocol).
- Most senior/experienced person undertakes the procedure.
- Think about other causes for the arrest: is a pelvic binder on?

There will be a thoracotomy pack in resus. The procedure itself needs to be done *quickly*. To begin with, all that is required is:

- A scalpel
- Big metal forceps
- Tuff cut scissors

This is often enough to get the chest open across the middle. Sometimes a Gigli Saw (looks like a wire with handles) is needed to get through the sternum.

At this point, some more equipment is needed:

- Sterile scissors
- Fancy gauze (celox or similar – the stuff that stops bleeding)
- Sutures
- Packing material

This procedure will be done very rarely, and the success rate is about 1%.

16 - Abdominal Injuries

Author: Dr John Glen, Consultant Intensivist, YGC

Originally written: April 2017

Most recent review: March 2019

Introduction

Abdominal trauma can be difficult to manage. Patients often have other injuries too, and the initial presentation of abdominal injuries can be non-specific. Being in a Trauma Unit, far from a Major Trauma Centre, adds problems of its own.

The challenges in the ED relate to:

- Level of intervention required prior to imaging
- Rapid diagnosis (i.e. CT)
- Decision on disposition. The options are:
    - Damage Control Surgery in Trauma Unit (then further decision on whether to send to Stoke).
    - Transfer straight to Stoke.
    - Manage in Trauma Unit (with or without surgery).

These decisions need to be made at a senior level, with input from the TTL at Stoke.

Injury Patterns and diagnosis

Similarly to chest trauma, abdominal trauma can be blunt or penetrating.

Penetrating trauma

Patients who have been stabbed are at high risk of requiring surgery. There are some signs right off the bat that suggest someone will need an operation. These include being shocked, having peritonitis, having blood come out of the mouth or anus, or having bowel visible. Most surgeons want a CT before they operate these days.

The big worry with penetrating trauma is that it might involve the chest as well. A knife can start in the abdomen and end up almost anywhere.

Blunt trauma

The key question here is about mechanism, and energy transfer. Someone who has been punched is unlikely to have avulsed a kidney, while someone who has been in a high-speed car accident could have seriously damaged just about anything in the abdomen that you can name. For us, the key is to get high-risk patients to CT as quickly and safely as possible.

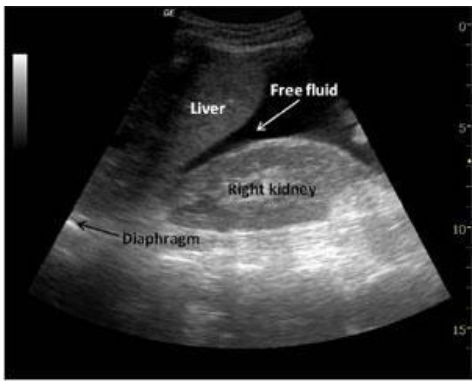
What about ultrasound, DPL etc.?

Nobody does DPL these days. (Diagnostic Peritoneal Lavage)

Ultrasound may be useful if:

- there is a delay in CT (although it may be better to work on getting the CT faster!)
- there are multiple patients, and you want to triage them. Works best with a radiologist.

Source: Google



Treatment

This really depends on the injury and its severity. Some injuries can just be observed, including low-grade liver, spleen and kidney injuries.

The reasons for going to theatre are:

Damage Control Surgery.

Sorting haemorrhage is the absolute priority. Some things can be removed (spleen, a kidney), some things can be packed with gauze (liver), vessels are tied off, and bowel problems are temporarily sorted by chopping out bad bits and clamping off the free ends. Patients can be OK for a few days with the abdomen left open on ICU. See Chapter 4 for more details on taking patients to theatre.

To restore anatomy

This can happen later on once the patient is stable.

Abdominal Trauma Summary

Priorities

1. Look after the patient's Airway, Breathing, and Circulation as per any trauma
2. CT will usually give the diagnosis so work on getting patient to CT ASAP.
3. **Senior** surgical and anaesthetic decision making is needed over whether to transfer out, or keep on site. Liaise with the TTL (locally and at Stoke). Patients who are so unstable that they are unlikely to survive the transfer to Stoke should definitely be operated on in the Trauma Unit.

The patient may have other injuries – particularly in penetrating trauma.

Final thought

The key to success here is getting a senior surgeon to the ED to help with the decision. If they are scrubbed in theatre with a phone held to their ear it just isn't going to work.



References / Further Reading:

1. ATACC Manual 2014
2. Abdominal Injuries - blunt and penetrating trauma. Midlands Critical Care and Trauma Networks. March 2019.
3. ATLS manual 9th Edition
4. http://www.uptodate.com

17 - Pelvic Fractures

Author: Dr John Glen, Consultant Intensivist, YGC

Originally written: April 2017

Most recent review: March 2019

Introduction

Pelvic fractures mustn't be missed. If there is any suspicion of a pelvic fracture you should treat. This is because people can bleed to death from it really quickly, and because poor patient handling can make the situation irretrievable. Most patients with a confirmed fracture will be referred to the MTC.

It takes a lot of force to break the pelvis. This means falling from a height, getting hit by a car, or crashing your motorbike.

Types of Injury

The most important type of fracture from an immediate ED perspective is a ring fracture, so after a few words about the others, we will concentrate on that.

The others

Acetabular Fracture

Where the femur meets the pelvis – there is a deep socket, and this can get broken into bits. Often a dislocated hip will complicate things. Can bleed a lot. Needs to go to the MTC.

Pubic Ramus Fracture

Seen most often in the elderly. Stable but painful.

Sacrococcygeal Fracture

Tailbone fracture – usually stable

Ring Fracture – the big one

The pelvis is a ring, a bit like a big Polo mint. When it breaks, it often breaks in more than one place (again, like a Polo mint). This means that there are at least two pieces which can move around. When the broken bits move apart, they create a lot of space for blood to accumulate. There is also a network of veins plastered against the inside of the pelvis, which are easily torn and can bleed a lot. Finally, an artery can be injured, and there are plenty to choose from in this area. So a pelvic ring fracture is bad news.

Diagnosis

In a patient with a suspicious mechanism, treat without a diagnosis.

Do not examine/rock the pelvis – this might disturb a blood clot and lead to disaster.

CT will almost always show the fracture (but see below).

Treatment

All patients with suspected pelvic fracture should have a pelvic binder applied. If it's on when they arrive, check it is in the right position. If it isn't on, then put it on the trauma mattress before the patient is transferred, so that they will be on it when you break the scoop.

You will be shown how to do this on the course.

The pelvic binder:

- Helps to restore the anatomical position, making bleeding less severe.
- Reduces volume in the pelvis.
- Prevents gross movement of broken bits.

It is sometimes so good that the CT appears normal! This is pretty rare, but if you really think the pelvis might be broken, and the CT is normal, then be super careful. Position an x-ray machine over the patient. Undo binder and take the picture. Do up the binder again, and wait for the x-ray result.

If the patient is bleeding from a vessel in the pelvis, this is best treated with interventional radiology. In most cases this means going to the MTC. But if you are lucky there may be a interventional radiologist on duty in your hospital: check on the day.



Summary

Priorities

1. Look after the patient's Airway, Breathing, and Circulation as per any trauma
2. Pelvic binder comes under (Catastrophic Haemorrhage) i.e. first.
3. CT will usually give the diagnosis so work on getting patient to CT ASAP.
4. Pelvic fractures generally go to the MTC.

Put the pelvic binder on the trauma mattress before the patient is moved onto it!

References / Further Reading:

1. ATACC Manual 2014
2. ATLS manual 9th Edition
3. http://www.uptodate.com

18 - Head Injuries

Author: Dr Gareth Rhys Thomas, ST3 Anaesthetics, YGC

Lead Author: John Glen, Intensivist, YGC

Originally written: April 2017

Most recent review: March 2019

Introduction

Head injury is the commonest cause of death in all patients under the age of 24 in the UK. Of the patients that do survive severe head injuries many will be disabled for life.

When patients with head injury arrive at North Wales hospitals, we can make a huge difference to their chances, even though we don't do brain surgery here. Following initial treatment and stabilisation, a lot of these patients will go to our designated Major Trauma Centre in Stoke.

Classification of brain injury

Brain injury can be **primary** or **secondary**. Primary brain injury is the damage that is done at the time of injury. The only thing we can do about this is prevention – like bike helmets and road safety.

Secondary brain injury what happens if we don't get it right in the ED.

Specific injuries

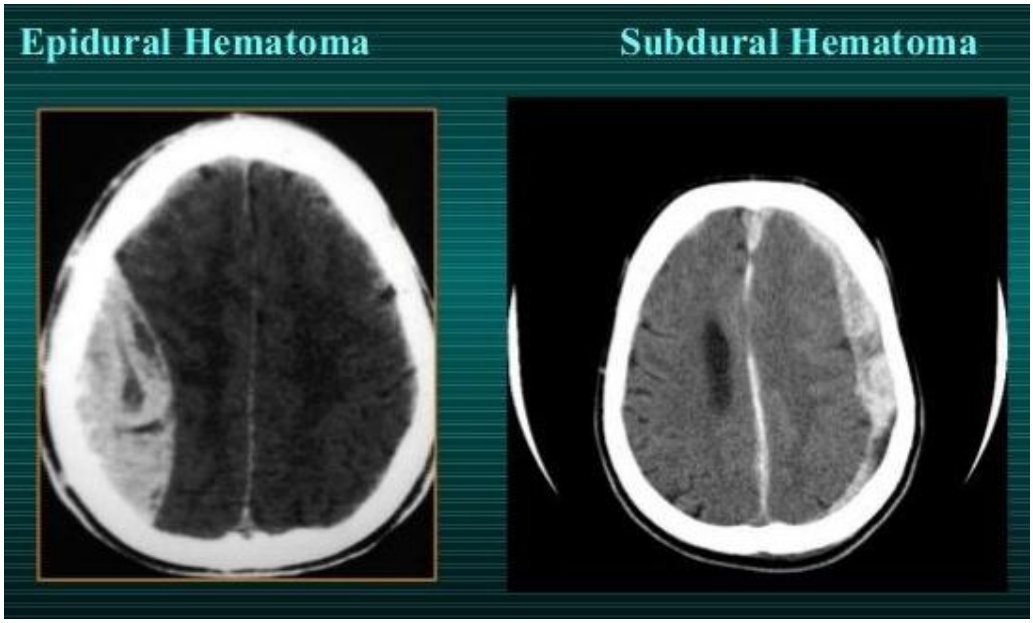
Skull fractures: Give an indication that the patient had a serious injury.

Contusions (bruising): are caused by the brain bouncing off the inside of the skull. Generally occur with high forces such as in traffic accidents or falls. Often treated with observation.

Extradural (or epidural) bleed: often caused by a torn artery next to the skull. Patient may get knocked out, wake up again, then gradually lose consciousness as the bleed gets bigger. This is called 'Talk and Die'. On CT the bleed is biconvex (a bit like an egg – egg-stradural!). Needs urgent surgery in Stoke.

Subdural bleed: caused by tearing of veins in the space between the brain and its covering. This space is often bigger in the elderly and alcoholics, so they are at higher risk for the injury. However, they can often compensate as there is more space for swelling. Younger people are often more clinically serious with coma from the outset. On CT the haematoma is concave (new-moon shaped). Needs urgent surgery in Stoke.

Diffuse Axonal Injury: In a serious impact, the brain can rattle around and twist inside the skull. This stretches and tears really important nerve fibres. The mildest form is called 'concussion'. Sometimes you don't see this injury on the first CT. But if it is severe, it can cause serious disability and death. The brain can swell up over a few hours, cutting off its own blood supply. Treatment is difficult.



Intracranial Pressure

The skull is a rigid box of bone. It contains brain tissue, blood, and the fluid around the brain (CSF). If something starts taking up extra room in the box, then something else has to be squeezed out, and eventually the pressure in the box starts to go up (this is called the Monro-Kellie doctrine). The pressure in the box is called **intracranial pressure**.

In head injury two things take up extra room: the swollen brain, and the bleeding inside the skull. Initially, CSF can be displaced into the spinal canal. After this blood vessels get squeezed until very little blood can get in. Finally the brain gets squashed out through the hole at the bottom and the patient dies (this is called **coning**).

If the pressure in the box is higher than the blood pressure, then blood can't get in to feed the brain. So the blood pressure needs to be high enough. The height of the blood pressure above the pressure in the box is called the **Cerebral Perfusion Pressure (CPP).**

CPP = Mean Arterial Pressure (MAP) minus **Intracranial Pressure (ICP)**

Assessment

Get as much history as possible to get an idea of the trend of neurological status. Make a note of:-

- GCS (particularly motor score) BEFORE sedation/anaesthesia (see chapter 12 or www.glasgowcomascale.com)
- Pupils

Initial Management

Avoiding the secondary brain injury described above is called **Neuroprotection**. It consists of:

Airway Protection: Anyone with GCS<8 requires intubation for airway protection, to ensure safety and manage the O_2 and CO_2 levels.

Prevent high (or low) CO_2: you want a 'low normal' arterial CO_2 of 4.5-5kPa on blood gas. If it is higher, then the ICP can go up due to the blood vessels being too big. If it is lower, then the blood vessels get too narrow and blood can't get in to the brain.

Prevent Low O_2: Episodes of hypoxia can be very damaging. Sats should be > 95% and $Pa0_2$ >13kPa.

Prevent Hypoglycaemia: Brain needs glucose to function.

Treat Seizures: Seizures increase oxygen requirements of the brain, so make any damage worse.

BP: Generally target a mean arterial pressure (MAP) of 80-90 mmHg or a systolic BP $\geq$ 110mmHg (any episodes of hypotension can be extremely damaging). But if the patient is bleeding out from somewhere that may have to take priority and a slightly lower BP accepted until this is controlled.

Intracranial Pressure: In addition to altering CPP there are other ways that ICP can be lowered:-

Good Venous Drainage: Patient should be head up at 30 degrees. Any c-spine collar should not be tight and ETT should be taped (not tied) so that venous drainage is not impaired.

Sedation/Paralysis: Patient should be well sedated and paralysed to prevent coughing or straining causing surges in ICP. The effect on BP of heavy sedation can be offset with vasopressors.

Rescue Measures

If there is evidence of the patient rapidly deteriorating with suggestion that they may be coning such as a blown pupil the patient should be discussed with neurosurgery and sometimes they will suggest ways to lower ICP further:-

Osmotic Agents: Mannitol or hypertonic saline can be administered to further lower ICP.

CO_2: The patient can be ventilated more aggressively aiming for a $PaC0_2$ 4-4.5kPa.

Don't let the patient become hypotensive or hypoxic. Even short episodes can be hugely detrimental to their recovery.

These patients are time critical. Every effort should be made to get them to CT, discussed with neurosurgery and transferred to Stoke as quickly (& safely) as possible.

Summary

- Management is targeted at preventing further brain damage (secondary brain injury)
- Initial management should always be the standard trauma assessment of ABC approach.
- Intubate if severe head injury requiring transfer or GCS <8.
- Don't let patient become hypoxic, hypotensive, hypoglycaemic.
- Treat seizures if they occur.
- Aim to transfer patient as quickly as is safely possible – early neurosurgical intervention can safe lives and prevent lifelong disability.

References

1. UptoDate: Management of acute severe traumatic brain injury
2. Brain Trauma Foundation guidance; 2016

19 - Spine Injuries

Authors: Dr Nicola Boyer, Anaesthetic SpR, YGC & Ms Juliet Clutton, Orthopaedic SpR.

Lead author: Dr John Glen, Consultant, ICM, YGC.

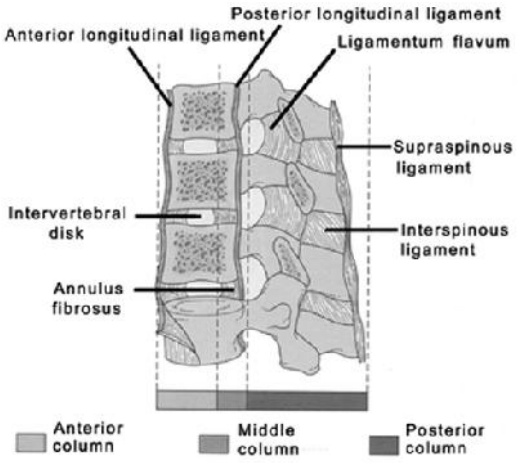
Originally written: April 2017

Most recent review: March 2019

Introduction

More than 500 people in the UK get a spinal injury every year. Half of these injuries involve the neck. About 1 in 20 of the multi-trauma patients you see will have a significant spinal injury.

North Wales does not have neurosurgeons, so patients with spinal **cord** injuries will need to be transferred. Most unstable **bony** injuries will also need transfer out.



Anatomy

The spine is made of 26 bones (vertebrae), held together by very strong ligaments at the front, middle, and back. There are: 7 cervical (C1-C7); 12 Thoracic (T1-T12); 5 Lumbar (L1-L5), a Sacrum and a Coccyx. The sacrum and coccyx are generally considered part of the pelvis.

The spine is divided into thirds, or "columns": front, middle and back. As long as only one of these is injured the spine is still stable. If two or three columns are injured in the same place then the spine is unstable and must be protected.

Injury can be to the **bones**, to the **spinal cord**, or both. In addition, people are keen to differentiate stable from unstable injuries as this has implications for how the patient is nursed.

Injury may occur due to penetrating or blunt trauma, and is classified according to the structures injured. Cord injury due to penetrating trauma is extremely rare, but may occur without bony injury.

The most important injury types are discussed below.

Bony Injury

C-spine

Atlas (C1)

- Atlanto-occipital dislocation (skull separates from neck) - rarely survives to hospital

- Jefferson's Fracture - axial loading (landing on your head) causes the ring of C1 to be compressed and "burst" into fragments - this is potentially unstable.

Axis (C2)

- Odontoid fracture - fracture of the peg that sticks upward and articulates with C1. Fragments can move and compress the cord. This is unstable.

- Hangman's fracture - fracture of the connection between the body and spinous process due to hyperextension. This allows movement, crushing the cord.

C3-C7

- C5-C6 is the most mobile part of this region of the C-spine, therefore fractures are more likely to be here than anywhere else in the C3-C7 region.

T-spine

- Anterior wedge compression fractures are most common, and are stable injuries.

- Chance fracture - transverse through vertebral body, often due to lap belts and associated with retroperitoneal and abdominal injuries

Thoracolumbar junction (T11-L1)

- Commonly injured area due to the change in mobility between the thoracic and lumbar spine. May cause damage to the tip of the spinal cord resulting in bladder and bowel dysfunction.

L-spine

- Similar pattern to T-spine

Injuriy in one area of the spine increases the likelihood of an injury in a different area.

Search for the second injury!

Cord Injury

Cord injury may be primary (occurs at time of trauma) or secondary. Secondary injury may be due to hypoxia, hypotension or movement of unstable bony fragments. This can be prevented by proper resuscitation and immobilisation of the spine.

Classified according to **level** and **completeness**.

Level

The area above which there is normal sensory and motor function. There may be a small region below this with partial function, this is called the zone of partial preservation.

Completeness

Any sensation or movement below the level of injury is due to an incomplete injury.

Spinal Cord Syndromes

Some patients get distinct patterns of motor and sensory loss that occur due to injury to a specific part of the spinal cord. Rare.

Spinal Cord Injury Without Radiographic Abnormalities (SCIWORA)

This is also rare: there are signs and symptoms of cord injury without any fracture. It usually occurs in the C-spine region. It is thought to result from either bruising or temporary artery occlusion.

SCIWORA is more common in children, due to their heavy head and mobile ligaments. Basically they get an abnormal range of movement without breaking anything.

If it does happen in adults, it tends to happen in the elderly after relatively low-energy trauma.

Spinal Injury Management

Maintain immobilisation until a spine injury has been excluded, but this may need to be relaxed/abandoned in certain situations for the safety of the patient.

Most polytrauma patients will have a full-body CT, which includes the spine. All patients need a neurological assessment as part of the secondary survey. This should not delay treatment of urgent and life-threatening problems, but you must hand this over if it has not been done. The neurological assessment should examine tone, power of the main movements of the limbs, reflexes and sensation.

In patients not requiring CT use NEXUS or Canadian C-Spine Rules to evaluate the C-spine. (See Appendix A.) Click here for an overview and comparison of the two.

Log-roll to palpate the rest of the spine. Any areas of bony tenderness should be imaged.

C-spine collars

Properly fitted collars may reduce movement. But there is little evidence of benefit and increasing evidence of harm. Badly fitted collars block airways and blood vessels, cause pressure sores, do not reduce movement, and cause an abnormal position. They are also incredibly uncomfortable. So use only well-fitted collars and blocks, be mindful of the risks, and remove if causing trouble.

Compliant/unconscious patients

- Use properly fitting collar and blocks
- When moving the patient use manual, in-line stabilisation (MILS)
- If the patient is very uncomfortable consider releasing/removing the collar, while retaining the blocks and instructing the patient not to move
- If the patient needs to be anaesthetised, always release the collar and use MILS as collars can convert a normal airway into a “can’t intubate” situation
- Once anaesthetised, or if obtunded, consider loosening/removing the collar after blocks have been fitted to improve cranial venous drainage

Non-compliant patients

- Try reasoning with the patient: making them more comfortable or having someone sit with them may calm them down.
- If you are unable to reason do not use force: this will result in extreme and potentially dangerous movements. Remove the collar and allow them to adopt whatever position they find comfortable Patients with serious neck injuries will protect themselves.

Tips to improve compliance with collars

1. **Pad all pressure points, especially the back of collar impacting on the occiput**
2. **Ensure the patient's ears are in the "holes" in the blocks**
3. **Always "stand" directly over the patient when talking to them, instruct visitors to do this**
4. **If stable, tilt (not break) the bed to 45° so they can see the world around them**
5. **Try not to leave them alone and make sure they have a call bell**
6. **Offer anti-emetics, feeling sick while "tied" to a bed won't be tolerated**

Referral for adults:

Spine injury in association with other injuries –

Refer to Stoke TTL.

Isolated spine injuries

- Bony injury **without** cord involvement:
  - If stable injury discuss with local orthopaedic team
  - If unstable refer to Stoke TTL
- Bony injury **with** cord involvement
  - Discuss with Stoke TTL. These patients are managed by both Stoke and Oswestry, Stoke TTL will decide appropriate destination.

Summary

1. Manage using a <C>ABCD approach
2. Maintain immobilisation, if possible.
3. Clear spine/identify injuries clinically or radiographically as soon as possible
4. Examine neurologically to exclude SCIWORA

References / Further Reading:

1. ATACC Course Manual 8th Edition
2. ATLS 9th edition
3. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4262055/
4. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3949434/
5. http://www.heftemcast.co.uk/cervical-spine-collars-immobilisation/
6. Overview of NEXUS and Canadian C-Spine rules

Appendix A

20 - Limb Injuries

Author: Dr John Glen, Consultant Intensivist, YGC

Originally written: April 2017

Most recent review: March 2019

Introduction

Injuries to the limb can be alarming, and they have a significant impact on the patient. It is important to find and document all injuries in a polytrauma patient, even if it seems less important at the time. A missed extremity fracture can be the thing that sets back recovery and rehabilitation.

Some limb injuries need do urgent treatment – particularly those that are presenting a threat to perfusion, or which are bleeding heavily. Complex injuries are often sent to the MTC.

Diagnosis

The mechanism of injury often gives a clue. For example, a patient who has jumped from a height and landed on their feet may have broken a bone in their foot (as well as everything else!) The patient (if conscious) needs to be asked where they are hurt, and needs a top-to-toe examination. In a polytrauma patient this can be deferred until the patient is stable and there is time.

Specific problems

Haemorrhage

If there is significant bleeding from a limb, apply direct pressure. Place an appropriate bandage and apply pressure through it. Realign fractures. Tourniquets will buy time, but need to come off eventually. On the skills course, you will be shown how to apply the combat application tourniquet, and you will be shown novel haemostatic dressings which really help to stop bleeding..

Fractures and dislocations - closed

Used to be called *simple*. This is a broken bone or dislocation or both without any associated wound.

Management

Assess pulse/capillary refill/feeling further down the limb. If absent, one attempt to put in place by someone senior. Then:

- Splint in current position and x-ray (AP and Lateral)
- Put into place and re-splint
- X-ray again, assess pulse again.

Open fractures (used to be called compound)

This is a broken bone with a wound. The bone doesn't have to be sticking out – any break in the skin at or around the fracture site will do.

Management

- Document. Take a picture if possible before wound is dressed.
- Remove gross contamination (e.g. grass).
- Saline soak dressing.
- Tetanus status; Antibiotics as per Microguide.
- Manage fracture as above.

Patient with open fractures are likely to be referred to the MTC by the orthopaedic surgeons.

Amputations

- Stop bleeding as above.
- If possible, apply chlorhex/saline-soaked dressing to stump (not iodine – degrades CT image)
- Amputated part: irrigate with saline, wrap in saline gauze, put in a bag with ice (**Avoid direct contact with ice, and do not freeze**)

Compartment Syndrome

This is when injured muscles swell up, and cut off their own blood supply due to high pressure. It happens in places that have the muscles enclosed by strong tissue: the shin, the foot, the forearm the hand, the gluteals and the thigh are possible sites.

Diagnosis

You need to think of it. Things that suggest compartment syndrome are:

- Very painful indeed, and more pain when muscle stretched.
- Altered sensation.
- Muscle feels tense.

Loss of pulse is a very late sign, and you really want to have diagnosed it before this.

Diagnosis confirmed by measuring pressure through a needle: treatment is urgent surgery.

Limb Trauma Summary

Priorities

1. Arrest haemorrhage.
2. Ensure distal perfusion.
3. Splint and image.
4. Some patients need to go to the MTC.

Don't forget Tetanus status



References / Further Reading:

1. ATACC Manual 2014
2. ATLS manual 9th Edition
3. http://www.uptodate.com

21 - Maxillofacial Trauma

Author: Olly Ayles, Critical Care Practitioner, EMRTS Cymru

Lead Author: Dr John Glen, Consultant Intensivist, YGC

Originally written: April 2017

Most recent review: March 2019

Introduction

Maxillofacial injuries can be complex, involving several sub specialties, and can have a long-term impact on the patient. This makes them both hard to get right, and important to get right. By far the largest aetiology is assault (often involving alcohol), followed by road traffic collisions and sporting accidents. Severe facial injuries be distressing for those assessing and treating them, which can divert attention away from other serious injuries.

Types of Injury

Soft Tissue Injuries

Rarely life threatening, but bleeding can be profuse as the face and head have a good blood supply. Blood is a particular gastric irritant when swallowed: as little as 20ml can cause vomiting.

Maxillofacial Fractures

Mandibular

Fractures and dislocations of jaw occur with trauma to lower third of the face: suspicion should be high with displaced teeth. An unconscious patient with bilateral fractures is an airway risk.

Nasal

Most common facial fracture characterised by displacement, swelling and tenderness. Bleeding can come out of the nose or down the throat.

Zygoma and Orbital

Characterised by flattened cheek with possible sensation loss to surrounding area extending into lip. Signs include nosebleed, poor mouth opening, change in vision and paralysis of upward gaze.

Le Fort Fractures

Maxillary fractures of midface are associated with high energy transfer blunt trauma causing facial instability. Depending on where the fracture lines are, the whole face can be collapsed. Higher grade injuries with gross instability and haemorrhage will require intervention to stabilise in the prehospital or ED environment.

Management of facial injuries

- Address life threatening injuries, as per standard approach described elsewhere in this course.
- If conscious the patient should be nursed in a position comfortable for them: this will likely be sitting up to allowing postural draining.
- Airway management may be challenging. If there is bleeding into the airway then 2 suction catheters may be needed to clear the view.
- Prevention of infection: follow advice from ENT/Maxfax and antibiotic guidelines.
- Investigation should include at least 2 x-ray views and/or CT scan.
- Ensure early review by ENT or maxillofacial surgeons.
- Patient may need ophthalmology input.

Specific management of Le Fort fractures

When haemorrhage is torrential, and/or level of consciousness is low, a combination of airway protection, haemorrhage control and splinting is needed. This is achieved with RSI, bilateral nose packs, bite blocks and hard cervical collar.

Epistats are nasal catheters that contain 2 balloons inflated with saline which apply pressure at posterior nasal space and walls. Bite blocks are placed either side of mouth between molars that once a hard collar is in place apply pressure through lower jaw, bite blocks and hard palate.

Summary

Priorities

1. Assessment and protection of airway.
2. Establish mechanism of injury and consideration of other injuries.
3. Control haemorrhage by appropriate splinting.

Occlusion of airway from either tongue, teeth or blood is the greatest risk.

In torrential haemorrhage two suction catheters will be necessary for RSI. After RSI, bleeding can be controlled with epistats, bite blocks and cervical collars.

References / Further Reading:

1. P Ceallaigh et al. Diagnosis and management of common maxillofacial injuries in the emergency department. EMJ 2006; 23:796-7.
2. ATLS manual 9th Edition

22 - When to refer to the MTC and Options for Transport

Author: Dr Rhiannon Talbot - Consultant in Emergency Medicine, YG

Dr Mark Knights, Consultant Anaesthetist, YG & EMRTS.

Date: 05/19

Review Date: 05/21

Introduction

The National Institute for Health and Care Excellence (NICE) states that 'the optimal destination for patients with major trauma is usually a major trauma centre.' In many parts of the country the ambulance service use a triage tool to determine when to bypass the local trauma units and take the patient straight to the MTC. This is more practical when the distances to the MTC are not that great. However, due to the geography of North Wales, many trauma patients will attend our Emergency Departments for initial stabilisation. Once stabilised they may then require a secondary transfer to a MTC to receive specialist care and treatment.



Referral to MTC

It is impossible to have a guideline that will dictate who needs referring to the MTC that covers every circumstance. In general, the indications for transfer to an MTC are:

- Patients with suspected injuries that require specialist management not available at their current hospital,
- Patients with suspected injuries that require input from multiple different specialist teams.

Our MTC is Royal Stoke University Hospital for patients who have had their 16th birthday, or Alder Hey Hospital for children <16 yrs old.

Email rhiannon.talbot@wales.nhs.uk

The following are examples of indications for referral to an MTC

- Head injury with GCS <14, penetrating head injury or intracranial haemorrhage
- Injuries to large vessels (eg aortic dissection)
- Unstable pelvic injuries
- Severe chest injuries (see indications for rib fixation)
- Abdominal injuries such as liver lacerations that can't be managed locally (discuss with local surgeons)
- Open fractures with significant tissue loss (NB simple open fractures can be managed locally!)
- Unstable Spinal injuries

Patients with serious injuries that fall within the territory of the on-site specialists at your hospital may be appropriate for local management, without transfer to MTC. This needs to be discussed with senior decision makers for the relevant specialty teams in your hospital, as well as discussion with the TTL at the MTC. These patients may be re-discussed at any time with the TTL at the MTC if there are any ongoing concerns about managing the patient locally.

Isolated limb, chest wall or intra-abdominal injuries (eg isolated bowel or spleen injuries) are often managed in the trauma unit. If you refer patients with intra-abdominal bleeding it is very important that a senior surgeon (middle grade or consultant) has seen them before a final decision is made whether or not to send them.

Sometimes when our departments are busy it feels like it is the easier option to transfer the patient to the MTC, even if they could be managed locally. Make sure you don't fall in to this trap. Patients should only be transferred if indicated and it is in their best interests!

It is important to document all decision making, including reason for transfer and reasons for choosing a specific transfer option.

Imaging Transfer

In patients who are likely to require discussion with the MTC TTL, all images need to be urgently transferred via the Image Exchange Portal (IEP). The radiographers will be able to arrange this to the relevant MTC or receiving hospital.

Making the Call

If transfer to the MTC is indicated then the local TTL should make this referral to the TTL at Stoke for adults or Alder Hey for children. These phone numbers should be part of your local trauma documentation and easy to find. Before making the referral ensure you have written down the ATMIST with all important injuries, and latest vital signs. RSUH have a form for the TTL's which they

complete when accepting a referral. A copy of this is in the Appendix at the end of this chapter. It obviously helps them greatly if the information is presented to them in the same headings and sequence. If you look at it carefully you will see it is not that different to an ATMISTER!

If there is a head injury you will need to tell them the latest GCS (or pre-RSI), pupil size, reaction and any focal neurology. Make sure you ask the TTL if they want any further treatments (eg Mannitol) and record any advice that is given by the TTL.

RSUH's operational policy dictates that they will make a decision whether or not to accept within 30 minutes. Often they will just tell you at the time to send the patient ASAP.

Paediatric Patients (before their 16th birthday)

For patients under the age of 16, the North West Children's Major Trauma Guidelines should be followed. The procedure is similar to the adult patients, except that the point of contact will be the TTL at Alder Hey Children's Hospital.

When transfer is likely to be futile

Sometimes the injuries are so severe and, taking in to account comorbidities, you consider the prognosis so poor that it is inappropriate to send the patient to the MTC. We do the patient, the family, the transferring team and ourselves no favours by transferring them hours away from home for life sustaining treatment to be withdrawn.

If you feel that the patient might be in this category then a consultant in your unit must be involved in the decision-making. **Don't expect a TTL in Stoke who hasn't seen the patient to make the decision for you**; If you refer a patient, their default will be to accept them. You can still discuss with the TTL at Stoke, but make it clear that you are calling for advice and not referring.

The aim should be for senior (ie consultant) decision making, sharing the risk.

> *eg "I am calling about an 88 year old female who, 4 hours ago has fallen down the stairs and was found by relatives unconscious. She has a large intracerebral bleed and a fracture of C1. Her signs are X, She had a GCS of 3 prior to RSI. We think her prognosis is poor and our intention is to admit to our ITU for 24 hours and withdraw care if there is no improvement. We have sent her CT scans electronically for your neurosurgeons to look at. I am calling to to check whether you think this is a reasonable plan and whether you have any suggestions"*
>
> or words to that effect.

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Burns Patients

Adult patients with major burns should be discussed with doctor on-call for the Burns Unit at Whiston Hospital. Adults with burns and trauma who require the services of both an MTC and a Burns Unit should be discussed with the TTL at RSUH in the first instance, who will arrange admission to another centre should this be required.

Spinal Injuries

Certain patients with spinal fractures or cord injuries may be referred directly to the Midlands Centre for Spinal Injuries (MCSI) in Oswestry *after* they have been admitted to the inpatient specialty in the local hospital. All acutely presenting patients should be referred to the TTL at RSUH in the first instance.

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Transport Arrangements

Planning for the transfer should start as early as possible (even prior to CT) as the logistics take significant time to arrange.

Once a patient has been accepted for transfer to the MTC, the modality of transfer needs to be confirmed. The decision to transfer and the modality should be considered carefully, and must be a consultant decision involving the TTL at the local unit, the receiving team and the transfer team. All risks and benefits must be evaluated. All those who are involved with organizing the transfer of patients or conducting the transfer must be familiar with the *Welsh guidelines for the transfer of the critically ill adult*. Selection of transport mode should also consider a number of key factors:

- Urgency of the transfer
- Condition of the patient
- Geographical factors
- Weather conditions
- Traffic
- Availability of modality
- Suitable landing sites at destination, and any secondary transfers from those landing sites
- Distance
- Availability of personnel



Transport Options

Broadly speaking, the options are:

1. EMRTS transfer (8am - 8pm). Contact the Air Support Desk on 0300 1232301. EMRTS will take all critically ill / injured patients who are time critical or at high risk of deterioration. If in doubt call the Air Support Desk and the EMRTS top cover consultant will call back to discuss. EMRTS may not necessarily be able to fly and so may also need a road ambulance to be booked. They will generally advise if this is the case.

2. Road transfer (WAST Urgent Care Service Ambulance) with doctor escort (Level 3 critical care patients (ie intubated), or Level 2 patients at risk of deterioration on route). Contact Ambulance Control directly. (Don't phone 999!)
3. Road transfer without doctor escort (Stable level 1 patients)
4. HM Coastguard (S-92 SAR) with or without doctor escort. This should be in very exceptional circumstances only! (ie time critical transfer **and** A55 is closed or significant delays by road **and** EMRTS Helimed unavailable). HMCG primary duty is search and rescue. At night this option is likely to take longer than a road transfer. Consultant decision only.

Do not request a paramedic crew if there is a doctor or nurse escorting. Using a paramedic crew will remove that valuable resource from the community, and their ability to respond to 999 calls. If a nurse or doctor is going as an escort, then an urgent care or non-paramedic crew can transfer the patient without a paramedic.

Urgency of Transport

When discussing with the transferring agency (WAST, EMRTS Air Support Desk or HMCG), Ambulance transfers need to be categorised as follows;

Immediate	Urgent	Elective
Ambulance crew will be at the patient's bedside **within 30 minutes** from the time of the call.	Ambulance crew will be at the patient's bedside **within 2 hours**; at a time specified by you in the hospital.	Ambulance crew will endeavour to arrive at a time specified by the hospital.These will usually be repatriations.

For time critical cases with a doctor / nurse escort you should ask for "Immediate UCS vehicle"



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Regardless of the method of transfer the TTL will need to co-ordinate the team to expedite this process as much as possible, whilst still keeping overall situational awareness of the patient's condition. The transferring team members are likely to become task focused.

Priorities:

1. Early Discussion with the MTC TTL and transfer if in the patients best interests
2. Early discussion with EMRTS Air Support Desk
3. Contact local senior decision makers for relevant specialty e.g. Surgical Registrar or Consultant for isolated splenic injuries.
4. Document all conversations and their timings
5. Discuss urgency of transfer, and the most appropriate transport options
6. Inform the MTC TTL when patient leaves department

Don't forget to document: times of calls, who was spoken to and the outcome

Document the rationale for your decision to transfer and your choice of transport

Summary

- Major trauma patients should be transferred to an MTC if they require specialist input after discussion.
- There are multiple different transfer methods, and they all have advantages and disadvantages.
- Documentation is important, including decision making about how and when to transfer.

References / Further Reading:

1. Major trauma: service delivery. NICE Guidelines. February 2016.
2. Your departmental operational guidelines for management of trauma and transfers.
3. Designed for Life: Welsh Guidelines for the transfer of the critically ill adult. *(Ed Farley-Hills and Sue O'Keeffe.) Revision 2016.*

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Appendix - MTC referral acceptance form:

TTL TRAUMA PATIENT REFERRAL FORM

REFERRER'S DETAILS			
NAME	HOSPITAL	CONTACT DETAILS	DATE & TIME OF REFERRAL
PATIENT DETAILS			
NAME	GENDER	DOB/AGE	HOSPITAL No
BACKGROUND / MOI			
ANATOMICAL INJURIES / AIS BODY REGION			
HEAD	THORAX	UPPER EXTREMITY	

FACE	ABDO & PELVIS	LOWER EXTREMITY
NECK	SPINE	EXTERNAL (Burns and other)

PHYSIOLOGICAL PARAMETERS

RR	HR	GCS
SaO_2	BP	$ETCO_2$
PUPILS	OTHER (LACATE/VBG)	

IMPORTANT INFORMATION

WARFARIN / NOAC ? REVERSAL NEEDED ?	TXA GIVEN ? 2nd TXA STARTED ?
CHEST DRAIN OUTPUT ?	BLOOD PRODUCTS GIVEN
ANTIBIOTIC COVER ?	

TREATMENTS GIVEN

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SUMMARY OF IMMEDIATE NEEDS	
1.	2.
3.	4.
SPECIALITY INVOLVEMENT AT UHNM: (NOTE SPECIALITY eg NEUROSURGERY)	
ADVICE GIVEN/FREE TEXT	
CT IMAGES LINKED: YES NO (BY "BLUELIGHT" PACS TRANSFER)	**PATIENT TRANSFER: YES NO**

TRANSFER DECISION TIME:				**ETA AT UHNM:**		
PATIENT LOCATION FOR F/U:						

CONTACT NUMBERS	NIC: 017826474632	RESUS: 74456	SWITCH: 715444
SPECIALITIES	Radiology Reg: 75880 Max Fax: 684 ENT: 265	Neuro Surg: 71925 General Surg: via switch RIC: 72972 or 181	T&O: 71928 O&G: 688 or 689 RAC: 78181/79033 or 130
WARDS:	ARTU: 76227	X-RAY Reception: 75900	

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23 - Packaging for transfer of the Critically Injured Adult

Lead author: Dr Ed Farley-Hills, Consultant in Anaesthetics and Critical Care. YG, Bangor

Date: May 2019

Review Date: May 2021

Introduction

Packaging a severely injured patient for a transfer is the bringing together of the patient and equipment into a mobile self-sustaining unit. Everyone involved with either organising, preparing for or conducting transfers (so pretty much all of us) should ideally do the Part A and Part B of the *All Wales Transfer Course* and must adhere to the document: *'Designed for Life; All Wales Guidelines for the Transfer of the Critically Ill Adult*

http://www.wales.nhs.uk/sites3/Documents/753/Guidelines for the transfer of the critically ill adult_Approved.pdf

Intra-hospital transfer (eg transfer to CT / ICU)

This mode of transfer takes place within the hospital building. For trauma, it is most likely to be a trip from ED to the CT scanner and back. Good packaging is essential but the risks of transfer are less than inter-hospital transfer due to the short distances (metres), the slow speed (walking pace), and ability to get help quickly should this be necessary. However, self-sufficiency is still essential in case of problems (for instance being stuck in a lift). Using a standard ED trolley is acceptable in such cases.

If spinal injury is suspected, care must be taken to immobilise the spine and minimise movement. Spinal boards should never be used. Your department should have clear guidelines on the preferred method for example using a scoop or a padded mattress such as the Wolverson mattress. The C-spine should be secured with a well-fitted cervical collar, blocks and tape. The patient can then be moved relatively easily into and out of the CT scanner and onwards on an inter-hospital transfer protecting the spine at all times.

The minimum standard of monitoring acceptable for transferring a patient in the hospital is:

- ECG
- Non-invasive BP on a regular cycle <5 minutes
- SpO_2
- $ETCO_2$

Equipment that should be taken with the patient at all times include:

- Airway equipment: BVM, Guedel, Supraglottic Airway, ET tube, Laryngoscope, Bougie
- Portable suction unit
- Oxygen – ensure cylinders have enough (& ventilator if required)
- Drugs: Sedation, muscle relaxant, vasopressors, emergency drugs (eg adrenaline)
- Fluids (or blood products if required)

Always ensure you have 2 working access (iv or intra-osseous) and you can easily access them.

Inter-Hospital transfer

Transferring a critically injured patient from one hospital to another involves a significant risk. We only accept this risk because there is a clear benefit of transferring them to a place where on-going specialist care can be provided that is unavailable locally. In order to reduce the risk and facilitate a rapid and safe transfer, the process requires careful co-ordination, communication, attention to detail and sound clinical decision-making.

General principles of all transfers.

- A patient who is unstable and requires life or limb saving time critical interventions that can be done locally in the trauma unit (i.e. Damage Control Surgery) is not suitable for transfer.
- Discuss with the Trauma Team Leader in Stoke and assess degree of urgency. Urgency can be described as Immediate (leave ASAP), Urgent (within 2 hours) or Elective (not time critical)
- Consider modes of transfer and package accordingly. If suitable for EMRTS, call the Air Support Desk early. For WAST, discuss with Ambulance control state the degree of urgency (see All Wales Guidelines)
- All personnel must conform to standards set by the AAGBI and in the All Wales Guidelines
- All equipment i.e. ventilators, pumps etc. should be secured to BSEN 1789 regulations on inter-hospital transfers. Equipment should always be charged when not in use, and checked daily.
- Use the pre-transfer checklist (Appendix)
- Minimum monitoring standards are as for intra-hospital transfer. Invasive blood pressure monitoring should be used, especially in head injury cases. Temperature should also be monitored; airway pressures and pupil sizes should be recorded regularly.
- Take enough oxygen. CD Cylinders contain 460L. Calculate likely consumption and take at least twice times this volume.
- Minimise infusion pumps-no more than 3.
- Make sure that wires and tubing are tucked away to minimise the risk of snagging.
- Have at least 2 iv access, and package in such a way that you can easily get to them.
- Keep the patient warm. Especially important if transfer is by air.
- Always phone the receiving hospital when you depart with an estimated time of arrival.

- Have a method of communication with your consultant and the TTL at the receiving hospital.

There are three possible modes of transfer for inter-hospital transfer:

1) EMRTS (Emergency Medical Retrieval & Transfer Service)
2) Road Transfer in Welsh ambulance (WAST)
3) Air transfer by HM Coastguard SAR (Bristow)

The method for packaging will be slightly different for each.

EMRTS Cymru
Emergency Medical Retrieval & Transfer Service

EMRTS

EMRTS is a professional transfer and retrieval service that will have their own kit and ways of packaging. If EMRTS are doing the transfer it is still helpful to assist them to package the patient if possible but they will lead and be responsible for it.

When they arrive they will require a formal handover (use the ATMIST format) with full description of injuries and treatments given so far. They will then lead the team to assist in transferring on to their packaging. Don't just walk off and leave them to it; there is still a lot that needs to be done and many hands make light work.

- The patient will need to be transferred on to an EMRTS scoop.
- All the monitoring will need to be transferred to EMRTS monitoring.
- Any infusions will need to be transferred on to EMRTS syringe drivers
- Intubated patients will need to be transferred on to the EMRTS ventilator

Access to the patient in the helicopter is limited. It is helpful to have iv access in the left arm, with a three way tap on extension.

Sometimes EMRTS will opt to transfer by road in a WAST ambulance. They may still use EMRTS equipment or they may decide to use your local transfer trolley and equipment.



WAST

Land transfer is the most common form of critical care transfer. It is still often the quickest method for time critical cases, especially outside of EMRTS operational hours.

The transfer trolley is now an all Wales standard and is specially designed to be mounted on the footplate of a standard WAST Mercedes 515 Ambulance. For spinal patients, the Vacuum Mattress can be mounted on the trolley for road transfers.

The transfer trolley will be designed to securely hold your ventilator, syringe drivers, monitor and portable suction unit. It also holds 2 oxygen cylinders.

Normally one of your anaesthetists (plus an ODP or nurse) will escort intubated patients and unstable level 2 patients who have a risk of deterioration en route. They should have adequate competencies and experience to do such transfers and should have completed the All Wales Transfer Course.

Access to the patient in the ambulance is limited so ensure they are well packaged prior to departure and you have everything you might need for the journey. A pre-departure checklist is very helpful. An example is included in the Appendix.

It is helpful to have iv access in the right arm, with a three way tap and extension for easy access to the patient.

Bristow

HM Coastguard (Search and Rescue S-92)

Air transfer by HM Coastguard is very rare and is most often used in transfers from Bangor when there are significant traffic disruptions on the A55, which would result in a very long road transfer. It must not be used routinely and may in fact delay your transfer to the tertiary site due to land transfers at either end of the journey. The larger S-92 helicopter is unable to land on the helipad outside the hospital at Stoke.

Their primary duty is search and rescue, so availability is not guaranteed due to the possibility of being tasked to other duties. At night it is unlikely that there will be any significant time benefit by using HM Coastguard unless there are reasons why road transfer would be longer than usual. THINK before asking for HM Coastguard air transfer. Indications include:

a) Life and limb saving interventions which are time critical AND
b) EMRTS are unavailable AND
c) Heavy traffic or road works making the road journey unacceptably long OR
d) Major roads en route (eg A55) are closed for any reason

The Vacuum Mattress is the best and safest mode for packaging and can be safely used on spinal patients.

Don't forget to make every effort to keep the patient warm. Getting cold is bad in major trauma. Awake patients should be given anti-emetics.

Patient Considerations before transfer

C-Spine

Cervical collars must be properly sized and well fitted. They may cause harm otherwise. In head injuries they can result in reduced venous drainage and worsened cerebral perfusion pressure. In anaesthetised patients blocks and tape or the vacmat can be used as an alternative.

Airway (see chapter 14 on airway management)

Refer to senior anaesthetist for review of need to intubate if:

- The airway is compromised or is likely to become compromised en route.
- GCS of 8 or less definitely needs intubation. Strongly consider with any deterioration of conscious level [GCS <14] or fluctuating conscious level.
- Agitated or combative patients.
- Traumatic brain injury where tight control of CO_2 is required.
- Acute burns around the airway. Worrying signs include voice change, feeling of tightness, stridor, sore throat, difficulty swallowing saliva, tongue swelling.
- Trauma to the face or neck which may result in airway compromise
- Breathing is compromised

Breathing

- Always treat simple and tension pneumothoraces with a chest drain. Do not clamp them. Chest drains with a one way flutter valve (e.g. Heimlich valves) are ideal.

- Optimise oxygenation and ventilation with head up posture, and use PEEP. Aim for tidal volumes of 6-8ml/kg.
- Check an ABG on the transport ventilator before departure.
- In awake patients with chest injuries, ensure pain is well controlled.

Circulation

- Hypovolaemia is tolerated poorly during transfer due to the acceleration/deceleration of the ambulance. Normovolaemia *should* be aimed for prior to departure (where possible).
- Hypotensive trauma patients who do not respond, or transiently respond to fluids, are bleeding until proven otherwise.
- This must be excluded or definitively treated before transfer. Get a senior surgeon to review the patient.
- Blood can be issued in a transfer box to take with you if you feel this may be needed. Discuss with haematology. Only open the box if you intend to give the blood.

Disability

- In head injuries, remember neuroprotective principles to manage Cerebral Perfusion Pressure [see Chapter 18] . Regularly monitor pupil sizes and take mannitol or hypertonic saline in case of deterioration.

Everything else

- Make sure all the immediately life threatening problems which are within the capability of the transferring unit to stabilise, are stabilised.
- Photocopy all notes and summarise injuries and treatments given. Ensure the patient has a wristband with ID corresponding with the notes.
- Inform the relatives of the reason for the transfer and give directions of where to go and contact numbers. Don't give them unrealistic expectations.

Contact the receiving hospital with your ETA on departure

Appendix A – TTC pre transfer checklist

See: Designed for Life: Welsh Guidelines for the transfer of the critically ill adult

Pre-transfer Checklist. (For transfers to another hospital)



1. VENTILATED PATIENTS *(non ventilated → proceed to section 2)*
☐ Check ETT position and secured.
☐ $ETCO_2$ sampling after HME filter in circuit.
☐ Check PEEP and Tidal volume *(target 6-8ml/kg).*
☐ Check ABG on transfer ventilator *(adjust rate to keep $PaCO_2$ in target).*
☐ Thoracostomies or Chest Drains required? *(All suspected or confirmed pneumothoraces)*
☐ Oesophageal temperature probe in situ. Target temperature =
☐ Check pupils. Protect eyes *(as appropriate).*
☐ Neuroprotection (if applicable) as per guidelines: MAP >85mmHg, $ETCO_2$ 3.0-4.5kPa.
2. ALL PATIENTS
☐ Patient ID wristband x 2 and **visually checked**.
☐ Accepted by receiving unit specialty and ward / ITU.
☐ Transfer team appropriately dressed and briefed.
☐ Patient warming as appropriate with Blizzard blanket or alternative.
☐ Notes, investigation results (+/- referral letter) copied & patient belongings packaged.
☐ Transfer documentation started.
☐ Vascular access x 2 – all flushed, secured and accessible with 3 way tap & extension *(Helicopter → Left side, Road transfer → Right side).*
☐ Snag risk minimised (lines and fluids).
☐ NIBP in situ on regular cycles.
☐ Arterial line and transducer accessible *(as appropriate).*
☐ Monitor/syringe drivers/ventilator secured appropriately.
☐ Check position of all chest drains, lines, ET tube, ng tube *(visual or on CXR / CT as appropriate)*
☐ Catheter checked and bag emptied *(as appropriate).*
☐ Anti-emetic given *(as appropriate).*
3. EQUIPMENT TO HAVE IMMEDIATELY ACCESSIBLE
☐ Emergency Anaesthesia Drugs.
☐ Intubation pack including BVM + LMA or iGel.
☐ Primary response bag & monitoring: ECG, SaO_2, NIBP, $ETCO_2$, pen-torch for Pupils.
☐ Muscle relaxant *(double predicted amount drawn up).*
☐ Sedation *(double predicted amount drawn up).*
☐ Vasopressors *(Infusion plus bolus if required).*
4. EQUIPMENT & DRUGS
☐ Visually check battery on all equipment being used.
☐ Sufficient oxygen *(double predicted amount).*
☐ Blood box (if applicable) plus paperwork & blood bank informed.
☐ Mannitol or hypertonic Saline *(as appropriate).*
5. COMMUNICATION
☐ Air or road crew – discuss transport plans including returning to base.
☐ Team have phone with numbers of consultant, receiving unit & Money.
☐ Relatives briefed; information and directions given.
☐ Receiving hospital informed of ETA. ***Remember to update ETA approx 20 minutes pre-arrival.***

24 - Handover at the MTC

Author: Dr Mark Knights, Consultant Anaesthetist, YG & EMRTS

Date: 05/19

Introduction

On arrival at the MTC there must be a formal handover of the patient from the transferring team to the receiving Team. This is a very important handover. It is the first time the MTC will have seen the patient and it will be their only face-to-face interaction with the team from the trauma unit. Get it wrong and not only will vital information be missed potentially leading to the wrong mental model and suboptimal care of the patient, but the impression of all of the care given to the patient up to that point may be tainted. In other words, all our hard work can be undone by one poorly conducted handover. We need to make sure we get it right. For consistency, use the ATMIST format. Keep it brief. Details can be passed on individually later.

Pre-departure

Team leaders are responsible for ensuring that they have given a formal handover to the transferring team. This should be done as an ATMIST verbally, with a written summary, including all injuries found (prioritising the important ones) and all treatments given to date. It should also include relevant parts of the "AMPLE" history (Allergies, Medications, Past Medical History), investigation results and also any interventions that have not been done (e.g. secondary survey, suturing of wounds, reduction of fractures, etc.).

Don't do this whilst the transfer team are busy packaging the patient. Set aside a time to handover and make sure you have their full attention.

During transfer

During the transfer your prime concern should be looking after the patient. Your secondary job is to keep a good record of the patient's vital signs and any interventions en route. You should also have some time en route to make sure you know all of the patient's injuries and interventions to date. Write down an ATMIST and rehearse it before you arrive.

On Arrival

On arrival, unless the patient is unstable or decompensating, resist the temptation to get in to resus (or ITU) as quickly as possible. It is worth just spending a minute to do a quick check of the patient:

1. Check the sedation is adequate, and top up the muscle relaxant if necessary.
2. Check the ventilation is fine and the $ETCO_2$ is within the required range.
3. Check the Blood Pressure is appropriate.
4. Check the pupils.
5. Check all your lines are still patent and your infusions are running.

This is time well spent. Better to arrive in resus nice and calm with a well packaged patient than rushing in to find that the cannula that the propofol was running through has fallen out and the patient is waking up.

If you are met at the ambulance or helicopter don't be pressured in to giving the handover until you are in Resus with the whole teams attention, but if the patient is unstable and you need assistance obviously say so.

ATMIST Handover

Expect there to be lots of people in the resus / ITU bay. Don't be intimidated. (Easier said than done, but they are all on your side).

Ask who is the team leader and, if the patient is stable, ask if they would like to transfer the patient across first before handover.

When ready for handover, get out your written prompt and make sure everyone is listening.

Tips:

- Stand where the whole team can see you.
- Speak loud, clear and confidently.
- You are delivering the handover to the whole team, but make eye contact with the Team Leader.
- Start by getting everyone's attention "OK everyone, listen in for an ATMIST"
- Aim to complete the ATMIST in 30 - 60 seconds.
- Read out each subtitle ie "**Age** – 25 year old female, **Time of Injury –** 3 hours ago, **Mechanism** – Cyclist hit by car, 30mph, etc…"
- **Injuries sustained** – start at head and work down to feet. No need to include normal findings. Just stick to what the positive findings are.
- **Signs and symptoms** – work through as ABCDE but summarise, rather than full details of all vital signs eg "On arrival airway was compromised, saturations were low and he was significantly shocked with heart rate of 150, etc…"

- Describe the GCS rather than read out numbers eg" Eyes - open to pain, Motor – localises to pain, voice – incomprehensible sounds giving a GCS of 8." - No-one is any good at translating numbers on the spot.
- **Treatments so far** – include all pre-hospital and hospital interventions. Work through CABCDE. Eg "Pre-hospital the Patient has had pelvic binder, 1g TXA and 250ml crystalloid. In hospital she has had RSI, ventilated on 40% oxygen, chest drain on the left side, 2 x units of packed red cells, 2 x 14G cannulae and an arterial line on the right arm, TXA 1g infusion commenced, hypertonic saline at 16:00, splint to right leg, iv augmentin for open fracture"
- Remember to add any relevant **Allergies, Medications and Past Medical History** (keep it short – it doesn't need to be a comprehensive list)
- Ask the team if they have any **questions**.

After your handover hands all the photocopied notes to the team leader / scribe and ask the scribe if they need any details or clarification. Collect all your kit and thank the team. Make sure you leave a copy of your transfer documentation, and bring a copy back to your unit for the patient records and for audit purposes.



Every handover of care involves a risk to the patient. All of our good work can be compromised by a poorly conducted handover.

Summary

- Team leaders must take the time to handover to the transfer team properly.
- The transferring team must have a good understanding of the patients' injuries and treatments.
- Take the time to ensure you arrive at your destination controlled with a well-packaged patient.
- Deliver your handover confidently and succinctly.
- Allow opportunity for questions.

25 - Trauma in children

Author: Dr Pete Williams Consultant in Paediatric Emergency Medicine & Pre-hospital Emergency Medicine, Ysbyty Gwynedd & EMRTS.

Date: 05/2019

Review Date: 05/2021

Introduction.

Trauma in children is stressful and emotive for all involved. Thankfully severe trauma in children is rare, even in dedicated paediatric Major Trauma Centres, but this unfamiliarity can lead to stress, delays, poor management, and poor outcomes. Our nearest paediatric MTC is Alder Hey in Liverpool. Local ambulance crews and parents with an injured child are not going to take them straight there. They will almost certainly come to your hospital first. Although thankfully rare, we are professionally responsible to ensure we are familiar with the principles of managing sick and injured children and know our local policies and processes very well.

Differences in children.

Children range from very small to adult size. This complicates management by necessitating drug doses and equipment are specific to the size of that child (compared to most adult medicine). Use a pre-prepared aide memoir wherever possible. This will minimise errors from incorrect calculations. Our local paediatric retrieval service (NWTS) have one on their website:

http://www.nwts.nhs.uk/documentation/crashcall

The “Paediatric Emergencies” app is also a good resource.

Communication ability varies with age. Gain the child’s trust by rapidly providing effective analgesia, communicating at an appropriate level that they can understand and involve the child’s caregiver at the bedside. Don’t lie to a child, they can be very astute; if something is going to hurt don’t say it won’t.

Assessment & management.

On receipt of a standby call from the ambulance service a resuscitation room space should be made ready and a paediatric trauma call announced. Staff should take predetermined roles and prepare their equipment, drugs and paperwork. Failure to assemble the trauma team before the child arrives usually leads to a period of chaos which should obviously be avoided. (See Chapter 9)

Handover from the pre-hospital team should be conducted in a standardised manner with all members of the trauma team listening to it in order to create a shared understanding of what has happened so far (as per Chapter 11) Only massive haemorrhage, critical airway or imminent arrest should prevent this.

It is commonplace for seriously injured children to be brought directly to hospital by their caregiver leaving no time to prepare. In this instance clear team leadership and staff who know their roles are required to rapidly gain control of the situation and start treatment.

Catastrophic compressible haemorrhage: This should be addressed by pre-hospital services but as already stated, many seriously injured children are brought to the emergency department by their caregiver, bypassing the ambulance service. Direct pressure and elevation should be applied to bleeding limbs, if this is inadequate, tourniquets should be applied. Heavily bleeding scalp wounds should have manual pressure applied. Penetrating injuries not controlled by direct pressure should be packed with haemostatic gauze, such as Celox®. Beware of large dressings or towels applied pre-hospital, they can absorb a large volume of blood and hide ongoing critical haemorrhage.

Airway, cervical spine control, breathing & ventilation: Children have a number of anatomical differences which make airway management different to adults. It is outside the remit of this text to cover all aspects of managing injured children. Beware the child with noisy breathing as this indicates partial airway obstruction and requires immediate assessment by an experienced clinician. If a cervical spinal injury is possible, given the mechanism of injury, then the head should be controlled with either manual stabilisation or blocks and tape. Should the child be resisting immobilisation, allow them to adopt their own position of comfort.

Examine the chest thoroughly for any injury, asymmetry or reduction in air entry. Provide supplemental oxygen to all injured children via a well fitting non-rebreathing mask. Children with head injuries and ineffective breathing should be intubated and ventilated at the earliest opportunity to control their carbon dioxide.

Circulation, haemorrhage control and vascular access: The smaller the child the less haemorrhage they will be able to tolerate. Make a systematic assessment of the circulatory status (just as for adults) and compare vital signs to age adjusted normal ranges. Give tranexamic acid early (*Tip - the dose is the same as for paracetamol 15mg/kg*). A thorough yet rapid examination should be conducted specifically searching for concealed haemorrhage which should be treated as it is found, for example delegating a member of staff to apply manual traction to a femoral shaft fracture. Two points of vascular access should be secured. If there are no obvious veins then intraosseous access should be gained. Check with your local laboratory whether they will accept marrow samples for biochemical measurement and crossmatching.

Disability & neuroprotection: Brain injury is a serious and common cause of not only mortality but also long term disability. We can do little to treat damage which happened at the time of impact but we must prevent any further damage to the brain or spinal cord once we receive the child. Hypotension, hypoxia, hypoglycaemia and fever should all be treated aggressively. The child with a brain injury and altered conscious level should be anaesthetised, intubated and have their

ventilation controlled rapidly. A cervical collar may be omitted in favour of manual stabilisation or blocks and tape as a collar will increase intracranial pressure. If haemodynamics permit, the bed should be tilted head up 15-20 degrees. The spine should be kept in neutral alignment until a spinal injury has been excluded. Should a brain injury which requires urgent neurosurgery be identified, the Trauma Team Leader will declare the child to be time critical and every effort must be made to expedite resuscitation, packaging and transfer to Alder Hey Hospital.

Exposure & environmental control: It is necessary to completely expose patients in order to thoroughly assess for hidden injuries. However, trauma patients lose heat easily and the smaller the patient, the quicker they lose heat. Hypothermia causes shivering which is painful for broken bones and considerably increases oxygen utilisation. Hypothermia adversely effects coagulation so a cold patient will bleed more. Children are very good at compensating for blood loss and will restrict blood flow to non-core tissues, this reduction in blood flow will make those tissues acidotic. Acidosis also adversely affects the ability for blood to clot. Acidosis and hypothermia ***must*** be treated aggressively to avoid the lethal triad.



Safeguarding.



Summary.

- Apply the same trauma assessment structure.
- Be thorough and meticulous in your assessment.
- Treat shock aggressively – remember the smaller the child the less blood volume they have!
- Safeguard every child, every time.
-

References / Further Reading:

1. ABC of Major Trauma 4th Edition. Skinner & Driscoll.

2. Oxford Handbook of Emergency Medicine 4th Edition. Wyatt & Illingworth.

26 - Trauma in the Elderly

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Date: 05/2019

Review Date: 05/2021

Introduction

Trauma in older patients is becoming increasingly common as people live longer. TARN data analysed from 1990 to 2013 suggested that a "profound change in demographics has occurred" as the majority of trauma received during 1990 was predominantly young males involved in road traffic collisions. We now treat many older people with significant injuries sustained as a result of a low mechanism of injury such as a fall from standing height.[1]

The traumatic episode for the elderly patient is further complicated by factors surrounding the individual's peri-morbid functional status. Many elderly patients have complex comorbidities and subsequently a decreased physiological reserve for survival of any significant trauma.

In other words, what may seem like relatively trivial mechanism may result in major consequences. It is easy to underestimate the risk and consequently undertreat.



Picture: Google.

Pathophysiology

The standard trauma triage rules that we currently use were designed to identify trauma in young adults. These rules do not accurately predict major trauma in the elderly and whilst new triage tools are being developed clinicians must retain an awareness of the pitfalls and special considerations when managing older patients. There are several important factors for which the elderly trauma patient's physiological response may be underestimated. This could lead to misdiagnosing and under treating a potentially life threatening injury.

Predisposition to trauma: Frailty, poor balance and loss of vision all contribute towards increasing the risk of a fall in an elderly person. **The best predictor of a fall in the elderly patient is a previous fall.**

Airway and C-spine protection: Edentulous elderly patients and those with poor fitting dentures, pose more difficult airway maintenance issues and aspiration risks. Cervical spine injuries are relatively common with trauma in advancing age due to various orthopaedic complications such as osteoporosis and ankylosing spondylosis.

Breathing and Ventilation: Chest muscle weakness will have a negative impact upon ventilation and cause early fatigue. This will also affects the patient's ability to cough and clear secretions from their airway. Loss of elasticity within the bronchioles and alveoli results in poor ventilation and gas exchange making the patient prone to hypoxic events and more sensitive to hypercapnia.

Cardiovascular: Myocardial demand significantly increases during trauma. Ischaemia of both myocardium and tissue is induced relatively easily following hypo-perfusion and/or acute anaemia from blood loss. A persistently elevated serum lactate, despite resuscitative fluid treatment, is a poor prognostic indicator.

Atherosclerosis and loss of elasticity make blood vessels less responsive to any acute changes in blood pressure. Patients are more prone to syncopal events and have a poor cardiovascular response to hypovolemic shock. Chronic illness such as congestive cardiac failure or hypertension and their subsequent pharmacotherapy, make the initial assessment of the elderly trauma patients cardiovascular status more complicated. Due to this delayed response and its complications, a 'normotensive' patient may in fact be quite 'hypotensive for them' and a trend in blood pressure is often more reliable. Also, patients in this category do not tolerate "permissive hypotension" well, and may require a more aggressive resuscitation approach.

Cognition: As the number of functioning neuronal pathways decreases, an elderly patients pain perception is likely to be diminished and their localising of pain less accurate. The amount of brain tissue is reduced and as atrophy occurs the relative intracerebral space increases. Fragile veins are therefore stretched and more prone to damage. The flip side of this however is that even large volumes of intracranial haematoma (eg subdural or intracerebral bleed) can be relatively well tolerated without significant rise in ICP or brain damage. This makes it very difficult to determine prognosis on the basis of CT scans.

Never attempt to predict a bleed as "unsurvivable" on the basis of a CT scan.

Loss of cerebral mass and function also poses a difficulty with assessing cognition. Acute vs Chronic confusion should not deter the clinician from a thorough assessment of neurologic function and discussions surrounding capacity are essential.

Musculoskeletal: As mentioned above, osteoporosis and other bone thinning disorders make the elderly population more prone to fractures. Neck of femur fractures and pelvic injuries are commonly seen following a fall from standing height (<2m).

Decrease in connective tissue integrity can conceal bleeding. Loss of the 'tamponade capability' of a cavity will disguise the normal stretch and nociceptive response. An abdominal examination in elderly trauma patients often lacks sensitivity and significant injuries can be missed.

Poor skin healing is common amongst the elderly population as a whole. Trauma patients undergoing a catabolic state will further decrease their rate of wound healing. Early removal from the scoop or extrication device is advised to prevent pressure areas developing.

Loss of subcutaneous fat makes it more difficult for some patients to retain heat, this can lead to hypothermia which impairs coagulation and worsens trauma mortality.

Polypharmacy: Many older patients are on multiple medications. A full drug history is essential, particularly regarding anticoagulants or drugs affecting the cardiovascular system such as beta blockers. Be wary of drugs which might cause harm when the patient is hypotensive or hypovolaemic such as non-steroidal anti-inflammatory drugs or ACE inhibitors as these can cause acute kidney injury.

Recovery: Multiple comorbidities, polypharmacy, and frailty make managing trauma in older patients challenging. Recovery can take much longer than in younger patients and may be complicated by poor social and care support necessitating long stays in hospital or transfer to a nursing facility.

Reduced physiological reserve gives far less room for error; for example both under and over fluid resuscitation will cause harm. However, if meticulous attention is paid to individualising medical treatment and nursing care, many of our older patients can make an excellent recovery and return to independent living.

When aggressive treatment or transfer is likely to be futile.

Clearly there will be times when the combination of age, frailty, co-morbidities and extent of injuries strongly suggest that the injuries are non-survivable or that the quality of life after treatment would be unacceptable to most people. We do the patient, the family and ourselves no favours by transferring them hours away from home for their life sustaining treatment to be withdrawn. We have already stated this in Chapter 22, but it cannot be overstated - and we still transfer patients inappropriately.

If you feel that the patient might be in this category then a consultant in your unit must be involved in the decision-making. **Don't expect a TTL in Stoke who hasn't seen the patient to make the**

decision for you; If you refer a patient, their default will be to accept them. You can still discuss with the TTL at Stoke, but make it clear that you are calling for advice and not referring.

The aim should be for senior (ie consultant) decision making, sharing the risk.

> *eg "I am calling about an 88 year old female who, 4 hours ago has fallen down the stairs and was found by relatives unconscious. She has a large intracerebral bleed and a fracture of C1. Her signs are X, She had a GCS of 3 prior to RSI. We think her prognosis is poor and our intention is to admit to our ITU for 24 hours and withdraw care if there is no improvement. We have sent her CT scans electronically for your neurosurgeons to look at. I am calling to to check whether you think this is a reasonable plan and whether you have any suggestions"*
>
> or words to that effect.

Summary.

1. Beware the 'normotensive' patient.
2. Beware the 'mildly confused' patient.
3. Beware the elderly patient found collapsed in the hallway …did they fall down a flight of stairs?
4. 'Chicken or egg'…did the trauma cause the medical issue or did their acute deterioration cause the trauma.
5. Have a much lower threshold for CT imaging.
6. Don't try and determine the prognosis from the CT scan.
7. Polypharmacy – Obtain a full medication history.

The art of managing trauma in elderly patients is balancing rapid and aggressive management with a gentle, realistic and humanitarian approach. Key to this is adjusting our standard treatment regimes to provide highly patient specific, personalised medcal and nursing care.

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2. Bonne S, Schuerer DJE. Trauma in the older adult: Epidemiology and evolving geriatric trauma principles. *Clin Geriatr Med*. 2013;29(1):137–150
3. Dalton,T. Rushing,M. Escott,M. Complexities of Geriatric Trauma Patients. Journal of *Emergency Medical Services Online*. Nov 2015 *www.jems.com/articles/print/volume-40/issue-11*
4. https://lifeinthefastlane.com/what-the-elderly-should-say/
5. Skinner,D.V. Driscoll,P.A. (2013) ABC of Major Trauma. Wiley-Blackwell; 134-139

27 - Trauma in the Pregnant Patient

Author: Dr Pete Williams Consultant in Paediatric Emergency Medicine & Pre-hospital Emergency Medicine, Ysbyty Gwynedd

Date: 05/2019

Review Date: 05/2021

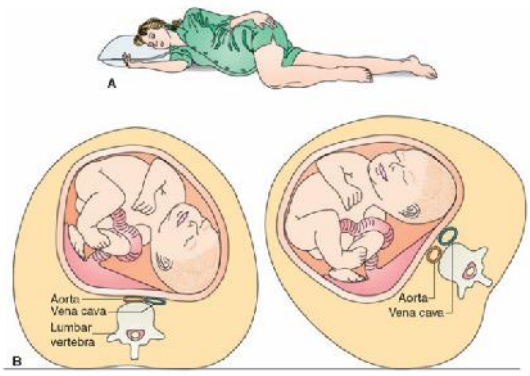
Introduction

Trauma in pregnancy, although relatively rare, complicates one in 12 pregnancies and is the leading cause of non-obstetric death in pregnant women in developed countries.

The structure of the trauma assessment and resuscitation are identical in pregnant and non-pregnant women however anatomical and physiological changes which occur in pregnancy must be taken into consideration

Changes in pregnancy

- As the uterus enlarges both it and the bladder rise out of the pelvis and both are at increased risk of injury.
- As the baby grows, the diaphragm is pushed higher into the chest and the developing baby has an ever increasing oxygen requirement. This makes pregnant patients desaturate quickly.
- Increased intra-abdominal pressure makes regurgitation and aspiration more likely.
- Cardiac output increases by 40%.
- Plasma volume increases by 50% which allows a pregnant patient to lose a lot of blood before she shows clinical signs.
- After 20 weeks (or when the uterus is higher than the umbilicus) the uterus compresses the inferior vena cava when the woman is laying flat. This will reduce venous return to the heart and therefore reduce cardiac output. Pelvic venous compression will increase bleeding from lower limb injuries.
- The bony pelvis is less prone to fracture but retroperitoneal haemorrhage may be torrential due to increased pelvic vascularity.
- The pituitary gland doubles in size and is at risk of infarction in untreated shock.



Assessment

The assessment of the pregnant patient follows the same CABCDE assessment of any other trauma victim however the above changes in pregnancy must be taken into consideration. The mother will compensate for blood loss and will preserve her own circulation at the expense of blood flow to the developing baby so signs of haemorrhage must be looked for carefully and managed aggressively.

In addition to our usual trauma blood tests it is important to know the mother's blood group in order to give Rhesus immunoglobulin to Rhesus negative mothers and a Kleihauer test should always be performed to assess for feto-maternal haemorrhage. O negative blood is safe to give, but as with any patient, fully cross-matched blood is preferable.

The pregnancy should not delay essential imaging but the Trauma Team Leader should discuss imaging requirements with the Consultant Radiologist and the patient to determine the best imaging strategy whilst minimising radiation exposure to radio-sensitive tissues such as the breasts as well as the developing baby. Ultrasound is useful for assessing the foetal heart rate and if there are concerns cardiotocography should be brought to the resuscitation room by the obstetric team.

Management

Involve the obstetric team early. Call them at the same time as the trauma team. Assign a member of staff to PULL the fundus off the midline to the patient's left side to reduce IVC compression; this must be continuous whilst the patient is supine.

Give oxygen. If RSI is indicated, take care to position optimally and be prepared for a potential difficult airway.

Secure large bore vascular access and have a low threshold for activating the major haemorrhage policy. Give tranexamic acid early and use ROTEM to guide blood product administration. Consider Fibrinogen / Cryoprecipitate early.

If chest drains are required insert them one or two spaces higher than normal, preferably under ultrasound guidance.

Abdominal injury, hypovolaemia or foetal distress may require emergency laparotomy. A consultant surgeon should be involved early.

If the woman is peri-arrest prepare for a resuscitative caesarean section. Call for a paediatric arrest team. This procedure must be performed in a timely manner in order to resuscitate the mother.

https://www.youtube.com/watch?v=Y0jQt2OVQtk

https://www.youtube.com/watch?v=hMvp5WsHY5A

Burns are managed in the same way as non-pregnant patients. Electrical injuries have a very high foetal mortality.

Priorities:

1. Pregnant women must be managed on a tilt or continuous manual displacement of the uterus.
2. The primary survey is identical to that of the non-pregnant patient.
3. Resuscitation of the mother is the best way to resuscitate the foetus.
4. Blood volume loss is often underestimated due to physiological compensation in pregnancy and signs of shock may appear late.
5. Involve the obstetric team in all injured pregnant patients
6. Resuscitative caesarean section in peri/cardiac arrest improves maternal survival: Don't forget to call the senior paediatricians.

Domestic violence is common in pregnancy and should not be forgotten as a cause of trauma.

Summary

- Apply the same trauma assessment structure.
- Call the Obstetricians as part of the team.
- Be thorough and meticulous in your assessment.
- Treat shock aggressively.
- All pregnant patients who have sustained an injury should be assessed by the obstetric team and even those who appear uninjured should be admitted for at least four hours observation.

References / Further Reading:

1. ABC of Major Trauma 4th Edition. Skinner & Driscoll.
2. https://coreem.net/podcast/episode-34-0-trauma-in-pregnancy/
3. http://www.trauma.org/archive/resus/pregnancytrauma.html

28 - Debriefs and STAG meetings

Author: Dr Rob Perry, EM Consultant, YG

Date: 05/19

Review Date: 05/21

Introduction

Debriefing is a valuable tool after a trauma call. It enables team members to discuss what went well, what did not, and how the trauma response could be improved on subsequent trauma calls.

Each North Wales hospital site holds regular Serious Trauma Audit Group (STAG) meetings. At these meetings, serious trauma cases are discussed and audited (including all patients transferred to the MTC). The performance of the trauma team is reviewed and avoidable delays flagged up by use of the DATIX system (for local issues) or the TRID (Trauma Related Issues Database) system (for issues involving the trauma network).

Debriefing

Holding an effective debriefing following a trauma call is difficult. Once the patient has left the resuscitation room, several key team members may either have left with them, or have been stood down earlier in the process. Nonetheless, if at all possible, an attempt should be made to have at least a short debrief with the team members who are available.

Debriefs should be non-hierarchical, with any team member able to voice their opinion as to how things went and how they could have been improved. The TTL is responsible for collating these opinions and feeding them back to the hospital trauma lead. They can then be discussed at the STAG meeting.

A debrief should ideally be held as soon as possible after the trauma call. However, in certain cases (particularly especially severe or complex traumas) it may be appropriate to hold a more formal meeting at a later date, when more team members can be involved.

STAG meetings

Each hospital site holds a STAG meeting approximately once a month. The meeting is chaired by the site's lead trauma consultant.

The main purpose of the STAG meeting is to review significant trauma cases seen in the hospital. All

patients with an ISS of 15+ should be reviewed, and also all patients transferred to the MTC. It is considered good practice that all trauma deaths should also be reviewed, and if possible also all patients with an ISS of 12 or more who are not transferred to the MTC.

Examples of good or poor practice are identified and fed back to the team as required. Avoidable delays, or problems with the transfer process, are documented by means of DATIX (for internal issues) or TRID (for issues involving the trauma network).

The STAG meeting also serves as the local level of trauma management. It is the appropriate forum for issues of significance regarding trauma management to be discussed. These can be fed up to the BCU Trauma Board, or to the Trauma Network, as felt appropriate.

STAG meetings are generally open to all staff involved in trauma. Contact your hospital trauma lead if you are interested in participating. If you wish to discuss a particular case, provide them with the details so that the notes can be made available for the meeting.



References / Further Reading:

1. http://stemlynsblog.org/good-to-talk-debrief-in-the-emergency-department/

29 - Trauma Audit and Research Network (TARN).

Author: Sue O'Keeffe, Critical Care & Trauma Network Manager, Wales

Date: 05/17

Review Date: 05/19

Introduction

The Trauma Audit and Research Network (TARN) is a national organisation that collects and processes data on moderately and severely injured patients in England, Wales and Northern Ireland. In doing so, it allows Networks, Major Trauma Centres, Trauma Units, ambulance services and individual clinicians to benchmark their trauma service with other providers across the country.

Background

One of the key recommendations of the NHS Clinical Advisory Group's report of September 2010, 'Regional Networks for Major Trauma' is *"all components of the Trauma Network including Pre-hospital services should submit data for all Major Trauma patients to a National Trauma Dataset (currently TARN)"*. It is also a mandatory requirement for all NHS Wales hospitals.

TARN is based at Salford Royal NHS Foundation Trust and is part of the Academic Health Science Centre at the University of Manchester. It is a national registry that collects prospectively entered data on the epidemiology, treatment and outcomes from major trauma. TARN serves two major functions: it publicly reports its findings and it allows trauma researchers to use its large dataset to conduct research.

A requirement for Peer Review, it is recommended that submission of full, accurate data to TARN should be achieved by all trauma care providers for successful performance management and clinical governance. The results of the audit should be discussed at Network audit meetings at least annually and distributed to all constituent teams in the Network.

TARN is able to benchmark an institution's mortality performance against validated large scale trauma outcome datasets and care processes against Key Performance Indicators (KPIs) respectively. KPI data is used to drive Quality Improvement processes at every level of the Trauma System.

Definitions

TARN Case Ascertainment %

- Measure of Expected vs. Submitted number of cases per Hospital
  TARN Data Accreditation %
- Measure of frequency of 'core' data field completion

TARN Inclusions and Exclusions

See full TARN Procedures Manual:

https://www.tarn.ac.uk/content/downloads/53/Complete%20manual%20-%20August%202014.pdf

In summary however:

INCLUSION CRITERIA

A. ALL TRAUMA PATIENTS IRRESPECTIVE OF AGE

B. WHO FULFILL THE FOLLOWING LENGTH OF STAY CRITERIA

DIRECT ADMISSIONS	PATIENTS TRANSFERRED IN
Trauma admissions whose length of stay is 72 hours or more OR Trauma patients admitted to a High Dependency Area regardless of length of stay OR Deaths of trauma patients occurring in the hospital including the Emergency Department (even if the cause of death is medical) OR Trauma patients transferred to other hospital for specialist care or for an ICU/HDU bed.	Trauma patients transferred into your hospital for specialist care whose combined hospital stay at both sites is 72 hours or more OR Trauma admissions to a ICU/HDU area regardless of length of stay OR Trauma patients who die from their injuries (even if the cause of death is medical) OR Patients transferred in for rehabilitation only do not need to be submitted to TARN.

C. AND WHOSE INJURIES MEET THE FOLLOWING CRITERIA

INJURY	INCLUDED	EXCLUDED
BURN	Any full thickness burn. Partial or superficial burn >10% body surface area.	Partial or superficial burn <10% body surface area.
FACIAL FRACTURE	Documented as displaced, open, compound or comminuted.	Simple, stable fracture.
FEMORAL FRACTURE	Shaft, condyle, supracondylar or head. Neck of femur <65 years old.	Neck of femur >65 years.
FOOT OR TOE: JOINT OR BONE	Massive destruction or crush injury.	Any combination of the following: Foot fracture, any number of fractured toes, metatarsals &/or tarsals, dislocated phalanges or inter-phalangeal joints, subtalar, transtarsal or transmetatarsal joints.
HAND OR FINGER: JOINT OR BONE	Massive destruction or crush injury.	Any combination of the following: Fractured hand, any number of fractured fingers, carpal or metacarpal, carpel-metacarpal, metacarpal-phalangeal or interphalangeal dislocation.
INHALATION	All included	
JOINTS (Excluding hands or feet)	Dislocation, with or without fracture.	Sprain, contusion or laceration to joint.
LIMB FRACTURE (Excluding femur)	Documented as displaced, open, compound or comminuted.	Simple stable fracture.
MUSCLE, TENDON OR LIGAMENT		All injuries to muscles, tendons & ligaments, alone or in combination with other muscles, tendons, ligaments injuries.
NERVE	Any injury to sciatic, facial, femoral or cranial nerve.	All other nerve injuries, single or multiple.

PELVIS	Sacrum, coccyx, acetabulum, ilium, ischium, sacro-ilium or symphysis pubis or pubic rami fractures.	Single pubic rami fracture >65 years old.
SKIN	Laceration or penetrating skin injuries with blood loss >20% (1litre) or with damage to underlying vessel, bone or organ. Degloving injury.	Simple skin lacerations or penetrating injuries with blood loss < 20% (1litre) and no damage to internal organs or vessels: single or multiple. Contusions or abrasions: single or multiple.
SPINE	Cord injury, fracture, dislocation or nerve root injury.	Spinal strain or sprain.
VESSEL	Any injury to femoral, neck, facial or cranial vessel. Transection or major laceration. Any other vessel injury with blood loss >20%.	Intimal tear or superficial laceration or perforation of any other vessel with <20% blood loss.

The above examples are shown for clarity and apply to injuries in isolation (except where highlighted in red) or where accompanied by skin injuries only.

Injury severity score (ISS)

Seriously injured adults and children are described as having suffered from major trauma. This is measured on a scale known as the Injury Severity Score (ISS) which scores injuries from 1 to 75, the latter being the most serious. Patients who have;

- an ISS>15 are defined as having suffered from major trauma
- an ISS of 9-15 have moderately severe trauma.

It is not possible to determine the ISS at the time of injury as it requires a full diagnostic assessment and often surgical intervention in hospital. For these reasons a system of triage is used which identifies those patients who are most likely to have had major trauma, these patients are referred to as “candidate major trauma” patients.

Pre-hospital emergency services have developed major trauma decision protocols for use by crews to determine the most appropriate destination of injured patients. Those with potential major trauma injuries (“candidate” major trauma patients) will be taken directly to a Major Trauma Centre (MTC) where travel times allow, otherwise to the nearest Trauma Unit (TU) for rapid stabilisation and transfer to the MTC where those injuries exceed the capability of a Trauma Unit and in line with local protocols.

The exact score will not be known until the entire patient data is submitted once discharged to the Trauma Audit Research Network (TARN) database for ISS coding.

- Injury coding is performed centrally at TARN
    - Ensuring coding uniformity
    - Coders check injury descriptions & imaging reports

Probability of survival

TARN now takes into account any pre-existing medical conditions, and also patient survival at 30 days after discharge (previously the model didn’t).

The biggest difference will be noticed within the elderly group. Previously, elderly patients with multiple pre-existing medical conditions often had high probabilities of survival, and any deaths would really bring down the overall survival figures. Now it is more appropriately weighted.

For example:
An 80 year old female with a GCS of 15 and a small subdural haematoma would previously have had a probability of survival of **92.6%,** regardless of pre-existing conditions. Under the improved model, the same patient's probability of survival would decrease if they had for example:

- a history of alcohol abuse, or bone cancer, or dementia, or coagulopathy, etc would reduce the probability of survival to 88.3%
- a history of dementia and coagulopathy would reduce the probability of survival to 82.6%
- a history of liver disease would reduce the probability of survival to 71.5%

TARN further group patients into seven probability of survival "bands" and those within the 80-90% band have a carry-over fraction around ***half*** that of those within the 90-95% band. This means that, if there are lots of elderly trauma deaths who had pre-existing conditions, their deaths now will not reduce the overall Ws figure as much as it used to i.e. the figure will more accurately reflect true performance.

Key Publications

TARN Procedures Manual:

https://www.tarn.ac.uk/content/downloads/53/Complete%20manual%20-%20August%202014.pdf

Summary

- TARN is a national registry that collects prospectively entered data on the epidemiology, treatment and outcomes from major trauma.
- TARN is mandatory for all NHS England and NHS Wales eligible patients.
- TARN measures completeness; expected vs. submitted cases and accreditation; frequency of 'core' data field completion.
- Injury Severity Score (ISS) scores injuries from 1 to 75, it is a retrospective score.

